

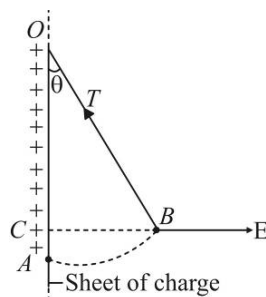
Solutions to Final Step | Part - B | Physics

Electrostatics

1.(A)

$$\tan \theta = \frac{F_e}{mg} = \frac{qE}{mg};$$

$$\tan \theta = \frac{q\sigma/2\epsilon_0}{mg} = \frac{\sigma q}{2\epsilon_0 mg}$$



2.(C) $v = u + at, u = 0; v = \frac{qE}{m}t; \frac{1}{2}mv^2 = \frac{q^2 E^2 t^2}{2m}$

3.(D) Since, the surface densities are equal, hence

$$\frac{q_1}{4\pi r^2} = \frac{q_2}{4\pi R^2} \quad (\text{where } (q_1 + q_2 = Q)) \quad \text{Or} \quad \frac{q_1}{r^2} = \frac{q_2}{R^2} = \frac{q_1 + q_2}{r^2 + R^2} = \frac{Q}{r^2 + R^2}$$

$$\therefore \quad q_1 = \frac{Q}{r^2 + R^2} \times r^2 \quad \text{and} \quad q_2 = \frac{Q}{r^2 + R^2} \times R^2$$

So, potential at the common centre,

$$V = \frac{q_1}{4\pi\epsilon_0 r} + \frac{q_2}{4\pi\epsilon_0 R} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} + \frac{q_2}{R} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{R^2 + r^2} \times \frac{r^2}{r} + \frac{Q}{R^2 + r^2} \times \frac{R^2}{R} \right] = \frac{1}{4\pi\epsilon_0} \frac{Q(R+r)}{(R^2 + r^2)}$$

4.(CD) Only option C and D are true

5.(B) In air, $F_1 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$; In medium (k), $F_2 = \frac{1}{4\pi\epsilon_0 k} \frac{q^2}{r^2} \Rightarrow \frac{F_1}{F_2} = \frac{k}{1}$

6.(D) Let, the third charge $-q$ is placed at a distance of 'x' from the charge $2q$. Then P.E. of the system is :

$$U = k \left[\frac{(2q)(8q)}{r} - \frac{(2q)(q)}{x} - \frac{(8q)(q)}{(r-x)} \right]$$

U is minimum, when $\frac{2q^2}{x} + \frac{8+q^2}{r-x}$ is maximum ; Solve for $\frac{dy}{dx} = 0 \Rightarrow x = \frac{r}{3}, \frac{d^2y}{dx^2} = +ve$

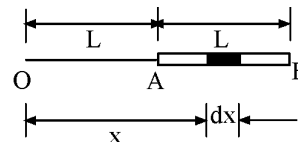
i.e. at $x = \frac{r}{3}$, y is minimum or U is maximum. So, there can't be any point between them where P.E. is minimum.

7.(C) When the two are joined by a metal wire, they become a single conductor. As charge can reside only on the outer surface of a conductor, the entire charge Q must flow to the outer sphere.

8.(D) Let an element of length dx , charge dq , at distance x from point O

$$dV = k \frac{dq}{x}$$

Where, $dq = \frac{Q}{L} dx$ $\therefore V = \int_L^{2L} \frac{k dq}{x} = \int_L^{2L} k \frac{\left(\frac{Q}{L}\right) dx}{x}$



$$= \frac{Q}{4\pi\epsilon_0 L} \int_L^{2L} \left(\frac{1}{x}\right) dx = \frac{Q}{4\pi\epsilon_0 L} [\log_e x]_L^{2L} = \frac{Q}{4\pi\epsilon_0 L} [\log_e 2L - \log_e L]$$

$$= \frac{Q}{4\pi\epsilon_0 L} \left[\log_e \frac{2L}{L} \right] = \frac{Q}{4\pi\epsilon_0 L} \ln(2)$$

9.(AB) Since $V \neq 0$, there must be charges on the surface or inside itself.

10.(C) With each rotation, charge Q crosses any fixed point P near the ring. Number of rotations per second $= \omega/2\pi$.

$$\therefore \text{Charge crossing } P \text{ per second current } \frac{Q\omega}{2\pi}$$

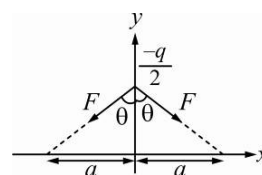
11.(C) $W = (-Q)(\Delta V) = -Q(V_f - V_i) = -Q(0 - V_i)$, $W = Q \cdot \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{Q\sqrt{2}}{a} \right) \times 4 = \frac{\sqrt{2}Q^2}{\pi\epsilon_0 a}$

12.(B) $f_{net} = 2F \cos \theta$

$$\therefore F_{net} = - \frac{2kq \left(\frac{q}{2} \right)}{\left(\sqrt{y^2 + a^2} \right)^2} \cdot \frac{y}{\sqrt{y^2 + a^2}}$$

$$F_{net} = - \frac{2kq \left(\frac{q}{2} \right) y}{\left(y^2 + a^2 \right)^{3/2}} \Rightarrow \frac{-kq^2 y}{a^3} \propto -y$$

[Negative sign indicate the net force is towards the mean position]



13.(A) $E_{(x)} = -\frac{dv}{dx}$, $E_x = (2 - 2x)$, at $x = 1$ $E_{(x=1)} = 0$

14.(BC) $E = -\frac{dv}{dr}$, since v is constant, $E = 0$ Since $E = 0$, there is no charge inside the region 15.(C)

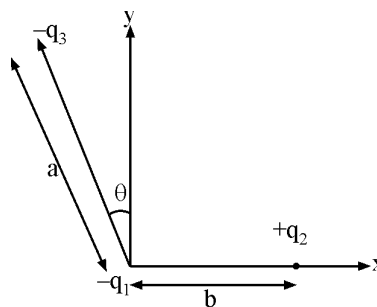
16.(B) Force on $-q_1$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{b^2} i + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{a^2} [\sin \theta i - \cos \theta j]$$

From above, x' component of force is :

$$f_x = \frac{q_1}{4\pi\epsilon_0} \left[\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta \right]$$

$$F_x \propto \left[\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta \right]$$



17.(D)

18.(C)

19.(C) Complete the cube, adding five other faces. $\phi_{\text{cube}} = \frac{q}{\epsilon_0}$, $\phi_{\text{face}} = \frac{q}{6\epsilon_0}$

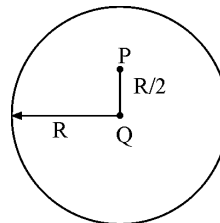
20.(C) At P due to shell, potential

$$V_1 = \frac{q}{4\pi\epsilon_0 R}$$

At P due to Q, potential

$$V_2 = \frac{Q}{4\pi\epsilon_0 \frac{R}{2}} = \frac{2Q}{4\pi\epsilon_0 R}$$

$$\therefore \text{Net potential at P, } V = V_1 + V_2 = \frac{q}{4\pi\epsilon_0 R} + \frac{2Q}{4\pi\epsilon_0 R}$$



21.(BC) Work done on an equipotential surface is zero also $W = - \int_A^B E \cdot dl$

22.(B) $V_A = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{R} + \frac{2Q}{2R} \right] = \frac{1}{4\pi\epsilon_0} \frac{3Q}{R}$ $V_B = \frac{1}{4\pi\epsilon_0} \frac{3Q}{2R}$, $V_A - V_B = \frac{Q}{8\pi\epsilon_0 R}$

23.(A) Using Gauss's theorem for radius r

$$\int \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} (Q + q)$$

$$\Rightarrow E \times 4\pi r^2 = \frac{1}{\epsilon_0} (Q + q) \quad \dots \dots (i)$$

q = charge enclosed between $x = a$ and $x = r$.

$$q = \int_a^r \frac{A}{x} 4\pi x^2 dx = 4\pi A \int_a^r x dx$$

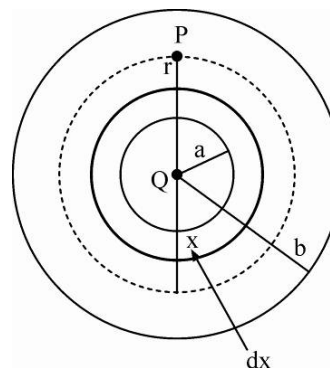
$$= 4\pi A \left[\frac{x^2}{2} \right]_a^r = 2\pi A (r^2 - a^2)$$

Putting the value of q in equation (i), we get

$$E \times 4\pi r^2 = \frac{1}{\epsilon_0} \left[Q + 2\pi A (r^2 - a^2) \right]$$

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r^2} + 2\pi A - \frac{2\pi A a^2}{r^2} \right]$$

E will be constant if it is independent of r



$$\therefore \frac{Q}{r^2} = \frac{2\pi A a^2}{r^2} \quad \text{or} \quad A = \frac{Q}{2\pi a^2}$$

24.(B) Consider a point in the emptied space whose position vector is $\vec{r}$ from the centre of the given sphere. Let the position vector of this point from the centre of cavity be $\vec{r}_2$. Let the position vector of the centre of cavity from the centre of the sphere be $\vec{r}_1$. Clearly $\vec{r}_1$ is constant. Consider the uniformly charged sphere without cavity. Electric field at $\vec{r}$ can be obtained using Gauss law.

$$E_{13} \times 4\pi r^2 = \sigma \times 4\pi r^3$$

$$E_1 = \frac{\sigma}{3} r$$

The direction of $\vec{E}_1$ vector is along $\vec{r}$, so we can write

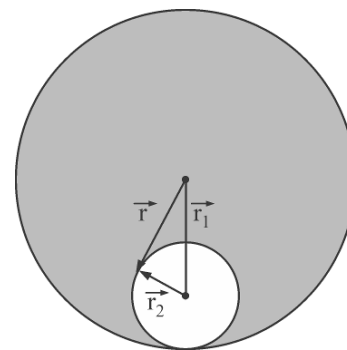
$$E_1 = -\frac{\sigma}{3} \vec{r}_2$$

Superposition of the two charges from the emptied space.

To get the net electric field due to actual charge distribution superimpose the above two fields.

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{3} \vec{r}_1$$

This implies that the electric field in the emptied region is non-zero and uniform.



25.(C) We know

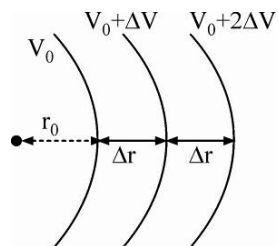
$$E = -\frac{dV}{dr}$$

Here, ΔV and Δr are same for any pair of surfaces.

So, $E = \text{constant}$

Now, electric field inside the spherical charge distribution,

$$E = \frac{\rho}{3\epsilon_0} r \quad ; \quad E \text{ would be constant if } \rho r = \text{constant} \Rightarrow \rho(r) \propto \frac{1}{r}$$



26.(BD) According to gauss law option B and D are correct

27.(C) The electric field lines around the cylinder must resemble that due to a dipole.

28.(C) Electric field due to complete disc ($R = 2a$),

$$E_1 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right] = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{\sqrt{4a^2 + h^2}} \right]$$

$$= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{2a} \right] \quad (\because h \ll a)$$

Electric field due to disc ($R = a$),

$$E_2 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{a} \right]$$

Hence, electric field due to given disc,

$$E = E_1 - E_2 = \frac{\sigma h}{4\epsilon_0 a} \quad \therefore \quad C = \frac{\sigma}{4a\epsilon_0}$$

29.(D) Let a particle of charge q having velocity v approaches Q upto a closest distance r and if the velocity becomes $2v$, the closest distance will be r' .

The law of conservation of energy yields, kinetic energy of particle = electric potential energy between them at closest distance of approach.

$$\text{Or } \frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r} \quad \text{Or } \frac{1}{2}mv^2 = \frac{kQq}{r} \quad \dots \dots \text{(i)} \quad \left[\because k = \text{constant} = \frac{1}{4\pi\epsilon_0} \right]$$

$$\text{and } \frac{1}{2}m(2v)^2 = \frac{kQq}{r'}$$

$$\text{Dividing equation (i) by equation (ii), we get : } \frac{\frac{1}{2}mv^2}{\frac{1}{2}m(2v)^2} = \frac{\frac{kQq}{r}}{\frac{kQq}{r'}} \Rightarrow \frac{1}{4} = \frac{r'}{r} \Rightarrow r' = \frac{r}{4}$$

30.(C) On outer surface there will be no charge.

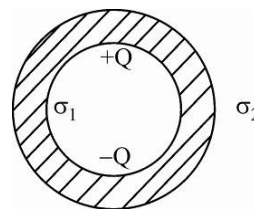
So, $Q_2 = \sigma_2 = 0$

On inner surface total charge will be zero but charge distribution will be there so

$Q_1 = 0$ and $\sigma_1 \neq 0$

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31.(AC) ACC. To Gauss law $\phi = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{Q-2Q}{\epsilon_0} = \frac{-Q}{\epsilon_0}$ no contribution of charge $5Q$ in the flux through Gaussian surface as it is not enclosed by it.

32.(B) Electric flux $\phi = \frac{q}{\epsilon_0}$

where q = total charge enclosed by closed surface

$$\begin{aligned} \therefore \phi &= \frac{1.25 + 7 + 1 - 0.4}{\epsilon_0} \\ &= \frac{8.85 C}{8.85 \times 10^{-12} C^2 N^{-1} m^{-2}} \\ &= 10^{12} N - m^2 / C \end{aligned}$$

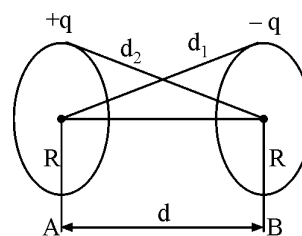
33.(B) V_A = Potential due to charge $+q$ on ring A + Potential due to charge $-q$ on ring B

$$= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{q}{d_1} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{q}{\sqrt{R^2 + d^2}} \right) \quad [\because d_1 = \sqrt{R^2 + d^2}] \quad \dots \dots (i)$$

$$\text{Similarly, } V_B = \frac{1}{4\pi\epsilon_0} \left(-\frac{q}{R} + \frac{q}{\sqrt{R^2 + d^2}} \right)$$

Potential difference, $V_A - V_B$

$$\begin{aligned} &= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{q}{\sqrt{R^2 + d^2}} \right) - \frac{1}{4\pi\epsilon_0} \left(-\frac{q}{R} + \frac{q}{\sqrt{R^2 + d^2}} \right) \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{R} + \frac{1}{4\pi\epsilon_0} \frac{q}{R} - \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{R^2 + d^2}} - \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{R^2 + d^2}} = \frac{q}{2\pi\epsilon_0} \left(\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right) \end{aligned}$$

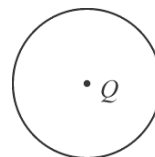


34.(ABCD)

35.(D) The flux passing through any enclosed area $= \frac{Q}{\epsilon_0}$

For a Gaussian surface, $\oint E \cdot ds = \frac{Q}{\epsilon_0}$

$$\text{or } E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \quad \text{or } E = \frac{Q}{\epsilon_0 \cdot 4\pi r^2}$$



As flux is the total number of lines passing through the surface, for a given charge, it is always the charge enclosed $Q/\epsilon_0 = \phi$. If area is doubled, the field will be half but the flux remains the same.

36.(C) In the given situation, it is a thick shell and not thin shell. So we have to consider line effect of induced charges.

$$\text{potential center } v_o = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{R} - \frac{q}{2r} + \frac{q}{3R} \right] = \frac{1}{4\pi\epsilon_0} \left(\frac{5q}{6R} \right)$$

37.(A)

38.(D)

39.(A)

40.(-80) As we know, potential difference $V_A - V_0$ is :

$$dV = -Edx$$

$$\int_{V_0}^{V_A} dV = - \int_0^2 30x^2 dx$$

$$V_A - V_0 = -30 \times \left[\frac{x^3}{3} \right]_0^2 = -10 \times [2^3 - (0)^3] = -10 \times 8 = -80 \text{ volt}$$

$$41.(2) \quad F = \frac{1}{4\pi\epsilon_0} \frac{q(Q-q)}{r^2}$$

$$\text{For maximum repulsion force } \frac{dF}{dq} = Q - 2q = 0$$

$$\frac{Q}{q} = \frac{2}{1} = 2$$

$$42.(0)$$

$$43.(0.25) \quad \text{Given } E = \frac{q}{4\pi\epsilon_0 x^2}$$

Hence, the magnitude of the electric intensity at a distance $2x$ from charge q is

$$E' = \frac{q}{4\pi\epsilon_0 (2x)^2} = \frac{q}{4\pi\epsilon_0 x^2} \times \frac{1}{4} = \frac{E}{4}$$

Therefore, the force experienced by a similar charge q at a distance $2x$ is $F = qE' = \frac{qE}{4}$

$$44.(-8)$$

$$45.(0.44) \quad \text{In vacuum, } F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} \quad \dots (i)$$

Suppose force between the charges is same when charges are r' distance apart in dielectric

$$\therefore F' = \frac{1}{4\pi\epsilon_0 k r'^2} \quad \dots (ii)$$

From eqns. (i) and (ii)

$$k r'^2 = r^2 \quad \text{or} \quad r = \sqrt{k} r'$$

In the given situation, force between the charges would be,

$$F' = \frac{1}{4\pi\epsilon_0} \frac{q^2}{\left(\frac{r}{2} + \sqrt{4} \frac{r}{2}\right)^2} = \frac{4}{9} \cdot \frac{q^2}{4\pi\epsilon_0 r^2} = \frac{4F}{9}$$

$$46.(-110) \quad \text{Change in potential in an electric field is given by :}$$

$$dV = -\vec{E} \cdot d\vec{r}$$

$$\int dV = - \int \vec{E} \cdot d\vec{r}$$

$$\text{Here, } d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\vec{E} = (25\hat{i} + 30\hat{j}) NC^{-1}$$

$$\int dV = - \int (25\hat{i} + 30\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$\int_0^V dV = - \left\{ \int_0^2 25dx + \int_0^2 30dy \right\}$$

$$V - 0 = - \left\{ 25[x]_0^2 + 30[y]_0^2 \right\}$$

$$V = [25 \times 2 + 30 \times 2]V = -110V = -110J/C$$

47.(0) $A(2, 2)$ and $(4, 1)$

$$V_B - V_A = - \int_{(2, 2)}^{(4, 1)} (y\hat{i} + x\hat{j}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = - \int_{(2, 2)}^{(4, 1)} (y dx + x dy) = - \int_{(2, 2)}^{(4, 1)} d(xy) = [-xy]_{2, 2}^{4, 1} = 0$$

$$W_{A \rightarrow B} = 0$$

48.(2) When joined by a wire, the two spheres attain common potential V .

$$\therefore \text{Intensity, } E_A = \frac{1}{4\pi\epsilon_0} \frac{q_A}{R_A^2}$$

$$= \frac{C_A V}{4\pi\epsilon_0 R_A^2} = \frac{4\pi\epsilon_0 R_A V}{4\pi\epsilon_0 R_A^2} = \frac{V}{R_A}$$

$$\text{Similarly, } E_B = \frac{V}{R_B}$$

$$\therefore \frac{E_A}{E_B} = \frac{R_B}{R_A} = \frac{2}{1}$$

49.(-18) Work done = QV where $Q = 2e$ for α -particle

$$= (2e) \times 25 = 2 \times 1.6 \times 10^{-19} \times 25 = 8 \times 10^{-18} J$$

50.(32)

DC Circuits

1.(B) Volume = $Al = 3 \Rightarrow A = \frac{3}{l}$

$$\text{Now } R = \rho \frac{l}{A} \Rightarrow l^2 = \frac{9}{\rho} \Rightarrow l = \frac{3}{\sqrt{\rho}}$$

2.(C) This is condition for balance wheatstone bridge

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \Rightarrow R_1 R_4 = R_2 R_3$$

3.(C) For series connection, $x = nR$

$$\text{For parallel connection, } y = \frac{R}{n}$$

$$\text{Therefore, } xy = nR \times \frac{R}{n} = R^2$$

4.(B) $\frac{16}{X} = \frac{4}{1/2}, X = 2$

5. (D) $P = \frac{V^2}{R}$

Resistance of the bulb,

$$R = \frac{120 \times 120}{60} = 240 \Omega$$

$$R_{eq} = 240 + 6 = 246 \Omega$$

$$\Rightarrow I_1 = \frac{V}{R_{eq}} = \frac{120}{246}$$

$$V_1 = I_1 R_1 = \frac{120}{246} \times 240 = 117.073 V$$

$$\text{Resistance of the heater} = \frac{V^2}{P} = \frac{120 \times 120}{240} = 60 \Omega$$

$$\text{As bulb and heater are connected in parallel. Net resistance} = \frac{240 \times 60}{300} = 48 \Omega$$

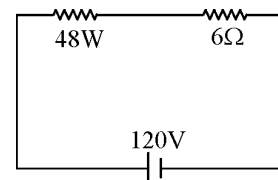
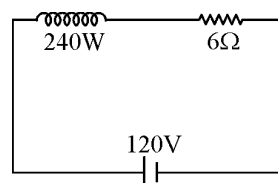
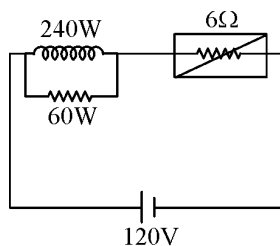
$$\text{Total resistance } R_2 = 48 + 6 = 54 \Omega$$

$$\text{Total current, } I_2 = V / R_2 = 120 / 54$$

Potential across heater = Potential across bulb

$$V_2 = \frac{120}{54} \times 48 = 106.66 V$$

$$V_1 - V_2 = 117.073 - 106.66 = 10.41 V$$



6.(A) Use formula for combinations of cells in parallel.

7.(D) $R_{\text{eff}} = R + \frac{r}{2} = \frac{2R+r}{2}, E_{\text{eff}} = E_m, I = \frac{E_{\text{eff}}}{R_{\text{eff}}} = \frac{2E_m}{2R+r}$

8.(A) Applying Kirchhoff's law, $I_3 = \frac{5}{11} A$

9.(B) $\frac{1/5}{R_{AJ}} = \frac{R_0}{R_{JB}}, R_Q = \frac{R_{JB}}{R_{AJ}} = \frac{2}{1} \times 1.5, R_Q = 3 \Omega$

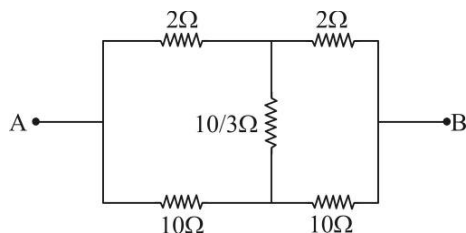
10.(C) In figure, $R_{\text{eff}} = \frac{4R}{5}$

11.(A) Only drift velocity of electrons determines the current in a conductor.

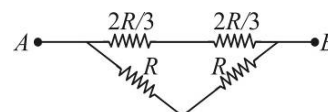
12.(C) For neutral temperature, $\frac{d\varepsilon}{dT} = 0 \Rightarrow \alpha + 2\beta T = 0$. Then, $T = -\frac{\alpha}{2\beta}$

13.(C)

14.(A)



$$\text{Equivalent circuit is balanced wheatstone bridge as shown } R_{AB} = \frac{10}{3} \Omega$$



15.(C) Resistance of the part AC

$$R_{AC} = 0.1 \times 40 = 4\Omega \text{ and } R_{CB} = 0.1 \times 60 = 6\Omega$$

$$\text{In balanced condition, } \frac{X}{6} = \frac{4}{6} \Rightarrow X = 4\Omega$$

$$\text{Equivalent resistance } R_{eq} = 5\Omega \text{ so current drawn from battery } i = \frac{5}{3} = 1A$$

16.(B)

17.(C)

$$18.(B) \quad V_0 + 10 - 5 + 10 = V_x \Rightarrow V_x = 15V$$

19.(A) Use symmetry

20.(A) Let the source voltage be V.

Equivalent resistance of the circuit when $r = fR$

$$R_{eq} = R + \frac{r \times R}{r + R} = R + \frac{fR}{f + 1} = \frac{(2f + 1)R}{(f + 1)} \quad \therefore \text{Current in the circuit, } I = \frac{V}{R_{eq}} = \frac{(2f + 1)R}{R(2f + 1)}$$

$$\text{Current in the resistance } r (= fR) \quad ; \quad I_2 = \frac{I}{f + 1} = \frac{V}{R(2f + 1)}$$

Now, heat generated per unit time in r

$$H = I_2^2 r = \frac{V^2}{2(2f + 1)^2} \quad ; \quad \text{For maximum } H, \frac{dH}{df} = 0$$

$$\Rightarrow \frac{V^2}{R} \left[\frac{1}{(2f + 1)^2} - \frac{4f}{(2f + 1)^3} \right] = 0 \quad \text{or} \quad 2f + 1 - 4f = 0 \Rightarrow f = \frac{1}{2}$$

21.(A) Let resistance be R_1 and R_2 . Then, $S = R_1 + R_2$ and $P = \frac{R_1 R_2}{R_1 + R_2}$

$$\therefore (R_1 + R_2) = \frac{n \times R_1 R_2}{R_1 + R_2} \quad [\because S = nP]$$

$$\text{Or } (R_1 + R_2)^2 = n R_1 R_2 \Rightarrow n = \left[\frac{R_1^2 + R_2^2 + 2R_1 R_2}{R_1 R_2} \right] = \left[\frac{R_1}{R_2} + \frac{R_2}{R_1} + 2 \right]$$

We know,

Arithmetic Mean $\geq$ Geometric Mean

$$\frac{\frac{R_1}{R_2} + \frac{R_2}{R_1}}{2} \geq \sqrt{\frac{R_1}{R_2} \times \frac{R_2}{R_1}} \Rightarrow \frac{R_1}{R_2} + \frac{R_2}{R_1} \geq 2$$

$$\text{So, } n(\text{minimum value}) = 2 + 2 = 4$$

22.(D) Here, $R(T) = R_0[1 + \alpha(T - T_0)]$

$$\text{At } T_0 = 300 \text{ K, } R_0 = 100\Omega \quad ; \quad T = 500 \text{ K, } R = 120 \Omega$$

$$\therefore 120 = 100(1 + \alpha(200)) \Rightarrow 200\alpha = \frac{6}{5} - 1 = \frac{1}{5} \Rightarrow \alpha = 10^{-3} \text{ } ^\circ\text{C}^{-1}$$

Temperature of the toaster is raised at constant rate from 300 K to 500 K is 30 s.

$$\text{So, incensement in the temperature in time } t = \frac{(500 - 300)}{30} t \quad ; \quad \Delta T = \frac{20}{3} t$$

Total work done in raising the temperature

$$= \int_0^t \frac{V^2}{R(t)} dt = \int_0^t \frac{V^2}{R_0(1 + \alpha \Delta T)} dt = \int_0^{30} \frac{(200)^2}{100 \left(1 + 10^{-3} \times \frac{20}{3} t \right)} dt = 400 \times \int_0^{30} \frac{dt}{\left(1 + \frac{1}{150} t \right)}$$

$$= 400 \times 150 \left[\ln \left(1 + \frac{t}{150} \right) \right]_0^{30} = 60000 \left[\ln \left(1 + \frac{30}{150} \right) - \ln 1 \right] = 60000 \ln \left(\frac{6}{5} \right) J$$

23.(B) $V = IR$

As $I = nev_d$ and $R = \frac{\rho l}{A}$ $\therefore V = nev_d \times \frac{\rho l}{A}$ or $\rho = \frac{V}{nev_d l}$

Here, $V = 5$, $n = 8 \times 10^{28} m^{-3}$, $vd = 2.5 \times 10^{-4} ms^{-1}$, $l = 0.1 m$, $e = 1.6 \times 10^{-19} C$

$$\therefore \rho = \frac{5}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4} \times 0.1} = 0.156 \times 10^{-4} \Omega m \approx 1.6 \times 10^{-5} \Omega m$$

24.(A) Applying KVL in loop PQCDP

$$-I_2 - 3I_1 + 9 - 2I_1 = 0 \Rightarrow 5I_1 + I_2 = 9 \quad \dots (i)$$

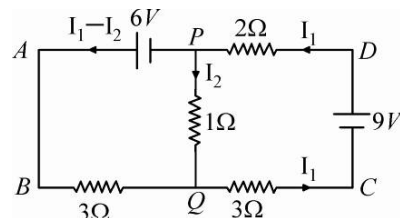
Applying KVL in loop PQBAP

$$-I_2 + 3(I_1 - I_2) - 6 = 0 \Rightarrow 3I_1 - 4I_2 = 6 \quad \dots (ii)$$

Solving eqn. (i) and (ii), we get :

$$I_1 = 1.83 A, I_2 = -0.13 A$$

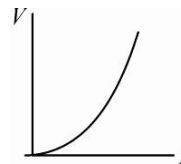
$\therefore$ The current in the 1Ω resistor is $0.13 A$, from Q to P .



25.(C) Given $v_d \propto \sqrt{E}$

We know, and $E = \frac{V}{l}$

or, $E \propto V$ So $I \propto \sqrt{V}$; $I^2 \propto V$

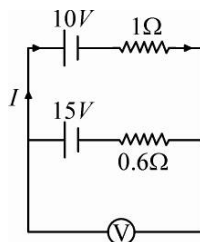


26.(C) Current in the circuit, $I = \frac{5}{1.6}$

$$= \frac{50}{16} = \frac{25}{8} A$$

Reading of the voltmeter

$$V = 15 - \frac{25}{8} \times 0.6 = 15 - \frac{15}{8} = 13.1V$$



27.(A) $\frac{R}{S} = \frac{L_1}{100 - L_1} \Rightarrow L_1 = \frac{100}{3} cm$

28.(C) Current in 4Ω is zero

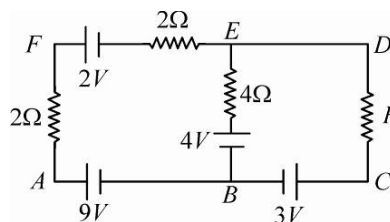
Applying KVL in loop EBCDE,

$$V_{EB} + V_{BC} + V_{CD} + V_{DE} = 0$$

$$-4 + 3 + V_{CD} + 0 = 0$$

$$V_{CD} = 1 \text{ volt}$$

$$\therefore V_A - V_D = 9 - 3 - 1 = 5V$$



29.(B) Let resistance of bulb filament be R_0 at $0^\circ C$, then from expression.

$$R_\theta = R_0[1 + \alpha \Delta \theta]$$

We have, $100 = R_0[1 + 0.005 \times 100]$

and $200 = R_0[1 + 0.005 \times x]$

where, x is temperature in $^{\circ}\text{C}$ at which resistance become 200Ω

$$\frac{200}{100} = \frac{1 + 0.005x}{1 + 0.005 \times 100} \Rightarrow x = 400^{\circ}\text{C} \quad \mathbf{30.(AD)}$$

31.(C) From $R_t = R_0(1 + \alpha t)$

$$\therefore 5 = R_0(1 + 50\alpha) \quad \dots(i)$$

$$\text{and } 6 = R_0(1 + 100\alpha) \quad \dots(ii) \quad \therefore \frac{5}{6} = \frac{1 + 50\alpha}{1 + 100\alpha} \quad \text{Or } \alpha = \frac{1}{200}$$

Putting the value of α in Eq. (i), we get : $5 = R_0 \left(1 + 50 \times \frac{1}{200} \right)$ or $R_0 = 4\Omega$

32.(8) In series, currents are equal. So, heat generated is proportional to resistance.

$$\Rightarrow R_1 = R_2 \Rightarrow \frac{\rho_1 L}{\left(\frac{\pi(2d)^2}{4} \right)} = \frac{\rho_2 (2L)}{\left(\frac{\pi d^2}{4} \right)} \Rightarrow \frac{\rho_1}{\rho_2} = 8$$

33.(6) The diagonal resistance of 2Ω is part of a balanced Wheatstone network, and hence can be removed.

$$\Rightarrow R_{AB} = 2.4 + \frac{(6)(9)}{6+9} = 6\Omega$$

34.(12) Current in the circuit, $i = \frac{E}{R+r}$

$$\text{Potential difference across the battery, } V_B = E - ir = E - \frac{Er}{R+r} = \frac{ER}{R+r}$$

Solving, we get $E = 12V$ and $r = 1.5\Omega$

35.(16) Current through 4Ω resistance, $i = \frac{12}{2+4} = 2A$

So, rate of heat dissipation, $H = i^2 R = (2)^2 (4) = 16W$

36.(285) Maximum measurable voltage for the voltmeter, $V = i_G(G + R)$

$$\Rightarrow 3 = (0.01)(15 + R) \Rightarrow R = 285\Omega$$

37.(20) When switch is open, net resistance of circuit, $R = r + 3$.

$$\Rightarrow i = \frac{60}{r+3} = 18A \Rightarrow r = \frac{1}{3}\Omega$$

$$\text{When switch is closed, net resistance of circuit, } R = r + \frac{8}{3} = 3\Omega \Rightarrow i = \frac{60}{3} = 20A$$

38.(25) $i = nAev_d$

$$\text{Also, } i = \frac{V}{\left(\frac{\rho l}{A} \right)} \Rightarrow V = n e \rho l v_d$$

$$\Rightarrow v_d = \frac{V}{n e \rho l} = \frac{96}{(8 \times 10^{28})(1.6 \times 10^{-16})(1.5 \times 10^{-8})(2)} = 25 \text{ mm/s}$$

39.(12) Current in branch containing P , $i_1 = \frac{48}{6} = 8A$

$$\text{Current in branch containing } Q, \quad i_1 = \frac{48}{8} = 6A$$

$$V_P - V_Q = +3(8) - 2(6) = 12V$$

$$40.(35) \quad R_2 = R_1(1 + \alpha \Delta T) \quad \Rightarrow \quad \alpha = \frac{1}{\Delta T} \left(\frac{R_2}{R_1} - 1 \right) = \frac{1}{10} \left(\frac{0.07}{20.00} \right) = 3.5 \times 10^{-4}$$

$$41.(12) \quad \text{In series, total power produced,} \quad P_1 = \frac{V^2}{R_1 + R_2}$$

$$\text{So,} \quad t_1 = \frac{H}{P_1} = \frac{H(R_1 + R_2)}{V^2}$$

$$\text{In parallel, total power produced,} \quad P_2 = \frac{V^2}{R_1} + \frac{V^2}{R_2} = \frac{V^2(R_1 + R_2)}{R_1 R_2}$$

$$\text{So,} \quad t_2 = \frac{H}{P_2} = \frac{H R_1 R_2}{V^2(R_1 + R_2)}$$

$$\text{Now,} \quad \frac{t_2}{t_1} = \frac{R_1 R_2}{(R_1 + R_2)^2} = \frac{\frac{R_2}{R_1}}{\left(1 + \frac{R_2}{R_1}\right)^2} = \frac{12}{169}$$

$$\text{Let } \frac{R_2}{R_1} = x \quad \Rightarrow \quad \frac{x}{(1+x)^2} = \frac{12}{169} \quad \Rightarrow \quad x = 12$$

$$42.(32) \quad \text{Potential difference across conductor A: } V_1 = \frac{E R_1}{R_1 + R_2} = \frac{E \rho_1 L_1}{\rho_1 L_1 + \rho_2 L_2} = \frac{E \rho L}{\rho L + \left(\frac{5}{4} \rho\right)(2L)} = \frac{2}{7} E = 8V$$

$$\text{Electric field in conductor A:} \quad E_A = \frac{V_1}{L} = \frac{8}{0.25} = 32 \text{ N/C}$$

$$43.(40) \quad \text{Let resistance of } AB = R$$

$$\text{Then, } i = \frac{40}{R+2}$$

$$\text{and } i \left(\frac{1 \times R}{1.2} \right) = 32 \quad \Rightarrow \quad \left(\frac{5R}{6} \right) \left(\frac{40}{R+2} \right) = 32 \quad \Rightarrow \quad R = 48 \Omega$$

So, resistance per unit length

$$= \frac{48}{1.2} = 40 \Omega/m.$$

$$44.(25) \quad \text{Let EMF of battery} = V$$

$$i_1 = \frac{V}{\frac{R_1(R_1 + 2R_2)}{2R_1 + 2R_2}} = 5.6; \quad i_2 = \frac{V}{\frac{R_2(R_2 + 2R_1)}{2R_1 + 2R_2}} = 10.5$$

Dividing the equations to eliminate V ,

$$\frac{R_2(R_2 + 2R_1)}{R_1(R_1 + 2R_2)} = \frac{8}{15} \quad \Rightarrow \quad \frac{1 + \frac{2R_1}{R_2}}{\left(\frac{R_1}{R_2}\right)\left(\frac{R_1}{R_2} + 2\right)} = \frac{8}{15}$$

$$\text{Let } \frac{R_1}{R_2} = x \quad \Rightarrow \quad 15 + 30x = 8x^2 + 16x$$

$$8x^2 - 14x - 15 = 0 \quad \Rightarrow \quad x = \frac{5}{2}$$

$$45.(3) \quad V_A - V_B = 4V \quad \Rightarrow \quad 2(2) - 8 + r(2) + 1(2) = 4$$

$$\Rightarrow \quad r = 3\Omega$$

$$46.(55) \quad R_A = \frac{220^2}{40} \quad \text{and} \quad R_B = \frac{220^2}{120}$$

In series, $i = \frac{220}{R_A + R_B}$

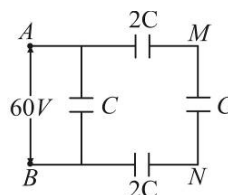
Potential difference across bulb B, $V_B = iR_B = 55V$

Capacitors

$$1.(C) \quad \frac{d_1}{K_1 A \epsilon_0} + \frac{d_2}{K_2 A \epsilon_0} = \frac{(d_1 + d_2)}{K_{eq} A \epsilon_0} \quad ; \quad K_{eq} = \frac{K_1 K_2 (d_1 + d_2)}{K_1 d_2 + d_1 K_2}$$

$$2.(D) \quad \text{Charge on capacitor } C \text{ between } MN \text{ is } = 60 \times \frac{C}{2} = 30C$$

$$\therefore V_{MN} = \frac{30C}{C} = 30V$$



$$3.(C) \quad \text{Distance between plates halved capacitance becomes doubled also charge becomes doubled,}$$

Force of attraction $\propto Q^2$. Force of attraction four times

$$4.(A) \quad \text{Electric field inside dielectric, } \frac{\sigma}{K \epsilon_0} = 3 \times 10^4 \quad \Rightarrow \quad \sigma = 2.2 \times 8.85 \times 10^{-12} \times 3 \times 10^4$$

$$= 6.6 \times 8.85 \times 10^{-8} = 5.841 \times 10^{-7} = 6 \times 10^{-7} C/m^2$$

$$5.(A) \quad \text{Since } \frac{3}{2} = \frac{6}{4}, \text{ the circuit is treated as balanced wheatstone bridge.}$$

$$\therefore C_{eq} = \frac{3 \times 6}{3+6} + \frac{2 \times 4}{2+4} = 2 + \frac{8}{6} = \frac{20}{6} = \frac{10}{3} \mu F$$

$$6.(B) \quad \text{When inner sphere is earthed } C_1 = \frac{4\pi \epsilon_0 ab}{b-a}$$

When outer sphere is earthed

$$C_2 = \frac{4\pi \epsilon_0 ab}{b-a} + 4\pi \epsilon_0 b = \frac{4\pi \epsilon_0 b^2}{b-a}$$

$$\frac{C_1}{C_2} = \frac{a}{b}$$

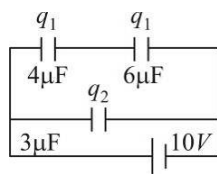
$$7.(ACD) \quad \text{Charge on capacitor } C_1 \text{ after } K_1 \text{ is closed is } C_1 E. \text{ After } K_1 \text{ is opened and } K_2 \text{ is closed, this charge gets redistributed between the capacitors such that the potential difference across both capacitors becomes equal.}$$

$$\Rightarrow \quad (i) \quad C_1 V_1 + C_2 V_2 = C_1 E \quad (\text{by conservation of charge})$$

$$(ii) \quad V_1 = V_2$$

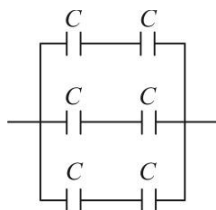
$$(iii) \quad \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

8.(C)



$$\frac{q_1}{4} + \frac{q_1}{6} = 10 \quad ; \quad q_1 = 24\mu C$$

9.(C)



Reduced circuit is, $C_{eq} = \frac{3}{2}C$

10.(D)
$$\frac{V_1}{V_2} = \frac{k_2 \frac{A \epsilon_0}{2d}}{k_1 \frac{\epsilon_0 A}{d}} = \frac{k_2}{2k_1} = \frac{3}{2}$$

11.(B) In steady state, current through battery $I = \frac{6}{2.8+1.2} = 1.5A$ $I_2 = \frac{3}{2+3} \times 1.5 = 0.9A$

12.(B) $d = 2cm = 2 \times 10^{-2}m, V = 10V, E = \frac{V}{d} = \frac{10}{2 \times 10^{-2}} = 500N/C$

13.(B) $Ce = 1 + \left(\frac{1 \times 1}{1+1} \right) = 1.5\mu F$

14.(B) Capacitors in series, $Ce = \frac{C_1 C_2}{C_1 + C_2} = \frac{C}{3}$

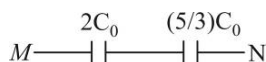
15.(CD) Basic Question

16.(A) $C_{parallel} = C_1 + C_2, C_{series} = \frac{C_1 C_2}{C_1 + C_2}$

Given $C_{parallel} = \frac{25}{6} C_{series}$ and $C_2 > C_1$

Solving $\frac{C_2}{C_1} = \frac{3}{2}$

17.(A) $\frac{1}{C_e} = \frac{1}{2C_0} + \frac{3}{5C_0} \quad C_0 = \frac{10}{11}C_0$



18.(C) $\Delta Q = (kCV - CV) = (k-1)CV = 2CV, W = \Delta QV = 2CV^2$

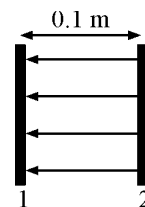
19.(A) Since, $V_2 > V_1$, so electric field will point from plate 2 to plate 1.

The electron will experience an electric force, opposite to the direction of electric field and hence move towards the plate 2.

Use work-energy theorem to find speed of electron when it strikes the plate 2.

$\therefore \frac{m_e v^2}{2} - 0 = e(V_2 - V_1)$ where, v is the required speed.

$\Rightarrow \frac{9.11 \times 10^{-31}}{2} v^2 = 1.6 \times 10^{-19} \times 20 \quad \text{or} \quad v = \sqrt{\frac{1.6 \times 10^{-19} \times 40}{9.11 \times 10^{-31}}} = 2.65 \times 10^6 ms^{-1}$



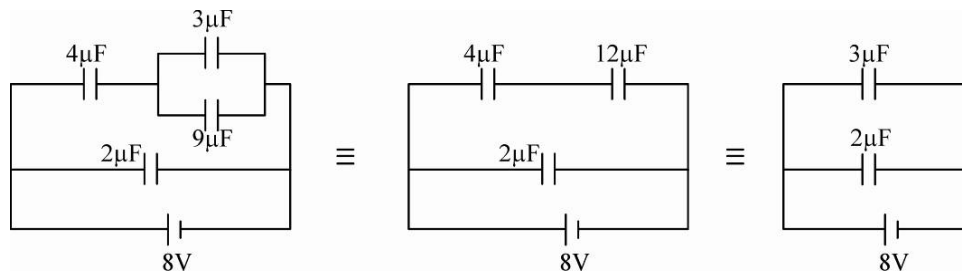
20.(C)

21.(B)

22.(B)

$$23.(A) \quad C = \frac{\epsilon_0 A}{d}, \frac{4C}{3} = \frac{\epsilon_0 A}{\left[d - \frac{d}{2} + \frac{d}{2k}\right]}, \frac{4}{3} = \frac{d}{\frac{d}{2} + \frac{d}{2k}}, 2d + \frac{2d}{k} = 3d, k = 2$$

24.(C) $3\mu F$ and $9\mu F$ are in parallel combination so that equivalent capacitance $= (3 + 9) = 12\mu F$



Now $4\mu F$ and $12\mu F$ are in series so their equivalent capacitance $= \frac{4 \times 12}{16} = 3\mu F$

Charge on $3\mu F = \times (8V) = 24 \mu C$

$\therefore$ charge on $4\mu F$ and $12\mu F$ are same ($24\mu C$) as they are in series.

Charge of $9 \mu F = \left(\frac{9}{9+3}\right) \times 24 \mu C = 18\mu C$

Required charge $Q = \text{Charge on } 4\mu F + \text{Charge on } 9\mu F$

$$Q = (24 + 18) \mu C = 42\mu C$$

Required electric field, $E = \frac{1}{4\pi\epsilon_0} \times \frac{Q}{r^2}$; $E = 9 \times 10^9 \times \frac{42 \times 10^{-6}}{(30)^2} = 420 \text{ NC}^{-1}$

25.(C) Energy stored in capacitor, $U = \frac{q^2}{2C} = \frac{1}{2C}(q_0 e^{-t/\tau})^2$

$$= \frac{q_0^2}{2C} e^{-2t/\tau} \quad [\text{where, } \tau = CR]$$

$$U = U_i e^{-t_1/\tau}$$

$$\frac{1}{2} U_i = U_i e^{-2t_1/\tau}$$

$$\frac{1}{2} = e^{-2t_1/\tau} \Rightarrow t_1 = \frac{\tau}{2} \ln 2$$

Now, $q = q_0 e^{-t/\tau}$

$$\frac{1}{4} q_0 = q_0 e^{-t_2/\tau}$$

$$t_2 = \tau \ln 4 = 2\tau \ln 2 \quad \therefore \quad \frac{t_1}{t_2} = \frac{1}{4}$$

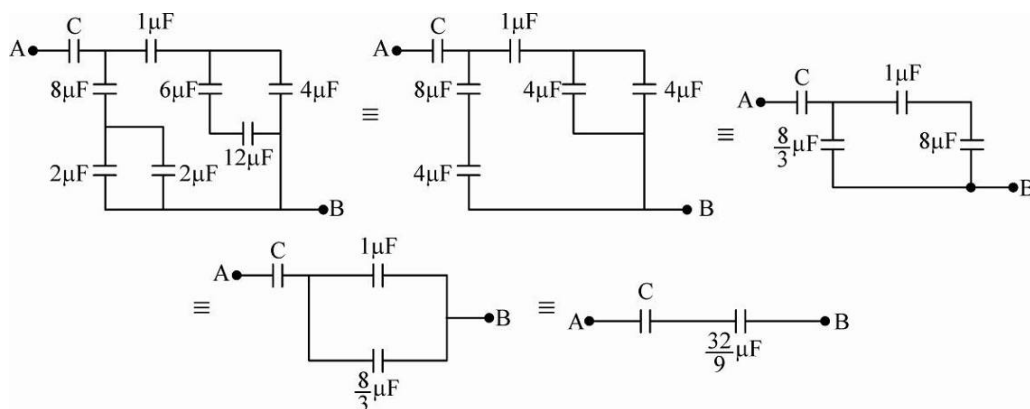
$$26.(D) \quad a: \frac{1}{C_a} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \Rightarrow C_a = \frac{4}{3} \mu F$$

$$b: C_b = 4 + 4 + 4 = 12\mu F$$

$$c: C_c = \frac{(4+4) \times 4}{(4+4)+4} = \frac{8}{3} \mu F$$

$$d: C_d = \frac{4 \times 4}{4+4} + 4 = 6\mu F$$

27.(A)



$$\text{Here } C_{AB} = 1 \mu F = \frac{C \times \frac{32}{9}}{C + \frac{32}{9}} \Rightarrow C + \frac{32}{9} = \frac{32C}{9} \Rightarrow \frac{23}{9}C = \frac{32}{9} \times C = \frac{32}{23} \mu F$$

28.(B) $W = \frac{1}{2}C[40^2 - 20^2] \Rightarrow C = \frac{W}{600}$

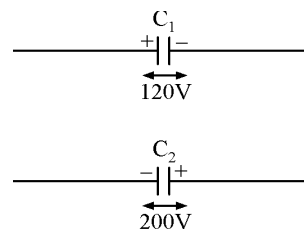
$$W' = \frac{1}{2}C[50^2 - 40^2] = 450C = 450 \left(\frac{W}{600} \right) = \frac{3}{4}W$$

29.(B) For potential to be made zero after connection, the charge of both capacitors are equal.

$$\therefore q_1 = q_2$$

$$\Rightarrow C_1 V_1 = C_2 V_2$$

$$\Rightarrow 120C_1 = 200C_2 \Rightarrow 3C_1 = 5C_2$$



30.(D) Equivalent capacitance of the circuit

$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{(1+2)} = \frac{1}{C} + \frac{1}{3} \text{ or } C_{eq} = \frac{3C}{C+3}$$

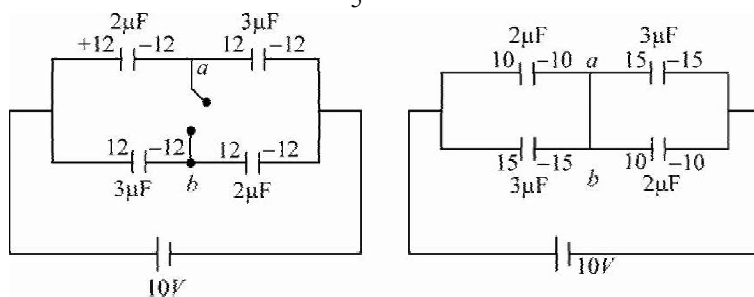
$$\text{Total charge in the circuit, } Q = C_{eq} E = \frac{3CE}{C+3}$$

Charge on the μF capacitor,

$$Q_2 = \frac{2}{3}Q = \frac{2}{3} \times \frac{3CE}{(C+3)} = \frac{2CE}{C+3} \text{ or } Q_2 = \frac{2E}{1+\frac{3}{C}} \text{ and } \frac{dQ_2}{dC} = \frac{6E}{(C+3)^2}$$

As C increase, Q_2 increases and slope of $Q_2 - C$ curve decreases. Hence graph (d) represents the correct variation.

31.(A) For upper and lower links, $C_{eq} = \frac{6}{5} \mu F$ $\therefore Q_{upper} = Q_{lower} = 12 \mu C$



On closing switch, charge on $2\mu F$ is $10\mu C$ and that on $3\mu F$ is $15\mu C$

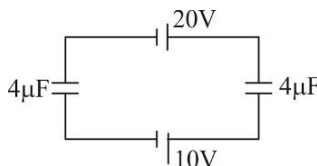
At a $q_1 = -12 + 12 = 0$

$q_f = 15 - 10 = 5\mu C$ $\therefore$ Charge $5\mu C$ flows from b to a

32.(A) The circuit can be reduced as :

$$V = \frac{10}{2} = 5 \text{ volt}$$

$$q = (4\mu F)(5v) = 20\mu C$$



33.(B) One capacitor is short circuited and the other two are in parallel.

34.(A) As initially charge is maximum $q = q_0 \cos \omega t$

$$\Rightarrow i = \frac{dq}{dt} = -\omega q_0 \sin \omega t$$

Given, energy stored in inductor $\left(U = \frac{1}{2} Li^2 \right)$ is equal to energy stored in capacitor $\left(U = \frac{q^2}{2C} \right)$

$$\therefore \frac{1}{2} Li^2 = \frac{q^2}{2C} \Rightarrow \frac{1}{2} L(\omega q_0 \sin \omega t)^2 = \frac{(q_0 \cos \omega t)^2}{2C}$$

$$\text{But } \omega = \frac{1}{\sqrt{LC}} \Rightarrow \tan \omega t = 1 \Rightarrow \omega t = \frac{\pi}{4} \Rightarrow t = \frac{\pi}{4\omega} = \frac{\pi}{4} \sqrt{LC}$$

35.(10) $i = i_0 e^{-t/RC}$

$$i = \frac{i_0}{e^2} \Rightarrow t = 2RC$$

$$= 2(10^3)(5 \times 10^{-6})$$

$$= 10 \text{ ms.}$$

36.(4) Let $C_A = C \Rightarrow C_B = 3C$

Now, in series, charge same

$$\Rightarrow C_A V_A = C_B V_B \text{ and } V_A + V_B = V_0 \Rightarrow \frac{V_0}{V_B} = 4$$

37.(5) After battery is removed, charge on capacitor $C_1 = Q = C_1 V$.

Now after connection of C_2

$$V_1 = V_2 \Rightarrow \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$\text{and } Q_1 + Q_2 = Q = C_1 V \Rightarrow Q_1 = \frac{C_1^2 V}{C_1 + C_2}$$

$$\text{So, } V_1 = \frac{Q_1}{C_1} = \frac{C_1 V}{C_1 + C_2} = \frac{V}{6} \Rightarrow \frac{C_1}{C_1 + C_2} = \frac{1}{6} \Rightarrow \frac{C_2}{C_1} = 5$$

38.(18) $i = \frac{V}{2+4} \Rightarrow v = 3 \times 6 = 18 \text{ volts.}$

$$39.(7) \quad i = i_0 e^{-t/RC}$$

$$i = \frac{i_0}{8} \Rightarrow t = \tau_1 = 3RC \log_e 2$$

$$i = \frac{i_0}{16} \Rightarrow t = \tau_2 = 4RC \log_e 2$$

$$\Rightarrow \frac{\tau_1}{\tau_2} = \frac{3}{4} \Rightarrow \frac{\tau_2 + \tau_1}{\tau_2 - \tau_1} = 7$$

$$40.(4) \quad C = \frac{C_0(2C_0)}{C + 2C_0} = \frac{2}{3}C_0$$

$$C' = \frac{C_0[(K+1)C_0]}{C_0 + (K+1)C_0} = \frac{(K+1)}{(K+2)}C_0$$

$$\frac{C'}{C} = \frac{5}{4} \Rightarrow \frac{K+1}{K+2} = \frac{5}{6} \Rightarrow K = 4$$

$$41.(2) \quad V = Ed = 10(0.2 \times 10^{-3}) \\ = 2 \times 10^{-3} V$$

$$42.(400) \quad V_c = V(1 - e^{-t/Rc}) \Rightarrow U_c = \frac{1}{2}CV_c^2 = \frac{1}{2}CV^2(1 - e^{-t/RC})^2$$

$$U_2 - U_1 = \frac{1}{2}CV^2 \left[\left(1 - \frac{1}{4}\right)^2 - \left(1 - \frac{1}{2}\right)^2 \right] \\ = \frac{5}{32}CV^2 = 400 \mu J$$

$$43.(6) \quad \text{Potential difference, } V = \frac{1}{2C} |Q_1 - Q_2| = \frac{1}{2C} \left(\frac{Q}{3} \right) = \frac{Q}{6C}$$

$$44.(1) \quad \text{Force between the plates of a capacitor, } F = \frac{Q^2}{2\epsilon_0 A}$$

Since the capacitors were identical, the area of the plates is same for both capacitors. And in series, charge on the capacitors is also equal. Hence, the force between the plates is same for both capacitors.

45.(200) When a capacitor of capacitance C is connected in series with a battery of EMF V and a resistance R , the charge on the positive plate of the capacitor is given by:

$$q(t) = CVe^{-\frac{t}{RC}}$$

Comparing with the given equation, we get $C = 10^{-5} F$ and $R = 200 \Omega$

46.(54) Let the common potential be V .

$$U_i = \frac{1}{2}CV^2 + \frac{1}{2}(2C)V^2 = \frac{3}{2}CV^2$$

Let the final charge on the capacitors be Q_1 and Q_2 .

$$\text{By charge conservation: } -Q_1 + Q_2 = -CV + 2CV$$

$$\text{Applying KVL in the loop: } \frac{Q_1}{C} + \frac{Q_2}{2C} = 0$$

$$\text{Solving, we get: } Q_1 = -\frac{CV}{3} \quad \text{and} \quad Q_2 = \frac{2CV}{3}$$

Total potential energy: $U_f = \frac{1}{2C}Q_1^2 + \frac{1}{2C}Q_2^2 = \frac{1}{2C}\left(\frac{CV}{3}\right)^2 + \frac{1}{2C}\left(\frac{2CV}{3}\right)^2 = \frac{5}{18}CV^2$

$$\Rightarrow \frac{U_i}{U_f} = \frac{27}{5}$$

47.(4) $U = \frac{1}{2}CV^2$

Heat dissipated in resistance = Loss in potential energy of capacitor

$$\Rightarrow H = \frac{1}{2}CV^2 - \frac{1}{2}C\left(\frac{V}{N}\right)^2 = \frac{1}{2}\left(\frac{N^2-1}{N^2}\right)CV^2 = \left(\frac{N^2-1}{N^2}\right)U$$

Now, $H = \frac{15}{16}U \Rightarrow N = 4$

48.(6) $C = \frac{\epsilon_0 A}{d} = 2\mu F$

$$\frac{1}{C'} = \frac{\left(\frac{d}{3}\right)}{2\epsilon_0 A} + \frac{\left(\frac{2d}{3}\right)}{4\epsilon_0 A} = \frac{d}{3\epsilon_0 A} \Rightarrow C' = \frac{3\epsilon_0 A}{d} = 3C = 6\mu F$$

49.(9) $U_S = 3\left(\frac{1}{2}C\left(\frac{V}{3}\right)^2\right) = \frac{1}{6}CV^2$

$$U_P = 3\left(\frac{1}{2}CV^2\right) = \frac{3}{2}CV^2 \Rightarrow \frac{U_P}{U_S} = 9$$

Magnetic Effect of Current

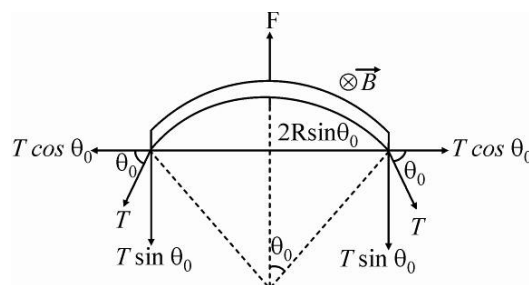
1.(D) $P_1 = P_2 \Rightarrow \frac{2\pi m_1 v_1}{q_1 B} + \frac{2\pi m_2 v_2}{q_2 B} = 0$

2.(A) Magnetic force on the circular arc

$$F = I(2R \sin \theta_0)B$$

For the arc to be in equilibrium, $F = 2T \sin \theta_0$

$$\therefore 2T \sin \theta_0 = 2IR \sin \theta_0 \quad BT = IRB$$



3.(A) $\vec{0} = q(\vec{V} \times \vec{B}) + q\vec{E} \Rightarrow -\vec{V} \times \vec{B} = \vec{E} \Rightarrow \vec{V} = \frac{\vec{E} \times \vec{B}}{B^2}$ 4.(A)

5.(D) When the currents are parallel, $I_1 I_2$ is positive and the force between them is attractive (i.e. negative). Similarly when currents are antiparallel, $I_1 I_2$ is negative and the force between them is repulsive (i.e. positive). So option (d) satisfies the condition.

6.(B) $F = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} \cdot \ell$ (i)

$$F' = -\frac{\mu_0}{2\pi} \frac{2I_1 I_2}{3d}$$
(ii)

$$\Rightarrow F' = -\frac{2}{3}F \quad 7.(A)$$

$$8.(B) \quad W = MB(1 - \cos \theta) \quad \text{or} \quad W = \frac{MB}{2} \quad \dots\dots(i)$$

$$\tau = MB \sin 60^\circ = (2w) \frac{\sqrt{3}}{2} = \sqrt{3}w$$

9.(B) Force exerted on a current carrying conductor,

$$F_{ext} = B(x) / L$$

$$\text{Average power} = \frac{\text{Work done}}{\text{Time taken}}$$

$$P = \frac{1}{t} \int_0^2 F_{ext} \cdot dx = \frac{1}{t} \int_0^2 B(x) / L dx = \frac{1}{5 \times 10^3} \int_0^2 3 \times 10^{-4} e^{-0.2x} \times 10 \times 3 dx$$

$$= 9 \left[1 - e^{-0.4} \right] = 9 \left[1 - \frac{1}{e^{0.4}} \right] = 2.967 = 2.97 \text{ W}$$

10.(C) The component $dl \cos \theta$ of element dl is parallel to the length of the wire-1. Hence force on this elemental

$$\text{component is } F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{r} (dl \cos \theta)$$

$$11.(D) \quad qE = \frac{mv_o^2}{r_1} \quad \therefore \quad r_1 = \frac{mv_o^2}{qE} \quad 12.(D)$$

$$r_2 = \frac{mv_o}{qB} \quad \therefore \quad \frac{r_1}{r_2} = \frac{Bv_o}{E}$$

$$13.(C) \quad \text{Energy of proton} = \frac{1}{2} mv^2 = qV$$

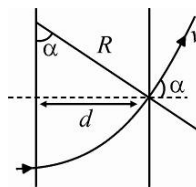
$$v = \sqrt{\frac{2qV}{m}}$$

$$\text{Magnetic force, } qv \sin 90^\circ = \frac{mv^2}{R}$$

$$R = \frac{mv}{qB}$$

$$\sin \alpha = \frac{d}{R} = \frac{dqB}{mv} = \frac{dqB}{m} \sqrt{\frac{m}{2qV}}$$

$$\sin \alpha = Bd \sqrt{\frac{q}{2mV}}$$



$$14.(A) \quad I = 12A, \vec{B} = 0.3\hat{k}T$$

$$A = 10 \times 5 \text{ cm}^2 = 50 \times 10^{-4} \text{ m}^2$$

$$M = I A \hat{n} = 12 \times 50 \times 10^{-4} \hat{n} \text{ Am}^2 = 6 \times 10^{-2} \hat{n} \text{ Am}^2$$

$$\text{Here, } \vec{M}_1 = 6 \times 10^{-2} \hat{i} \text{ Am}^2, \vec{M}_2 = 6 \times 10^{-2} \hat{k} \text{ Am}^2$$

$$\vec{M}_3 = -6 \times 10^{-2} \hat{j} \text{ Am}^2, \vec{M}_4 = -6 \times 10^{-2} \hat{k} \text{ Am}^2$$

$\vec{M}_2$ is parallel to $\vec{B}$, it means potential energy is minimum, therefore in orientation (2) the loop is in stable equilibrium.

$\vec{M}_4$ is anti-parallel to $\vec{B}$, it means potential energy is maximum, therefore in orientation (4) the loop is in unstable equilibrium.

- 15.(B) Magnetic field due to the loop is in $\otimes$ direction or along $-z$ direction. Now we can find the direction of magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$ from Fleming's left hand rule.

16.(A) $B_A = \frac{\mu_0 NI}{2R}$

$$B_B = \frac{\mu_0 N(2I)}{2(2R)} = \frac{\mu_0 NI}{2R} \therefore \frac{B_A}{B_B} = 1$$

- 17.(B) The magnetic field at the centre of circular coil is $B = \frac{\mu_0 I}{2r}$

$$r = \frac{\ell}{2\pi} \therefore B = \frac{\mu_0 I \pi}{\ell} \dots\dots(i)$$

When wire of length ℓ bends into a circular loops of n turns, then

$$\ell = n \times 2\pi r^1 \Rightarrow r^1 = \frac{\ell}{n \times 2\pi}$$

$$B^1 = \frac{\mu_0 n I}{2r^1} = n^2 B$$

- 18.(D) Let R be the radius of the loop. Then

$$A = \pi R^2 \quad \text{or} \quad R = \sqrt{\frac{A}{\pi}}$$

$$B = \frac{\mu_0 i}{2R} \quad \text{or} \quad i = \frac{2BR}{\mu_0}$$

And $M = iA = \frac{2BRA}{\mu_0} = \frac{2BA}{\mu_0} \sqrt{\frac{A}{\pi}}$

- 19.(A) Cyclotron is used to accelerate the charged particle.

- 20.(C) Magnetic needle placed in non-uniform magnetic field. It may experience force and torque both due to unequal forces acting on poles.

21.(B) $\frac{1}{2}mu^2 = qV$

- 22.(C) Formula based question

- 23.(D) Let the length of right wire be l , then its mass is λl .

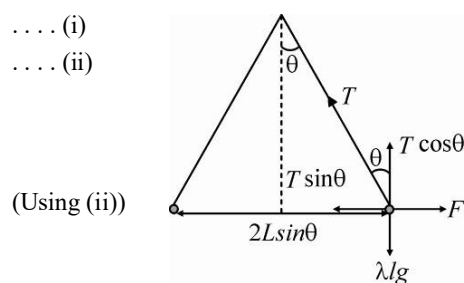
Force acting on this wire are tension (T), weight ($\lambda l g$) and force of repulsion due to other wire (F).

From figure, $T \cos \theta = \lambda l g$

$T \sin \theta = F$

Here, $F = \frac{\mu_0}{2\pi} \frac{I^2 l}{(2L \sin \theta)}$

Or $T \sin \theta = \frac{\mu_0}{2\pi} \frac{I^2 l}{(2L \sin \theta)}$



Or $\frac{\lambda \ell g}{\cos \theta} \sin \theta = \frac{\mu_0}{2\pi} \frac{I^2 \ell}{(2L \sin \theta)}$ (Using (i))

$$\Rightarrow I = \sqrt{\frac{4\pi L \lambda \sin^2 \theta g}{\mu_0 \cos \theta}} = 2 \sin \theta \sqrt{\frac{\pi \lambda g L}{\mu_0 \cos \theta}}$$

- 24.(B)** When particle describes circular path in a magnetic field, its velocity is always perpendicular to the magnetic force.

Power $P = \vec{F} \cdot \vec{V}$

Here $\theta = 90^\circ \quad \therefore \quad P = 0$

Also $w = P \cdot t \quad \therefore \quad w = 0$

25.(D)

- 26.(A)** Let the current which produces full scale deflection in the galvanometer be I_g ,

Then according to question

$$\frac{4}{5} I_g = \frac{V}{G + R} = \frac{2}{G + 2400} \quad \dots (i)$$

$$\frac{2}{5} I_g = \frac{2}{G + 4900} \quad \dots (ii)$$

From eqns. (i) and (ii), $\frac{4}{2} = \frac{G + 4900}{G + 2400} \Rightarrow G = 100 \Omega$

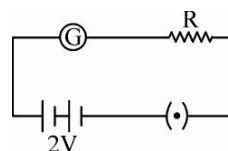
Putting G in eq. (i)

$$\frac{4}{5} I_g = \frac{2}{100 + 2400} \Rightarrow I_g = \frac{2 \times 5}{4 \times 2500} = 1mA$$

For a deflection of 10 divisions

$$\frac{1}{5} I_g = \frac{V}{G + R} \Rightarrow \frac{1}{5} \times 10^{-3} = \frac{2}{100 + R} \Rightarrow R = 9900 \Omega$$

Now, current sensitivity $= \frac{I_g}{n} = \frac{1mA}{50div} = 20 \mu A / \text{division}$



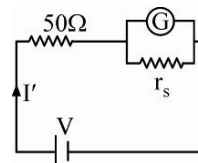
- 27.(D)** Current in the circuit without ammeter

$$I = \frac{V}{R} = \frac{5V}{50\Omega} = 0.1 A \quad \therefore \quad \text{Allowed current with ammeter, } I' = 0.099 A$$

Also $I' = \frac{V}{R_{eq}}$ where $R_{eq} = 50 + \frac{100r_s}{100 + r_s}$

$$\therefore \quad 0.099 = \frac{5}{50 + \frac{100r_s}{100 + r_s}} \quad \text{or} \quad 50 + \frac{100r_s}{100 + r_s} = \frac{5}{0.099}$$

$$\Rightarrow \frac{100r_s}{100 + r_s} = 0.5 \Rightarrow 100r_s = 50 + 0.5r_s \quad \text{or} \quad r_s = \frac{50}{99.5} = 0.5 \Omega$$



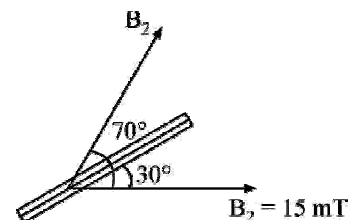
- 28.(A)** Neutron has no charge, others have charge

- 29.(B)** The magnetic dipole attains stable equilibrium under the influence of these two fields making an angle $\theta_1 = 30^\circ$ with B_1 and $\theta_2 = 75^\circ - 30^\circ = 45^\circ$ B_2 .

For stable equilibrium, net torque acting on

dipole mass be zero, i.e., $\tau_1 + \tau_2 = 0$ or $\tau_1 = \tau_2$

Or $mB_1 \frac{\sin \theta_1}{\sin \theta_2} = 15mT \times \frac{\sin 30^\circ}{\sin 45^\circ} = 15mT \times \frac{1}{2} \times \sqrt{2} = 10.6mT = 11mT$



30.(BCD)

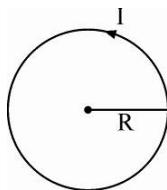
32.(D) $qE = -q(\vec{V} \times \vec{B})$ on solving

$$B = 10^3 \text{ Wb/m}^2$$

33.(D) Wire A is bent into a circle of radius R ,

$$l = 2\pi R \Rightarrow R = \frac{l}{2\pi}$$

$$B_A = \frac{\mu_0 I}{2R} = \frac{\mu_0 I}{2 \times \left(\frac{l}{2\pi}\right)} = \frac{\mu_0 \pi I}{l}$$



Wire B is bent into a square of side a

$$l = 4a \Rightarrow a = \frac{l}{4}$$

$$B_B = 4 \times \left[\frac{\mu_0 I}{4\pi(a/2)} (\sin 45^\circ + \sin 45^\circ) \right]$$

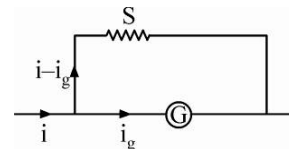
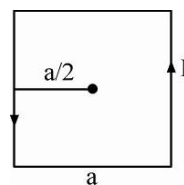
$$= \frac{2\mu_0 I}{\pi a} \times \frac{2}{\sqrt{2}} = \frac{16\mu_0 I}{\sqrt{2}\pi l}$$

$$\therefore \frac{B_A}{B_B} = \frac{\mu_0 \pi I / l}{16\mu_0 I / \sqrt{2}\pi l} = \frac{\pi^2}{8\sqrt{2}}$$

34.(A) Given $i_g = 1 \text{ mA}$, $G = 100 \Omega$, $i = 10 \text{ A}$, $S = ?$

$$(i - i_g)S = i_g G$$

$$\therefore S = \frac{i_g G}{i - i_g} = \frac{1 \times 10^{-3} \times 100}{(10 - 10^{-3})} = 10^{-2} \Omega = 0.01 \Omega$$



35.(D) $T = \frac{2\pi m}{q \times B}$

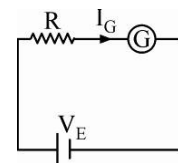
36.(A) Case I: $I_G = \frac{V_E}{R+G}$... (i)

Case II: $I = \frac{V_E}{R + \frac{GS}{G+S}}$... (ii)

$$I'_G = \frac{I_G}{2} = \frac{IS}{G+S} \dots \text{(iii)}$$

$$\therefore \text{From (i), (ii) and (iii), we get : } \frac{V_E}{2(R+G)} = \frac{(G+S)}{(RG+RS+GS)} \times \frac{S}{(G+S)}$$

$$\Rightarrow RG + RS + GS = 2RS + 2GS \Rightarrow RS + GS = RG \text{ or } S(R+G) = RG$$



37.(C) Both magnetic fields are perpendicular to each-other

$$\text{Resultant} = \sqrt{B_1^2 + B_2^2}$$

$$38.(400) B = \frac{\mu_0 I}{2\pi R} \Rightarrow I = \frac{0.08 \times 10^{-3}}{2 \times 10^{-7}} = 400 \text{ A}$$

$$39.(9) B = \frac{\mu_0 IR^2}{2(R^2 + x^2)} \text{ with } x = R$$

$$40.(2) \text{ Resistance} = \frac{\rho L}{\pi r^2} \therefore i = \frac{E \cdot \pi r^2}{\rho L} \text{ and } B = \frac{\mu_0 i}{2\pi R}$$

$$41.(24) M = \frac{er^2\omega}{2}$$

$$42.(10) \text{ For equilibrium } mg = I(B \sin \theta) \cdot 2\pi R \Rightarrow B = 0.64 \text{ T}$$

43.(0) B is independent of the distance from the "current sheet".

44.(400) $M = IA = 1 \times \pi \left(\frac{4}{\pi} \right) = 4 \text{ Am}^2 \quad \therefore \quad \tau = MB \sin 90^\circ = 4 \times 100 \times 1 = 40 \text{ Nm}$

45.(48) $B_p = \frac{\mu_0 I}{4\pi d} (\sin 37^\circ + \sin 53^\circ) = \frac{7\mu_0 I}{48\pi x}$

46.(1) $v = \frac{E}{B}$ and $E = \frac{V}{d}$

47.(6) $dM = \left(\frac{dQ}{T} \right) \pi x^2$ Or $M = \frac{1}{6} Q \omega L^2$

48.(98) $\frac{mg}{\ell} = \frac{\mu_0 i_1 i_2}{2\pi d}$

49.(19) $5G = 95s$

50.(1) $B_{in} = \frac{\mu_0 i r}{2\pi R^2}$ and $B_{out} = \frac{\mu_0 i}{2\pi r}$

51.(30) $\frac{1}{2}mv^2 = qV$

$\sin \theta = \frac{\ell}{R}$ and $R = \frac{mv}{qB}$

52.(6) $B = B_1 - B_2 = 420\mu_0$

$B = \frac{N\mu_0 i}{2R}$ for circular coils

Electromagnetic Induction

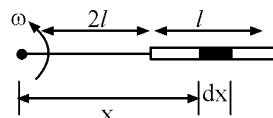
1.(C) $\phi = [B_o (2\hat{i} + 3\hat{j} + 4\hat{k})] \cdot [L^2\hat{k}] = 4B_o L^2 \text{ Wb}$

2.(D) $\therefore$ Induced emf is rate of change of magnetic flux,

$e = \int B v dx \quad [\because v = \omega x] \quad \Rightarrow \quad e = \int_{2l}^{3l} (\omega x) \cdot B \cdot dx$

Using $\int x^n dx = \frac{x^{n+1}}{n+1} \Rightarrow e = B\omega \left[\frac{x^2}{2} \right]_{2l}^{3l}$

$e = B\omega \frac{[(3l)^2 - (2l)^2]}{2} = \frac{5Bl^2\omega}{2}$

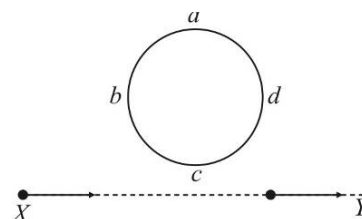


3.(D) As there is no change in flux, so induced current is zero.

4.(D) $\phi = (20t\hat{i} + 10t^2\hat{j} + 50\hat{k}) \cdot (2\hat{k})$

$\phi = 100 \quad \Rightarrow \quad \varepsilon = -\frac{d\phi}{dt} = 0$

5.(A) When the electron moves from X to Y, the flux linked with the coil *abcd* (which is into the page) will first increase and then decrease as the electron passes by. So the induced current in the coil will be first anticlockwise and will reverse its direction (*i.e.* will become clockwise) as the electron goes past the coil.



6.(C) $P.d. = B(2R)v = 2BRv$

Positive charge will accumulate at Q and negative charges at M .

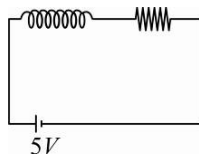
7.(D) Rise of current in L - R circuit is given by

where $I_0 = \frac{E}{R} = \frac{5}{5} = 1A$

Now, $\tau = \frac{L}{R} = \frac{10}{5} = 2s$

After $2s$, i.e., at $t = 2s$

Rise of current $I = (1 - e^{-1})A$



8.(BC)

9.(B)

10.(AD)

11.(D) $|V_L| = \left| L \frac{di}{dt} \right|$

12.(C) Choke coil should have high inductance and low resistance.

13.(C) Module based Question

14.(A) Induced emf , $e = -L \frac{di}{dt}$

For $0 \leq t \leq \frac{T}{4}$,

$i-t$ graph is a straight line with positive constant slope.

$\therefore \frac{di}{dt} = \text{constant}$

$\Rightarrow e = -ve$ and constant

For $0 \leq t \leq \frac{T}{4}$

For $\frac{T}{4} \leq t \leq \frac{T}{2}$ i is constant

$\therefore \frac{di}{dt} = 0$

$\Rightarrow e = 0$

For $\frac{T}{4} \leq t \leq \frac{T}{2}$

For $\frac{T}{2} \leq t \leq \frac{3T}{4}$

$i-t$ graph is a straight line with negative constant slope.

$\therefore \frac{di}{dt} = \text{constant}$

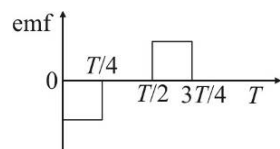
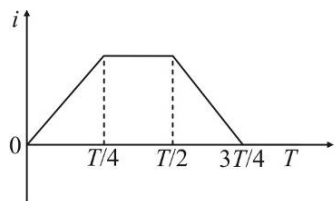
$\Rightarrow e = +ve$ and constant

For $\frac{T}{2} \leq t \leq \frac{3T}{4}$

For $\frac{3T}{4} \leq t \leq T$, i is zero $\therefore \frac{di}{dt} = 0$

$\Rightarrow e = 0$

For $\frac{3T}{4} \leq t \leq T$



From this analysis, the variation of induced emf with time as shown in figure below.

15.(C) $i = i_0 e^{-t/\tau_L}$

$18 = 20e^{-2/\tau_L}$

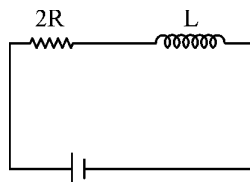
Solving this, we get: $\tau_L = \frac{2}{\ln(10/9)}$

16.(B) $\left[\frac{\text{Magnetic flux}}{\text{Electric flux}} \right] = \left[\frac{\text{Magnetic field}}{\text{Electric field}} \right] = \left[\frac{B}{E} \right]$

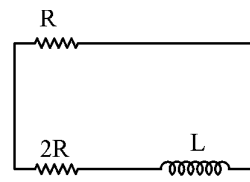
$\left[\frac{E}{B} \right]$ has the dimensions of velocity. i.e., $[LT^{-1}]$ $\therefore \left[\frac{\text{magnetic flux}}{\text{electric flux}} \right] = \left[\frac{1}{\text{velocity}} \right] = [TL^{-1}]$

17.(B) During charging $\tau_{L1} = \frac{L}{2R}$ and during discharging $\tau_{L2} = \frac{L}{3R}$

$\therefore \frac{\tau_{L1}}{\tau_{L2}} = \frac{L/2R}{L/3R} = \frac{3}{2}$



Charging



discharging

18.(B) $L = \frac{\mu_0 N^2 A}{\ell}$

19.(A) For $r \leq R$: $\oint \vec{E} \cdot d\vec{l} = \left| \frac{d\phi}{dt} \right|$ or $E(2\pi r) = (\pi r^2) \frac{dB}{dt}$ or $E \propto r$

i.e., E - r graph is a straight line passing through origin.

For $x \geq R$

$\oint \vec{E} \cdot d\vec{l} = \left| \frac{d\phi}{dt} \right|$ or $E(2\pi r) = (\pi R^2) \left(\frac{dB}{dt} \right)$

i.e., $E \propto \frac{1}{r}$ or E-r graph is a rectangular hyperbola.

20.(ABD)

21.(A) $e = -\frac{d\phi}{dt}$ $\therefore \text{Volt} = \frac{\text{Weber}}{S} = T - m^2 / s$

22.(A)

23.(ABC)

24.(D)

25.(C) During the entry and exit of the bar magnet in the coil, the polarity changes. The only option is (C).

26.(C)

27.(A)

28.(D)

29.(B) The flux associated with coil of area A and magnetic induction B is :

$\phi = BA \cos \theta = \frac{1}{2} B \pi r^2 \cos \omega t$ $\left[\because A = \frac{1}{2} \pi r^2 \right]$

$\therefore e_{\text{induced}} = -\frac{d\phi}{dt} = -\frac{d}{dt} \left(\frac{1}{2} B \pi r^2 \cos \omega t \right) = \frac{1}{2} B \pi r^2 \omega \sin \omega t$

$\therefore \text{Power } P = \frac{e_{\text{induced}}^2}{R} = \frac{B^2 \pi^2 r^4 \omega^2 \sin^2 \omega t}{4R}$

Hence, $P_{\text{mean}} = \langle P \rangle$

$= \frac{B^2 \pi^2 r^2 \omega^2}{4R} \cdot \frac{1}{2}$ $\left[\because \langle \sin^2 \omega t \rangle = \frac{1}{2} \right]$
 $= \frac{(B \pi r^2 \omega)^2}{8R}$

30.(A) Maximum field lines will pass in situation (i).

31.(A) I_1 is in the direction ba and I_2 in the direction dc

32.(C)

33.(B)

34.(D) $M = \frac{\mu_0 N_1 \times N_2 \times A}{l}$

where $N_1 = 300$ turns and $N_2 = 400$ turns,
 $A = 10 \text{ cm}^2$ and $l = 20 \text{ cm}$

Substituting the values in the given formula, we get $M = 2.4\pi \times 10^{-4} H$.

35.(B) Here, $B = B_0 e^{-t/\tau}$

Area of the circular loop, $A = \pi r^2$. Flux linked with the loop at any time t .

$$\phi = BA = \pi r^2 B_0 e^{-t/\tau}$$

Emf induced in the loop, $\varepsilon = -\frac{d\phi}{dt} = \pi r^2 B_0 \frac{1}{\tau} e^{-t/\tau}$

Net heat generated in the loop

$$= \int_0^\infty \frac{\varepsilon^2}{R} dt = \frac{\pi^2 r^4 B_0^2}{\tau^2 R} \int_0^\infty e^{-\frac{2t}{\tau}} dt = \frac{\pi^2 r^4 B_0^2}{\tau^2 R} \times \frac{1}{\left(-\frac{2}{\tau}\right)} \times \left[e^{-\frac{2t}{\tau}} \right]_0^\infty = \frac{-\pi^2 r^4 B_0^2}{2\tau^2 R} \times \tau(0 - 1) = \frac{\pi^2 r^4 B_0^2}{2\tau R}$$

36.(D) length of the line plane, $l = 20 \text{ m}$

Wing span, $l' = 15 \text{ m}$

Height of plane, $h = 5 \text{ m}$

Velocity of plane = 240 m s^{-1} towards east

$$\sin \theta = \frac{2}{3}, B = 5 \times 10^{-5} T, V_B = ?, V_w = ?$$

V_B = Voltage developed between lower and upper side of the plane

$$= v h B \cos \theta$$

$$= 240 \times 5 \times 10^{-5} \times \frac{\sqrt{5}}{3} = 44.72 \times 10^{-5} \times \frac{\sqrt{5}}{3}$$

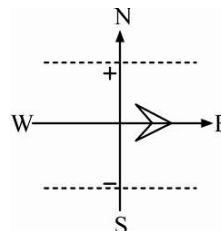
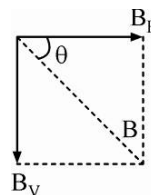
$$= 44.72 \times 10^{-3} V = 45 \text{ mV}$$

$$B_v = B \sin \theta$$

$$= 5 \times 10^{-5} \times \frac{2}{3} = \frac{1}{3} \times 10^{-4} T$$

V_w = Voltage developed between tips of the wings

$$= W_v l' v = \frac{1}{3} \times 10^{-4} \times 15 \times 240 = 1200 \times 10^{-4} = 120 \text{ mV}$$



37.(B) Magnetic force on electron in the metal sheet,

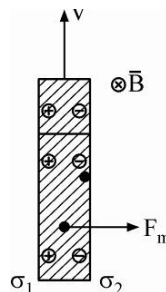
$$\vec{F}_m = -e(\vec{v} \times \vec{B})$$

At equilibrium,

$$F_m = F_e (\text{induced}) \text{ and } \sigma_2 = -\sigma_1$$

$$evB = e \frac{\sigma}{\varepsilon_0} = e \frac{\sigma}{\varepsilon_0} \Rightarrow \sigma = \varepsilon_0 vB = \sigma_1$$

$$\sigma_2 = -\varepsilon_0 vB$$

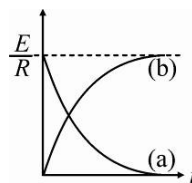


38.(A) For RC circuit :

$$i = \frac{E}{R} e^{-t/RC}$$

For RL circuit :

$$i = \frac{E}{R} (1 - e^{-t(L/R)})$$



39.(2) Use $\phi = \vec{B} \cdot \vec{A}$

40.(8) $i_1 = \frac{10}{6+4} = 1 \text{ A}$ ($i_L = 0$)

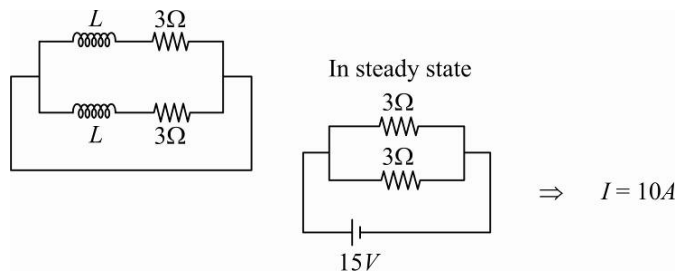
$$i_2 = \frac{10}{6+2} = \frac{10}{8} \text{ A} \quad \therefore \quad \frac{i_1}{i_2} = 0.8$$

41.(104) In step up,

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = 8$$

$$\therefore V_S = 8V_P \Rightarrow P_S = I_S(960) \text{ or } I_S = \frac{10^4}{960} = 10.4 \text{ A}$$

42.(10)



43.(16) Potential difference across coil is $V = L \frac{di}{dt}$ or $V = (2)(4) = 8 \text{ V}$

Now energy stored per unit time = Power = $Vi = (8)(2) = 16 \text{ J/s}$

44.(512) $v = L \frac{di}{dt} \Rightarrow 4t = L \frac{di}{dt}$

$$\frac{1}{L} \int_0^4 4t \, dt = \int_0^i di \Rightarrow i = 32 \text{ A}$$

$$\text{Energy stored, } U = \frac{1}{2} Li^2 = \frac{1}{2} \times 1 \times 32^2 = 512 \text{ J}$$

45.(4) The emf induced across the rod AB is $e = Bv_{\perp} l$

Here, $v_{\perp} = v \sin 30^\circ$ = component of velocity perpendicular to length

$$\therefore e = Bvl \sin 30^\circ = (2)(4)(1) \left(\frac{1}{2} \right) = 4 \text{ V}$$

The free electrons of the rod shift towards right due to the force $q(\vec{v} \times \vec{B})$. Thus, the left side of the rod is at higher potential.

or $V_A - V_B = 4 \text{ V}$

46.(51) $e_{AB} = \int_{0.07}^{0.1} (2)(10r) \, dr = 0.051 \text{ V}$

$$47.(10) \quad \varepsilon_i = -\frac{d\phi}{dt} = -20t + 50 \quad \text{At } t = 3s, \quad \varepsilon_i = -10V$$

$$48.(1) \quad \text{Use : } I = I_0(1 - e^{-t/\tau})$$

$$49.(167) \quad I_1 = 5A, I_2 = 2A$$

$$\Delta I = 2 - 5 = -3A$$

$$\Delta t = 0.1s$$

$$\varepsilon = 50V$$

$$\text{As, } \varepsilon = -L \frac{\Delta I}{\Delta t} \quad ; \quad 50 = -L \left(\frac{-3}{0.1} \right); 50 = 30L$$

$$L = \frac{5}{3} = 1.67 H$$

$$50.(8) \quad V_A - IR + E - L \frac{dI}{dt} = V_B \Rightarrow V_A - 5 \times 1 + 8 - 5 \times 10^{-3}(-10^3) = V_B \Rightarrow V_B - V_A = 8V$$

$$51.(6) \quad \text{From Energy conservation :}$$

$$\frac{1}{2} CV_0^2 = \frac{1}{2} CV^2 + \frac{1}{2} LI^2 \Rightarrow \frac{1}{2} \times 2 \times 10^{-6} \times 12^2$$

$$= \frac{1}{2} \times 10^{-6} \times 6^2 + \frac{1}{2} \times 6 \times 10^{-6} I^2 \Rightarrow I = 6A$$

$$52.(4) \quad e = M \frac{dI}{dt} = M(4t) - 20t$$

$$i = \frac{e}{R} = \frac{20t}{10} = 2t, q = \int Idt = \int_0^2 2t dt = 4C$$

$$53.(4) \quad I = 2 \sin^2 t A$$

$$L = 2H$$

$$dI = 2 \cos t^2 \cdot 2t dt = 4t \cos t^2 dt$$

$$\text{When } I = 0, t = 0$$

$$\text{When } I = 2, t = \sqrt{\frac{\pi}{2}}$$

$$E = \int_0^2 L I dI = L \int_0^2 2 \sin t^2 dt$$

$$= L \int_0^{\sqrt{\pi/2}} 8t \sin t^2 \cos t^2 dt$$

$$\text{Let } t^2 = x, \text{ so } 2t dt = dx$$

$$E = L \int_0^{\pi/2} 4 \sin x \cos x dx = 2L \int_0^{\pi/2} \sin 2x dx$$

$$= 2L \left(\frac{-\cos 2x}{2} \right)_0^{\pi/2} = \frac{2L}{2} = 4J$$

AC Circuits

1.(B) Use $I = I_0 \sin \omega t$

2.(B) $\tan \frac{\pi}{4} = \frac{X_C}{R} \quad \therefore \quad X_C = R$

$$\tan^{-1}(2) = \frac{X_L}{R} \quad \text{or} \quad \frac{X_L}{R} = 2 \quad \therefore \quad X_L = 2R$$

Now, in L - C - R circuit $Z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{2}R$

$\therefore$ Power factor $= \cos \theta = \frac{R}{Z} = \frac{1}{\sqrt{2}}$ 3.(C)

4.(A) $I_{rms} = \frac{12}{24} = \frac{1}{2} A$

$$I_{Peak} = \sqrt{2} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}} A$$

5.(A) Time constant τ is the duration when the value of potential drops by 63% of its initial maximum value (i.e., V_0/e). Here, 37% of 25 V = 9.25 V which lies between 150 s to 200 s in the graph.

6.(AD) LC or RLC

7.(C) The full cycle of alternating current consists of two half cycles. For one half, current is negative. Therefore, for an AC cycle, the net value of ammeter, read the average value. Hence, the alternating current cannot be measured by DC ammeter.

8.(CD) RC or C-only

9.(A) Fact based Question

10.(ABD)

11.(D) The emf generated would be maximum when flux (cutting) would be maximum i.e., angle between area vector of coil and magnetic field is 0° . The emf generated is given by [As a function of time]

$$e = NBA\omega \cos \omega t \quad \Rightarrow \quad e_{max} = NAB\omega$$

12.(ABC) Formula based Question

13.(A)

14.(C) In the condition of resonance $X_L = X_C$ or $\omega L = \frac{1}{\omega C}$

Since, resonant frequency remains unchanged, so

$$\sqrt{LC} = \text{constant} \quad \text{or} \quad LC = \text{constant}$$

$$\therefore L_1 C_1 = L_2 C_2 \quad \text{or} \quad L \times C = L_2 \times 2C \quad \text{or} \quad L_2 = \frac{L}{2} \quad \text{15.(D)}$$

16.(C) (A) In a circuit having C alone, the voltage lags the current by $\frac{\pi}{2}$.

(B) In circuit containing C and L , the voltage leads the current by $\frac{\pi}{2}$ or lags by $\frac{\pi}{2}$.

(C) In LR circuit, the phase difference between current and voltage can have any value between 0 to $\frac{\pi}{2}$ depending on the values of L and R but not $\frac{\pi}{2}$ or 0.

(D) In a circuit containing L alone, the voltage leads the current by $\frac{\pi}{2}$.

Hence, the option (C) is correct.

17.(AD) The power line is AC so average power is 0 and the line consists of RLC so phase difference is less than $\frac{\pi}{2}$

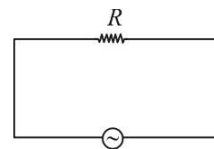
18.(C) Case I: $P = V_{rms} I_{rms} = V_{rms} \times \frac{V_{rms}}{R}$

$$P = \frac{V_{rms}^2}{R} \Rightarrow V_{rms}^2 = PR \quad \dots (i)$$

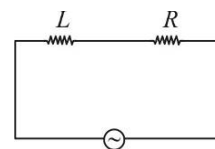
Case II: Power drawn in LR circuit

$$P' = V_{rms} I_{rms} \cos \phi = V_{rms} \times \frac{V_{rms}}{Z} \times \frac{R}{Z} = V_{rms}^2 \frac{R}{Z^2} = PR \times \frac{R}{Z^2}$$

$$P' = P \frac{R^2}{Z^2}$$



[Using equation (i)]



19.(B) Here,
Efficiency of the transformer, $\eta = 90\%$

Input power, $P_{in} = 3kW = 3 \times 10^3 W = 3000W$

Voltage across the primary coil, $V_p = 200V$

Current in the secondary coil, $I_s = 6A$

As $P_{in} = I_p V_p \quad \therefore \quad$ Current in the primary coil, $I_p = \frac{P_{in}}{V_p} = \frac{3000W}{200V} = 15A$

Efficiency of the transformer, $\eta = \frac{P_{out}}{P_{in}} = \frac{V_s I_s}{V_p I_p} \quad \therefore \quad \frac{90}{100} = \frac{6V}{3000} \text{ or } V_s = \frac{90 \times 3000}{100 \times 6} = 450V$

20.(C) When L is removed, the phase difference between the voltage and current is

$$\tan \phi_1 = \frac{X_C}{R}; \tan \frac{\pi}{3} = \frac{X_C}{R} \text{ or } X_C = R \tan 60^\circ \text{ or } X_C = \sqrt{3}R$$

When C is removed, the phase difference between the voltage and current is:

$$\tan \phi_2 = \frac{X_L}{R}; \tan \frac{\pi}{3} = \frac{X_L}{R} \text{ or } X_L = R \tan 60^\circ = \sqrt{3}R$$

As $X_L = X_C$, the series LCR circuit is in resonance.

Impedance of the circuit, $Z = \sqrt{R^2 + (X_L - X_C)^2} = R \quad (\because X_L = X_C) \quad$ Power factor, $\cos \phi = \frac{R}{Z} = \frac{R}{R} = 1$

21.(D) Frequency of LC oscillation $= \frac{1}{2\pi\sqrt{LC}}$

Or, $\frac{f_1}{f_2} = \left(\frac{L_2 C_2}{L_1 C_1} \right)^{1/2} = \left(\frac{2L \times 4C}{L \times C} \right)^{1/2} = (8)^{1/2} \quad \therefore \quad \frac{f_1}{f_2} = 2\sqrt{2} \Rightarrow f_2 = \frac{f_1}{2\sqrt{2}} \text{ or } f_2 = \frac{f}{2\sqrt{2}} \quad (\because f_1 = f)$

22.(D) $\tan \phi = \frac{X_C - X_L}{R}; \tan \left(\frac{\pi}{4} \right) = \frac{\frac{1}{\omega C} - \omega L}{R}$

$$R = \frac{1}{\omega C} - \omega L \text{ or } R + 2\pi f L = \frac{1}{2\pi f C} \quad \text{or} \quad C = \frac{1}{2\pi f (R + 2\pi f L)}$$

23.(D) From the relation, $\tan \phi = \frac{\omega L}{R}$

$$\text{Power factor, } \cos \phi = \frac{1}{\sqrt{1 + \tan^2 \phi}} = \frac{1}{\sqrt{1 + \left(\frac{\omega L}{R} \right)^2}} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

24.(A) In an A.C. circuit with inductance coil, the voltage V leads the current I by a phase difference of 90° . Or the current I lags behind the voltage V by a phase difference of 90° . Thus the voltage goes on decreasing with the increase in time as shown in the graph (a).

25.(D) In case of oscillatory discharge of a capacitor through an inductor, charge at instant t is given by $q = q_0 \cos \omega t$

Where, $\omega = \frac{1}{\sqrt{LC}}$... (i)

$\therefore \cos \omega t = \frac{q}{q_0} = \frac{CV_2}{CV_1} = \frac{V_2}{V_1} \quad (\because q = CV)$

Current through the inductor

$$I = \frac{dq}{dt} = \frac{d}{dt}(q_0 \cos \omega t) = -q_0 \omega \sin \omega t$$

$$|I| = CV_1 \frac{1}{\sqrt{LC}} [1 - \cos^2 \omega t]^{1/2}$$

$$V_1 \sqrt{\frac{C}{L}} \left[1 - \left(\frac{V_2}{V_1} \right)^2 \right]^{1/2} = \left[\frac{C(V_1^2 - V_2^2)}{L} \right]^{1/2} \quad (\text{Using (i)})$$

26.(C) $I_{mean}^2 = I_1^2 \overline{\cos^2 \omega t} + I_2^2 \overline{\sin^2 \omega t} + 2I_1 I_2 \overline{\cos \omega t \sin \omega t} = I_1^2 \times \frac{1}{2} + I_2^2 \times \frac{1}{2} + 2I_1 I_2 \times 0 = \frac{I_1^2 + I_2^2}{2}$

$$I_{mean} = \frac{\sqrt{I_1^2 + I_2^2}}{\sqrt{2}}$$

27.(D) Current in LR circuit is $I = \frac{V_0}{\sqrt{R^2 + \omega^2 L^2}} \sin\left(\omega t - \frac{\pi}{2}\right)$, i.e., it is sinusoidal in nature.

28.(B) At any time t , the equation of the given circuit is

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0 \quad \dots (i)$$

which is equivalent to that of a damped pendulum. The solution to eqn. (i) is

$$q = Q_0 e^{-Rt/2L} \cos(\omega' t + \phi)$$

Where $\omega' = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$

The square of maximum charge on capacitor at any time t

$$Q_{\max}^2 = Q_0^2 e^{-Rt/L} \cos^2(\omega' t + \phi)$$

$\therefore$ It decays exponentially with time.

For $L_2 (L_2 < L_1)$, the curve is more steep.

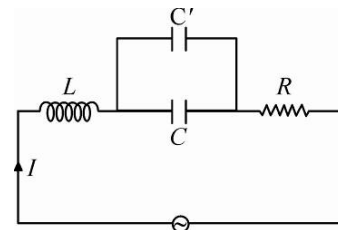
29.(D) Since power factor has to be made 1.

$\therefore$ Effective capacitance should be increased thus connecting in parallel.

$\therefore \cos \phi = 1 \quad \therefore \phi = 0$

$$I \omega L = \frac{I}{\omega(C + C')} \quad \text{or} \quad C + C' = \frac{1}{\omega^2 L} \quad \therefore \quad C' = \frac{1}{\omega^2 L} - C$$

$$C' = \frac{1 - \omega^2 LC}{\omega^2 L}$$



30.(0) For given circuit current is lagging the voltage by $\frac{\pi}{2}$, so circuit is purely inductive and there is no power consumption in the circuit. The work done by battery is stored as magnetic energy in the induction.

31.(1) Given, $L = 10 \text{ H}$, $f = 50 \text{ Hz}$

For maximum power

$$X_C = X_L \quad \text{or} \quad \frac{1}{\omega C} = \omega L \quad \text{or} \quad C = \frac{1}{\omega^2 L}$$

$$\therefore C = \frac{1}{4\pi^2 \times 50 \times 50 \times 10} \quad \text{or} \quad C = 0.1 \times 10^{-5} \text{ F} = 1 \mu\text{F}$$

32.(250) At resonance, $\omega L = \frac{1}{\omega C}$

$$\text{Current flowing through the circuit, } I = \frac{V_R}{R} = \frac{100}{1000} = 0.1 \text{ A}$$

So, voltage across L is given by $V_L = I X_L = I \omega L$

$$\text{But } \omega L = \frac{1}{\omega C} \quad \therefore V_L = \frac{I}{\omega C} = \frac{0.1}{200 \times 2 \times 10^{-6}} = 250 \text{ volt}$$

33.(2) Given : $I_p = 4 \text{ A}$, $N_p = 140$ and $N_s = 280$

$$\text{From the formula } \frac{I_p}{I_s} = \frac{N_s}{N_p} \quad \text{or} \quad \frac{4}{I_s} = \frac{280}{140} \quad \text{So, } I_s = 2 \text{ A}$$

34.(100) Power factor $= \frac{R}{Z}$

$$P_i = \frac{1}{2} \quad \therefore Z = 2R; \quad P_f = \frac{1}{4} \quad \therefore Z = 4R$$

35.(200) $V^2 = V_R^2 + (V_L - V_C)^2$

$$\text{Here, } V_L = V_C = 100 \text{ V} \quad \therefore V_R = V = 200 \text{ V}$$

36.(60) Power factor, $\cos \theta = \frac{R}{Z} = \frac{60}{\sqrt{(60)^2 + (100 - 20)^2}} = 0.6$

37.(2) In the LC circuit the energy oscillates between inductor (in the magnetic field) and capacitor (in the electric field).

$$U_{E_{\max}} [\text{Maximum energy stored in capacitor}] = \frac{Q^2}{2C}$$

$$U_{B_{\max}} [\text{Maximum energy stored in inductor}] = \frac{li_0^2}{2}$$

where, i_0 is the current at this time.

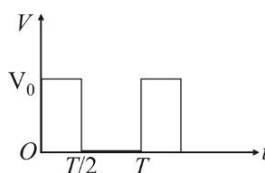
For the given instant, $U_E = U_B$

$$\text{i.e., } \frac{q^2}{2C} = \frac{Li^2}{2}$$

$$\text{From energy conservation } U_E + U_B = U_{E_{\max}} = U_{B_{\max}} \quad \Rightarrow \quad \frac{2q^2}{2C} = \frac{Q^2}{2C} \quad \text{or} \quad q = \frac{Q}{\sqrt{2}}$$

38.(8) Power factor = $\cos \phi = \frac{R}{Z} = \frac{12}{15} = \frac{4}{5} = 0.8$

39.(2) $V = V_0$ for $0 \leq t \leq \frac{T}{2}$ $V = 0$ For $\frac{T}{2} \leq t \leq T$

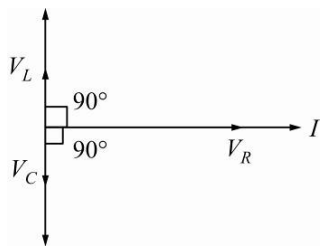


$$V_{rms} = \left[\frac{\int_0^T V^2 dt}{\int_0^T dt} \right]^{1/2} = \left[\frac{\int_0^{T/2} V_0^2 dt + \int_{T/2}^T (0) dt}{\int_0^T dt} \right]^{1/2} = \left[\frac{V_0^2}{T} [t]_0^{T/2} \right]^{1/2} = \left[\frac{V_0^2}{T} \left(\frac{T}{2} \right) \right]^{1/2} = \left[\frac{V_0^2}{2} \right]^{1/2} V_{rms} = \frac{V_0}{\sqrt{2}}$$

40.(4) $P = I^2 R = \left(\frac{I_0}{\sqrt{2}} \right)^2 R = \frac{I_0^2 R}{2}$

41.(0) In an LCR series AC circuit, the voltage across inductor L leads the current by 90° and the voltage across capacitor C lags behind the current by 90° .

Hence, the voltage across LC combination will be zero.



42.(50) $\nu = 25 \text{ Hz}$. In one cycle ac current becomes zero twice

43.(242) The given circuit is under resonance as $X_L = X_C$. Hence, power dissipated in the circuit is : $P = \frac{V^2}{R} = 242W$

44.(65) For a dc source

$$I = 10 \text{ A}, V = 80 \text{ V}$$

Resistance of the arc lamp.

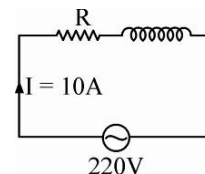
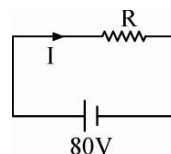
$$R = \frac{V}{I} = \frac{80}{10} = 8\Omega$$

For an ac source.

$$\mathcal{E}_{rms} = 220V$$

$$\nu = 50 \text{ Hz}$$

$$\omega = 2\pi \times 50 = 100\pi \text{ rads}^{-1}$$



Arc lamp will glow if $I = 10A$ $\therefore I = \frac{\mathcal{E}_{rms}}{\sqrt{R^2 + \omega^2 L^2}}$ or, $Z^2 = \left(\frac{\mathcal{E}_{rms}}{I} \right)^2$

or, $8^2 + (100\pi)^2 L^2 = \left(\frac{220}{10} \right)^2$ or, $L^2 = \frac{22^2 - 8^2}{(100\pi)^2}$ $\therefore L = \frac{\sqrt{30 \times 14}}{100\pi} = 0.065H$

Light & Interference

- 1.(B) According to lens maker's formula, when the lens is in the air.

$$\frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \Rightarrow \frac{1}{f} = \frac{1}{2x} \Rightarrow f = 2x$$

Here, $\frac{1}{x} = \frac{1}{R_1} - \frac{1}{R_2}$, x is positive for convex lens.

In case of liquid, where refractive indices are $\frac{4}{3}$ and $\frac{5}{3}$, we get :

Focal length in first liquid,

$$\frac{1}{f_1} = \left(\frac{\mu_s}{\mu_{s1}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f_1} = \left(\frac{3/2}{4/3} - 1 \right) \frac{1}{x} \Rightarrow f_1 \text{ is positive,}$$

Nature of lens does not change, $\frac{1}{f_1} = \frac{1}{8x} = \frac{1}{4(2x)} = \frac{1}{4f} = f_1 = 4f$

$$\frac{1}{f_2} = \left(\frac{\mu_s}{\mu_{s2}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \Rightarrow \frac{1}{f_2} = \left(\frac{3/2}{5/3} - 1 \right) \left(\frac{1}{x} \right) \Rightarrow f_2 \text{ is negative.}$$

Nature of lens change i.e. convex behave as diverging lens.

- 2.(B) As beam of light is incident normally on the face AB of the right angled prism ABC , so no refraction occurs at face AB and it passes straight and strikes the face AC at an angle of incidence $i = 45^\circ$.

For total reflection to take place at face AC ,

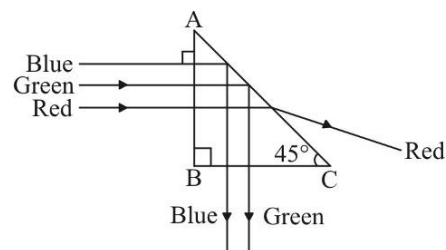
$$i > i_c \text{ or } \sin i > \sin i_c$$

Where i_c is the critical angle.

$$\therefore \sin 45^\circ > \frac{1}{\mu} \text{ or } \frac{1}{\sqrt{2}} > \frac{1}{\mu} \text{ Or } \mu > \sqrt{2} = 1.414$$

As $\mu_{red} (= 1.39) < \mu (= 1.414)$ while $\mu_{green} (= 1.44)$ and $\mu_{blue} (= 1.47) > \mu (= 1.414)$, so only red colour will be transmitted through face AC while green and blue colour will suffer total internal reflection.

So the prism will separate red colour from the green and blue colours as shown in the following figure.



- 3.(B) $I = 4I_0 \cos^2 \phi / 2$ $\therefore I_1 = 4I_0 \cos^2 0 / 2 = 4I_0$ ($\because \Delta x = 0$)

$$\text{Now, } \therefore \phi = \frac{2\pi}{\lambda} \times \Delta x = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} \Rightarrow \phi = \frac{\pi}{2}$$

$$\text{and } I_2 = 4I_0 \cos^2 \frac{\pi}{4} = 2I_0 \quad \left[\phi = \frac{2\pi}{\lambda} \Delta x \right] \therefore I_1 : I_2 = 2 : 1$$

4.(A) The situation is shown in the figure.

In figure A and B represent the edges of the slit AB of width a and C represents the midpoint of the slit.

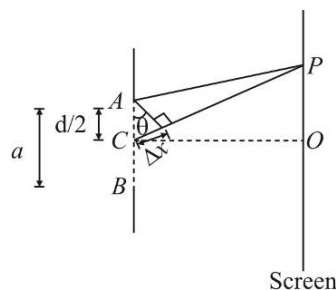
For the first minimum at P ,

$$a \sin \theta = \lambda \quad \dots(i)$$

Where λ is the wavelength of light.

The path difference between the wavelets from A to C is

$$\begin{aligned} \Delta x &= \frac{a}{2} \sin \theta = \frac{1}{2}(a \sin \theta) \\ &= \frac{\lambda}{2} \quad \text{(Using (i))} \end{aligned}$$



The corresponding phase difference $\Delta\phi$ is $\Delta\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$

5.(B) The situation is shown in the figure.

Let f_o and f_e be the focal lengths of the objective and eyepiece respectively.

For normal adjustment distance of the objective from the eyepiece (tube length) = $f_o + f_e$.

Treating the line on the objective as the object and eyepiece as the lens.

$$\therefore u = -(f_o + f_e) \text{ and } f = f_e$$

$$\text{As } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

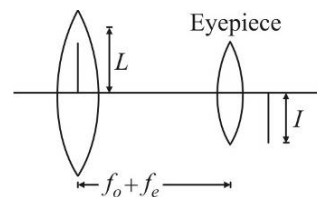
$$\therefore \frac{1}{v} - \frac{1}{-(f_o + f_e)} = \frac{1}{f_e}$$

$$\frac{1}{v} = \frac{1}{f_e} - \frac{1}{f_o + f_e} = \frac{f_o + f_e - f_e}{f_e(f_o + f_e)} = \frac{f_o}{f_e(f_o + f_e)}$$

$$\text{Or } v = \frac{f_e(f_o + f_e)}{f_o}$$

$$\text{Thus, } \frac{I}{L} = \frac{v}{|u|} = \frac{\frac{f_e(f_o + f_e)}{f_o}}{(f_o + f_e)} = \frac{f_e}{f_o}$$

$$\text{Or } \frac{f_o}{f_e} = \frac{L}{I} \quad \dots (i)$$



$$\therefore \text{The magnification of the telescope in normal adjustment is } m = \frac{f_o}{f_e} = \frac{L}{I} \quad \text{(Using (i))}$$

$$6.(C) \quad \Delta h_o = \left(1 - \frac{1}{\mu_1}\right) h_1 ; \quad \Delta h_k = \left(1 - \frac{1}{\mu_2}\right) h_2 \quad \therefore \quad \Delta h = \Delta h_o + \Delta h_k = \left(1 - \frac{1}{\mu_1}\right) h_1 + \left(1 - \frac{1}{\mu_2}\right) h_2$$

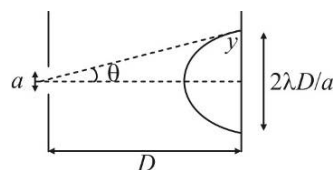
7.(C) Given situation is shown in the figure.

$$\text{For central maxima, } \sin \theta = \frac{\lambda}{a}$$

$$\text{Also, } \theta \text{ is very -very small so } \sin \theta \approx \tan \theta = \frac{y}{D}$$

$$\therefore \frac{y}{D} = \frac{\lambda}{a}, y = \frac{\lambda D}{a}$$

$$\text{Width of central maxima} = 2y = \frac{2\lambda D}{a}$$



8.(A)

- 9.(C) Scattering for blue light is largest and it is polarized by scattering.

Also for polarized light $I = I_0 \cos^2 \theta$

10.(D) As $\mu = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} \cot \frac{A}{2} = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad \left[\because \mu = \cot \frac{A}{2} \right]$

$$\frac{\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} \sin\left(\frac{\pi}{2} - \frac{A}{2}\right) = \sin\left(\frac{A}{2} + \frac{\delta}{2}\right); \frac{\pi}{2} - \frac{A}{2} = \frac{A}{2} + \frac{\delta}{2}$$

$$\delta = \pi - 2A = 180^\circ - 2A$$

- 11.(C) For double slit experiment,

$$d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}, D = 1 \text{ m}, \lambda = 500 \times 10^{-9} \text{ m}$$

$$\text{Fringe width } \beta = \frac{D\lambda}{d}$$

Width of central maxima in a single slit

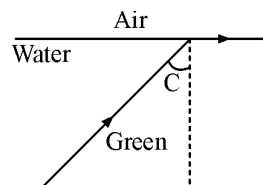
As per question, width of central maxima of single slit pattern = width of 10 maxima of double slit pattern

$$\frac{2\lambda D}{a} = 10 \left(\frac{\lambda D}{d} \right) \Rightarrow a = \frac{2d}{10} = \frac{2 \times 10^{-3}}{10} = 0.2 \times 10^{-3} \text{ m} = 0.2 \text{ mm}$$

- 12.(B) Here, $\sin C = \frac{1}{n_{\text{water}}}$ and $n_{\text{water}} = a + \frac{b}{\lambda^2}$

If frequency is less, then λ is greater and hence $RI_{n(\text{water})}$ is less and therefore critical angle increases.

So, they do not suffer total internal reflection and come out at angle less than 90° in the air medium.



- 13.(C) Intensity at any point on the screen is $I = 4I_0 \cos^2 \frac{\phi}{2}$

Where I_0 is the intensity of either wave and ϕ is the phase difference between two waves.

$$\text{Phase difference, } \phi = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$\text{When path difference is } \lambda, \text{ then } \phi = \frac{2\pi}{\lambda} \times \lambda = 2\pi$$

$$\therefore I = 4I_0 \cos^2 \left(\frac{2\pi}{2} \right) = 4I_0 \cos^2 (\pi) = 4I_0 = K \quad \dots (i)$$

$$\text{When path difference is } \frac{\lambda}{4}, \text{ then } \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2} \quad \therefore I = 4I_0 \cos^2 \left(\frac{\pi}{4} \right) = 2I_0 = \frac{K}{2}$$

- 14.(A) Note As refractive index for $z > 0$ and $z \leq 0$ is different X - Y plane should be boundary between two media.

$$\text{Angle of incidence, } \cos i = \frac{A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = \frac{1}{2}$$

$$\therefore i = 60^\circ$$

From Snell's law $\frac{\sin i}{\sin r} = \frac{\sqrt{3}}{2}$
 $\Rightarrow r = 45^\circ$

15.(D) Magnifying power of a microscope, $m = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$

Where f_o and f_e are the focal lengths of the objective and eyepiece respectively and L is the distance between their focal points and D is the least distance of distinct vision.

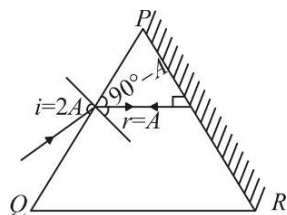
If f_o increases, then m will decrease.

Magnifying power of a telescope, $m = \frac{f_o}{f_e}$

Where f_o and f_e are the focal lengths of the objective and eyepiece respectively.

If f_o increase, then m will increase.

16.(B)



On reflection from silvered surface, the incident ray will retrace its path, if it falls normally on the surface.

By geometry, $r = A$

Applying Snell's law at surface PQ, $1 \sin i = \mu \sin r$ $\mu = \frac{\sin i}{\sin r} = \frac{\sin 2A}{\sin A} = 2 \cos A$

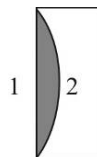
17.(B) Both statements I and II are correct but statement II does not explain statement I.

18.(A) The combination of two lenses 1 and 2 is as shown in figure.

$\therefore \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

According to lens maker's formula

$$\begin{aligned} \frac{1}{f_1} &= (\mu_1 - 1) \left(\frac{1}{\infty} - \frac{1}{-R} \right) = \frac{(\mu_1 - 1)}{R}; \frac{1}{f_2} = (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right) \\ &= (\mu_2 - 1) \left(-\frac{1}{R} \right) = -\frac{(\mu_2 - 1)}{R} \quad \therefore \frac{1}{f} = \frac{(\mu_1 - 1)}{R} - \frac{(\mu_2 - 1)}{R} \quad \frac{1}{f} = \frac{(\mu_1 - \mu_2)}{R} \\ f &= \frac{R}{(\mu_1 - \mu_2)} \end{aligned}$$



19.(A) Converging power of cornea, $P_c = +40D$

Least converging power of eye lens, $P_e = +20D$

Power of the eye-lens, $P = P_c + P_e = 40D + 20D = 60D$

Power of the eye lens $P = \frac{1}{\text{Focal length of the lens (f)}}$

$f = \frac{1}{P} = \frac{1}{60D} = \frac{1}{60}m = \frac{100}{60}cm = \frac{5}{3}cm$

Distance between the retina and cornea-eye lens = focal length of the eye lens $= \frac{5}{3}cm = 1.67cm$

$$20.(A) \quad \frac{1}{u} + \frac{1}{v} = \frac{1}{f} \quad \Rightarrow \quad -\frac{1}{u^2} \frac{du}{dt} - \frac{1}{v^2} \frac{dv}{dt} = 0 \quad \Rightarrow \quad \frac{dv}{dt} = -\frac{v^2}{u^2} \left(\frac{du}{dt} \right)$$

$$\text{But } \frac{v}{u} = \frac{f}{u-f} \quad \therefore \quad \frac{dv}{dt} = -\left(\frac{f}{u-f} \right)^2 \left(\frac{du}{dt} \right) = \left(\frac{0.2}{-2.8-0.2} \right)^2 \times 15 = \frac{1}{15} \text{ ms}^{-1}$$

$$21.(C) \quad 3\lambda_1 = 4\lambda_2 \quad \Rightarrow \quad \lambda_2 = \frac{3}{4}\lambda_1 = \frac{3}{4} \times 590 = \frac{1770}{4} = 442.5 \text{ nm}$$

22.(C) For the second minimum,
Path difference = 2λ

$$\text{Therefore, corresponding value of phase difference is } \Delta\phi = \frac{2\pi}{\lambda} \times \text{path difference} \quad \therefore \Delta\phi = \frac{2\pi}{\lambda} \times 2\lambda = 4\pi$$

23.(A) As the emergent ray emerges normally from the opposite face,

$$\therefore e = 0, r_2 = 0 \text{ as } r_1 + r_2 = A \quad \therefore r_1 = A$$

Applying Snell's law for incident ray

$$1 \sin i = \mu \sin r_1 = \mu \sin A \text{ or } \mu = \frac{\sin i}{\sin A}$$

For small angle, $\sin i \approx i, \sin A \approx A$

$$\therefore \mu = \frac{i}{A} \text{ or } i = \mu A$$

24.(D) By law of Malus i.e., $I = I_0 \cos^2 \theta$

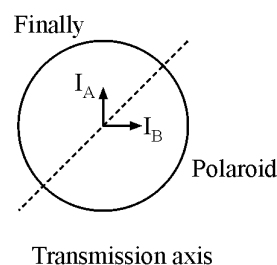
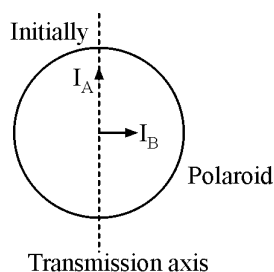
$$\text{Now, } I_{A'} = I_A \cos^2 30^\circ$$

$$I_{B'} = I_B \cos^2 60^\circ$$

$$\text{As, } I_{A'} = I_{B'}$$

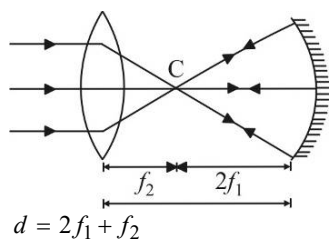
$$I_A \cos^2 30^\circ = I_B \cos^2 60^\circ$$

$$\Rightarrow I_A \frac{3}{4} = I_B \frac{1}{4} \Rightarrow \frac{I_A}{I_B} = \frac{1}{3}$$



$$25.(C) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f} = \text{constant, so (C) is the correct graph.}$$

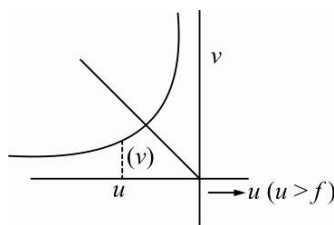
26.(C)



27.(A) It is possible when object kept at centre of curvature.

$$u = v$$

$$u = 2f, v = 2f$$



$$28.(B) \quad \text{As } \mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad \mu = \frac{\sin\left(\frac{A + A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin A}{\sin\left(\frac{A}{2}\right)} \quad (\because \delta_m = A \text{ (Given)})$$

$$= \frac{2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = 2 \cos\left(\frac{A}{2}\right)$$

As $\delta = i + e - A$

At minimum deviation, $\delta = \delta_m, i = e$

$$\therefore \delta_m = 2i - A; 2i = \delta_m + A$$

$$i = \frac{\delta_m + A}{2} = \frac{A + A}{2} = A \quad (\because \delta_m = A \text{ given})$$

$$i_{\min} = 0^\circ \Rightarrow A_{\min} = 0^\circ$$

$$\text{Then, } \mu_{\max} = \frac{\pi}{2} \Rightarrow A_{\max} = \frac{\pi}{2} \quad \therefore i_{\max} = \frac{\pi}{2} \Rightarrow A_{\max} = \frac{\pi}{2}$$

$$\text{Then, } \mu_{\min} = 2 \cos 45^\circ = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$$

So refractive index lies between 2 and $\sqrt{2}$

29.(C) Let $\mu = \frac{3}{2}$

According to lens maker's formula $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

For biconvex lens, $R_1 = +20\text{cm}, R_2 = -20\text{cm}$

$$\therefore \frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \text{ or } f = 20\text{cm}$$

According to thin lens formula, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Here, $u = -30\text{cm}$

$$\therefore \frac{1}{20} = \frac{1}{v} - \frac{1}{-30} \Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30} \Rightarrow v = 60\text{cm}$$

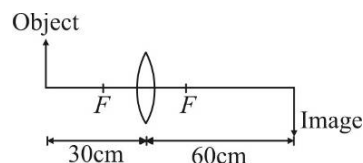
The image is formed at a distance of 60 cm on the right hand side of the lens. It is a real image.

$$\text{Magnification, } m = \frac{v}{u} = \frac{h_1}{h_0}$$

$$\frac{60\text{cm}}{-30\text{cm}} = \frac{h_1}{2\text{cm}} \Rightarrow h_1 = -4\text{cm}$$

-ve sign shows that image is inverted.

The image is real, inverted and height of 4 cm as shown in figure.

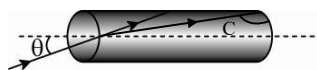


30.(D) $\sin C = \frac{\sqrt{3}}{2} \dots(i)$

$$\sin r = \sin(90^\circ - C) = \cos C = \frac{1}{2}$$

$$\frac{\sin \theta}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\sin \theta = \frac{2}{\sqrt{3}} \times \frac{1}{2} \quad ; \quad \theta = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$



31.(C) For dispersion without deviation $\delta_1 + \delta_2 = 0$

$$(\mu_1 - 1)A_1 + (\mu_2 - 1)A_2 = 0 \quad A_2 = -\frac{(\mu_1 - 1)A_1}{(\mu_2 - 1)}$$

$$\text{Substituting the given values, we get, } A_2 = -\frac{(1.5 - 1)15^\circ}{(1.75 - 1)} = -10^\circ$$

–ve sign shows that two prisms must be joined in opposition

$$32.(D) \quad \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$$

$$\text{As, } I = I_{\max} \cos^2\left(\frac{\phi}{2}\right) \quad \text{or} \quad \frac{I}{I_{\max}} = \cos^2\left(\frac{\phi}{2}\right) \quad \text{or} \quad \frac{I}{I_0} = \cos^2\left(\frac{\pi}{6}\right) = \frac{3}{4}$$

33.(C) For total internal reflection, $\sin i > \sin C$

Where,

i = angle of incidence, C = critical angle

$$\text{But, } \sin C = \frac{1}{\mu} \quad \therefore \quad \sin i > \frac{1}{\mu} \quad \text{or} \quad \mu > \frac{1}{\sin i} \quad \mu > \frac{1}{\sin 45^\circ} \quad (i = 45^\circ \text{ (Given)})$$

$$\mu > \sqrt{2}$$

$$34.(B) \quad \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}; \quad \therefore \text{Power } P = \frac{f_1 + f_2}{f_1 f_2}$$

35.(A)

$$36.(D) \quad \text{From figure, } \sin C = \frac{3}{\sqrt{(4)^2 + (3)^2}} = \frac{3}{5}$$

Where C is the critical angle.

$$\text{Also, } \sin C = {}^l\mu_a$$

$$\sin C = \frac{1}{{}^a\mu_l} \left[\text{since } {}^l\mu_a = \frac{1}{{}^a\mu_l} \right]$$

$$\text{Also } {}^a\mu_l = \frac{\text{velocity of light in air } (c)}{\text{velocity of light in liquid } (v)}$$

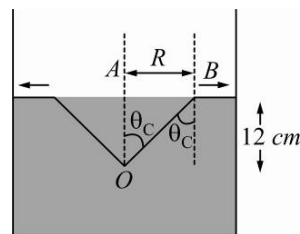
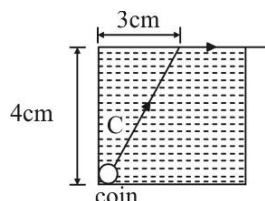
$$\therefore \quad \sin C = \frac{v}{c} = \frac{v}{3 \times 10^8} \quad \text{or} \quad v = 3 \times 10^8 \times \frac{3}{5} = 1.8 \times 10^8 \text{ ms}^{-1}$$

37.(B) The situation is shown in figure.

$$\sin \theta_C = \frac{1}{\mu}; \quad \tan \theta_C = \frac{AB}{OA}$$

$$\therefore \quad AB = OA \tan \theta_C$$

$$\text{or} \quad AB = \frac{OA}{\sqrt{\mu^2 - 1}} = \frac{12}{\sqrt{\left(\frac{4}{3}\right)^2 - 1}} = \frac{36}{\sqrt{7}}$$



$$38.(C) \quad R.P = 1/\Delta\theta \quad \text{The angular resolution, } \Delta\theta = \frac{1.22\lambda}{D} = \frac{1.22 \times 5000 \times 10^{-8}}{0.1} = 6.1 \times 10^{-4} \approx 10^{-4}$$

$$39.(C) \quad I = I_0 \left(\frac{\sin \theta}{\theta} \right)^2 \quad \text{and} \quad \theta = \frac{\pi}{\lambda} \left(\frac{\alpha y}{d} \right)$$

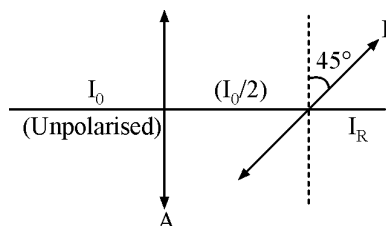
For principal maximum $y = 0$

$$\therefore \theta = 0$$

Hence, intensity will remain same.

40.(C) Relation between intensities is :

$$I_r = \left(\frac{I_0}{2}\right) \cos^2(45^\circ) = \frac{I_0}{2} \times \frac{1}{2} = \frac{I_0}{4}$$

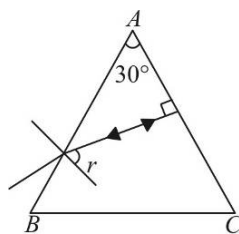


41.(A) $\angle r = 30^\circ$ (using law of triangle)

$$\Rightarrow \mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sqrt{2} \times \sin 30 = \sin i$$

$$\Rightarrow \sin i = \frac{1}{\sqrt{2}} \Rightarrow i = 45^\circ$$



42.(A) We know $\frac{y}{D} \geq 1.22 \frac{\lambda}{d} \Rightarrow D \leq \frac{yd}{1.22\lambda}$

$$\therefore D = \frac{10^{-3} \times 3 \times 10^{-3}}{1.22 \times 5 \times 10^{-7}} 5m \Rightarrow D_{max} = 5m$$

43.(C) Applying Snell's law of refraction at A, we get

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 45^\circ}{\sin r} \quad \therefore \sin r = 1/\sqrt{2}\mu$$

$$\therefore r = \sin^{-1}\left(\frac{1}{\sqrt{2}\mu}\right) \quad \dots (i)$$

Applying the condition of total internal reflection at B, we get

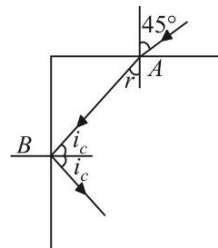
$$i_c = \sin^{-1}(1/\mu) \quad \dots (ii)$$

Where i_c is the critical angle.

Now, $r + i_c = 90^\circ = \pi/2$

$$\therefore \sin^{-1} \frac{1}{\sqrt{2}\mu} = \cos^{-1} \frac{1}{\mu} \quad \text{Or,} \quad \sin^{-1} \frac{1}{\sqrt{2}\mu} = \cos^{-1} \frac{1}{\mu}$$

$$\therefore \frac{1}{\sqrt{2}\mu} = \frac{\sqrt{\mu^2 - 1}}{\mu} \text{ or } \frac{1}{2} = \mu^2 - 1. \quad \therefore \mu = \sqrt{3/2}$$



44.(A) $D = (\mu - 1)A$ For blue light μ is greater than that for red light so $D_2 > D_1$.

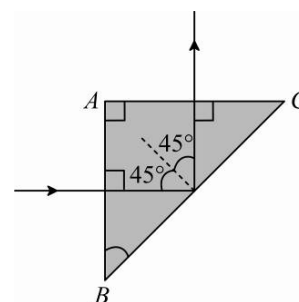
45.(B) For total internal reflection from glass-air interface, critical angle C must be less than angle of incidence.

i.e., $C < i$

or $C < 45^\circ$ ($\because \angle i = 45^\circ$)

$$\text{But } n = \frac{1}{\sin C} \Rightarrow = \sin^{-1}\left(\frac{1}{n}\right)$$

$$\text{or } \sin^{-1}\left(\frac{1}{n}\right) < 45^\circ \quad \text{or } \frac{1}{n} < \sin 45^\circ$$



$$\text{or } n > \frac{1}{\sin 45^\circ}$$

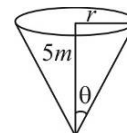
$$\text{or } n > \frac{1}{(1/\sqrt{2})}$$

$$\text{or } n > \sqrt{2}$$

46.(B) θ is the critical angle.

$$\therefore \theta = \sin^{-1}(1/\mu) = \sin^{-1}(3/5)$$

$$\text{Or } \sin \theta = 3/5 \quad \therefore \tan \theta = 3/4 = r/4 \text{ or, } r = 3m.$$



$$47.(A) \quad \frac{1}{F} = \frac{2}{f} + \frac{2}{R}$$

$$\dots(i) \quad (\because R = 2f_m)$$

$$\therefore \frac{1}{f} = (1.5 - 1) \left(\frac{1}{\infty} - \frac{1}{30} \right) \quad \text{or} \quad \frac{1}{f} = \frac{0.5}{30} = -\frac{1}{60} \quad \text{or} \quad f = -60cm$$

Hence, from equation (i)

$$\frac{1}{F} = \frac{2}{60} + \frac{2}{30} = \frac{6}{60}$$

$$F = 10 \text{ cm}$$

$$O = I$$

$$\therefore m = -\frac{v}{u} = \frac{I}{O} \Rightarrow \frac{v}{u} = -1 \quad \text{or} \quad v = -u$$

$$\text{Thus, from lens formula } \frac{1}{F} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{10} = \frac{1}{-u} - \frac{1}{u}$$

$$\frac{1}{10} = -\frac{2}{u} \quad \therefore u = -20 \text{ cm}$$

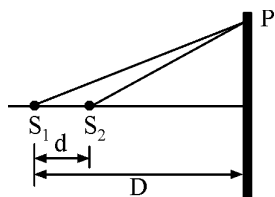
Hence, to get a real image, object must be placed at a distance 20 cm on the left side of lens.

48.(C) Total apparent depth, $y = y_1 + y_2 = 5 + 2 = 7cm$

If x is real depth = thickness of slab. Then as

$$\mu = \frac{\text{real depth}}{\text{apparent depth}} = \frac{x}{y} \text{ or, } x = \mu y = 1.5 \times 7 = 10.5cm.$$

49.(D) For bright fringes, $S_1P - S_2P = n\lambda$



So, fringes are concentric circle (centre of origin). [$\because$ Due to point source, wave fronts will be spherical]

50.(B) The minimum mirror length should be half of the height of man.

$$51.(A) \quad I = I_0 \cos^2 \theta$$

$$\text{Intensity of polarized light} = \frac{I_0}{2}$$

$$\therefore \text{Intensity of un-transmitted light} = \frac{I_0}{2}$$

- 52.(D) The particular angle of incidence for which reflected light is totally polarized for reflection from air to glass, is called the angle of polarisation (i_p) (Brewster's law)

Accordingly, $n = \tan i_p$

$$\Rightarrow i_p = \tan^{-1}(n)$$

where n is refractive index of glass.

- 53.(A) For lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $u = -30, f = 20, v = 60 \text{ cm}$

To have an upright image of the object, coincide with it, image should tend to form at centre of curvature of convex mirror. Therefore, the distance of convex mirror from the lens = $60 - 10 = 50 \text{ cm}$.

- 54.(C) For possible interference maxima on the screen the condition is :

$$d \sin \theta = n\lambda \quad \dots(i)$$

$$\text{Given, } d = \text{slit - width} = 2\lambda \quad \therefore 2\lambda \sin \theta = n\lambda \quad \text{or} \quad 2 \sin \theta = n$$

The maximum value of $\sin \theta$ is 1, hence, $n = 2 \times 1 = 2$

Thus, equation (i) must be satisfied by 5 integer value i.e., $-2, -1, 0, 1, 2$. Hence, the maximum number of maxima corresponding to $n = -2$ & $n = 2$ are practically not possible.

- 55.(C) $n > \frac{\sin r}{\sin i}$ i.e., $n > \frac{\sin 90^\circ}{\sin 45^\circ} \Rightarrow n > \sqrt{2}$

- 56.(C) Objective of compound microscope is a convex lens. Convex lens forms real and enlarged image when an object is placed between its focus and lens.

$$57.(B) \text{ Time} = \frac{\text{distance}}{\text{velocity}} = \frac{t}{v} = \frac{t}{c/\mu} = \frac{\mu t}{c}$$

$$58.(C) \quad n = \frac{3}{2} \quad \left[\because n = \frac{c}{v} \right]$$

$$3^2 + (R - 3\text{mm})^2 = R^2 \quad \Rightarrow \quad 3^2 + R^2 - 2R(3\text{mm}) + 3\text{mm}^2 = R^2$$

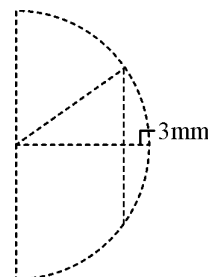
$$\Rightarrow R = 15\text{cm}$$

$$\therefore \frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{15} \right)$$

$$\left[\because \frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \right]$$

$$\text{and } R_1 = \infty, R_2 = -15$$

$$\Rightarrow f = 30 \text{ cm}$$



- 59.(C) By displacement method, size of object (O) = $\sqrt{I_1 \times I_2}$

$$\text{Therefore area of source of light } (A) = \sqrt{A_1 A_2}$$

- 60.(B)

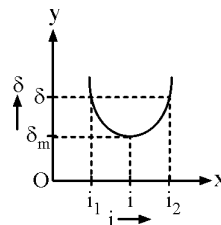
- 61.(A)

- 62.(D) Separations between the slits (d_1) = 16 cm and (d_2) = 9 cm

$$\text{Actual distance of separation } (d) = \sqrt{d_1 d_2} = \sqrt{16 \times 9} = 12 \text{ cm}$$

- 63.(C) $v_g = \frac{c}{\mu} = \frac{3 \times 10^8}{\frac{3}{2}} = 2 \times 10^8 \text{ m/s}$ $t = \frac{x}{v_g} = \frac{4 \times 10^{-3}}{2 \times 10^8} = 2 \times 10^{-11} \text{ s}$

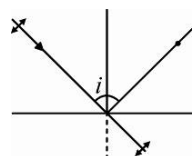
- 64.(C) We know that the angle of deviation depends upon the angle of incidence. If we determine experimentally, the angles of deviation corresponding to different angles of incidence and then plot i (on x -axis) and δ (on y -axis), we get a curve as shown in figure.



Clearly, if angle of incidence is gradually increased from a small value, the angle of deviation first decreases, becomes minimum for a particular angle of incidence and then begins to increase.

65.(A) $\beta' = \frac{\beta}{\mu} = \frac{0.4}{\frac{4}{3}} = 0.3 \text{ mm}$

- 66.(C) At Brewster's angle, $i = \tan^{-1}(\mu)$, the reflected light is completely polarized, whereas refracted light is partially polarized. Thus the reflected ray will have lesser intensity compared to refracted ray.



$\therefore I_{\text{reflected}} < \frac{I_0}{2}$

- 67.(D) Resolving power of an optical instrument is inversely proportional to λ i.e. $RP \propto \frac{1}{\lambda}$

$\therefore \frac{\text{Resolving power at } \lambda_1}{\text{Resolving power at } \lambda_2} = \frac{\lambda_2}{\lambda_1} = \frac{5000}{4000} = 5:4$

- 68.(D) Telescope resolves and brings the objects closer which is far away from the telescope. Hence for telescope with magnifying power 20, the tree appears 20 times nearer.

- 69.(C) Size of spot, b

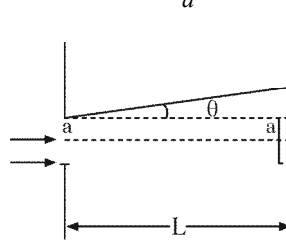
= Geometrical spread + diffraction spread

$\therefore b = a + L \frac{\lambda}{a}$

Now, value of b would be minimum if $\frac{db}{da} = 0$

$1 + L \left(\frac{-\lambda}{a^2} \right) = 0 \Rightarrow a^2 = \lambda L \Rightarrow a = \sqrt{\lambda L}$

$\therefore b_{\min} = \sqrt{\lambda L} + \frac{\lambda L}{\sqrt{\lambda L}} = 2\sqrt{\lambda L} = \sqrt{4\lambda L}$



- 70.(A) For convex lens.

$u_1 = -60 \text{ cm}, f_1 = 30 \text{ cm}$

$\frac{1}{v_2} = \frac{1}{f_2} + \frac{1}{u_1} = -\frac{1}{120} + \frac{1}{40} = \frac{2}{120} \text{ or } v_2 = 60 \text{ cm}$

For plane mirror, virtual object 10 cm behind the mirror. Hence, real image 10 cm in front of the mirror.

Now, again for concave lens, $u_2 = 40 \text{ cm}$ i.e., light rays from the object retrace their path after striking the plane mirror. Hence the final image is formed at the object itself.

- 71.(C)

72.(B) Here, $R = -10 \text{ cm}$

Object distance from mirror, $u = (10 - 6) = 4 \text{ cm}$

Focal length of mirror, $f = -\frac{R}{2} = -\frac{10}{2} = -5 \text{ cm}$

Image distance = $v = ?$

Using mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \Rightarrow \quad \frac{1}{v} + \frac{1}{-4} = \frac{1}{-5}$$

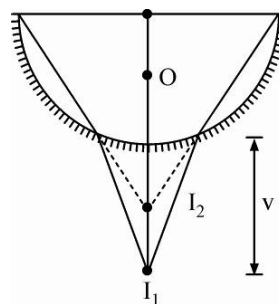
$$\Rightarrow \quad \frac{1}{v} = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

$$v = 20 \text{ cm}$$

Now, I_1 acts as object for plane glass surface,

$$\therefore \text{Apparent depth} = \frac{R + v}{\mu} = \frac{30}{1.5} = 20 \text{ cm}$$

Hence, the position of the image of the air bubble made by the mirror is seen 20 cm below the flat surface.



73.(A) Hence $\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}$

$$D = 30 \text{ cm} = 30 \times 10^{-2} \text{ m} = 0.3 \text{ m}$$

$$R = 10ly = 10 \times 9.46 \times 10^{15} \text{ m}$$

$$l = ?$$

The limit of resolution of a telescope

$$\Delta\theta = \frac{1.22\lambda}{D} = \frac{l}{R}$$

$$l = \frac{1.22\lambda R}{D} = \frac{1.22 \times 6 \times 10^{-7} \times 10 \times 9.46 \times 10^{15}}{0.3} = 2.31 \times 10^8 \text{ km}$$

74.(D) Given, wavelength of light, $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$. Least distance of distinct vision,

$$D = 25 \text{ cm} = 25 \times 10^{-2} \text{ m}$$

Radius of pupil, $r = 0.25 \text{ cm}$

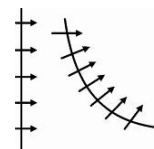
$$\therefore \text{Diameter of pupil, } d = 2r = 0.50 \text{ cm} = 0.50 \times 10^{-2} \text{ m}$$

$$= \frac{1.22 \times 500 \times 10^{-9}}{0.50 \times 10^{-2}} = 1.22 \times 10^{-4} \text{ rad} \quad \therefore \text{Minimum separation that eye can resolve,}$$

$$x = \Delta\theta D = 1.22 \times 10^{-4} \times 25 \times 10^{-2}$$

$$= 30.5 \times 10^{-6} \text{ m} \approx 30 \mu\text{m}$$

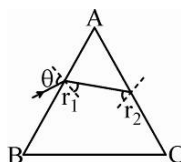
75.(B) Consider a plane wavefront travelling horizontally. As refractive index of air increases with height, so speed of wavefront decreases with height. Hence, the light beam bends upwards.



76.(C) According to Snell's law

$$\sin \theta = \mu \sin r_1$$

$$\Rightarrow \quad \sin r_1 = \frac{\sin \theta}{\mu}$$



$$\text{or } r_1 = \sin^{-1} \left(\frac{\sin \theta}{\mu} \right)$$

$$\text{Now, } A = r_1 + r_2 \quad \therefore \quad r_2 = A - r_1 = A - \sin^{-1} \left(\frac{\sin \theta}{\mu} \right) \quad \dots (i)$$

For the ray to get transmitted through the face AC, r_2 must be less than critical angle.

$$\text{i.e., } r_2 < \sin^{-1} \left(\frac{1}{\mu} \right)$$

$$\text{or } A - \sin^{-1} \left(\frac{\sin \theta}{\mu} \right) < \sin^{-1} \left(\frac{1}{\mu} \right) \quad (\text{Using (i)})$$

$$\Rightarrow \sin^{-1} \left(\frac{\sin \theta}{\mu} \right) > A - \sin^{-1} \left(\frac{1}{\mu} \right)$$

$$\Rightarrow \frac{\sin \theta}{\mu} > \sin \left(A - \sin^{-1} \left(\frac{1}{\mu} \right) \right) \Rightarrow \theta > \sin^{-1} \left[\mu \sin \left(A - \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$$

77.(C) Here, $u = 10 \text{ cm}$, $v = +15 \text{ cm}$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{15} - \frac{1}{10} = \frac{1}{f} = \frac{2}{R} \quad \text{or} \quad -\frac{5}{150} = \frac{2}{R} \quad ; \quad R = -\frac{300}{5} = 60 \text{ cm}$$

78.(C)

79.(D) Angular magnification $m = \frac{f_0}{f_e} = \frac{150}{5} = 30$

$$\text{So, } \frac{\tan \beta}{\tan \alpha} = 30$$

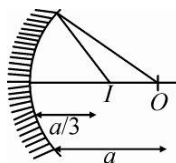
$$\tan \beta = \tan \alpha \times 30 = \left(\frac{50}{10000} \right) \times 30 \quad ; \quad = \frac{15}{10} = \frac{2}{w3} \quad ; \quad \beta = \tan^{-1} \left(\frac{3}{2} \right) \quad \therefore \quad \theta = \beta = 60^\circ$$

80.(B) This combination will behave like a mirror of power

$$P_{eq} = 2P_L + P_M$$

$$P_{eq} = 2 \frac{1}{f} + 0$$

$$f_{eq} = -\frac{f}{2}$$



So the behaviour of the system, will be like a mirror of focal length $-\frac{f}{2}$

$$\text{Using mirror equation } \frac{1}{v} + \frac{1}{u} = \frac{1}{f_{eq}}$$

$$\text{Here, } u = -a, v = -a/3, f_{eq} = -f/2$$

$$\frac{1}{-a/3} + \frac{1}{-a} = \frac{-1}{f/2}; \frac{4}{a} = \frac{2}{f} \quad \text{or} \quad a = 2f$$

81.(B) Intensity at any point on the screen is given by

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$I_{\max} = 4I_0$$

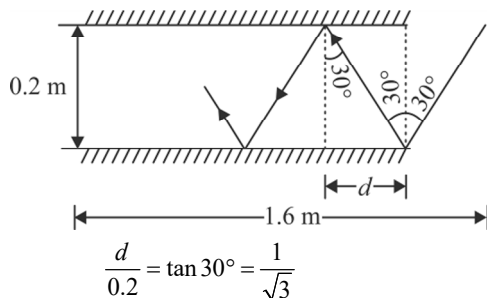
$$\text{Now, } \frac{I_{\max}}{2} = 2I_0 = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

$$\cos\left(\frac{\phi}{2}\right) = \frac{1}{\sqrt{2}} \quad \therefore \quad \frac{\phi}{2} = \frac{\pi}{4}; \phi = \frac{\pi}{2}$$

$$\text{Also } \frac{2\pi}{\lambda} \Delta x = \frac{\pi}{2}; \Delta x = \frac{\lambda}{4}$$

$$y \frac{d}{D} = \frac{\lambda}{4} \therefore y = \frac{\lambda D}{4d} = \frac{\beta}{4}$$

82.(14)

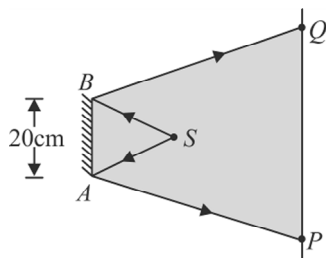


$$\therefore d = \frac{0.2}{\sqrt{3}}$$

$$N = \frac{1.6}{d} = 8\sqrt{3} = 13.85$$

Therefore, actual number of reflections required are 14.

83.(6)



Insect can see the image of source S in mirror, so far as it remains in field of view of image overlapping with the road.

Shaded portion is the field of view, which overlaps with the road upto length PQ .

By geometry we can see that, $PQ = 3AB = 60 \text{ cm}$

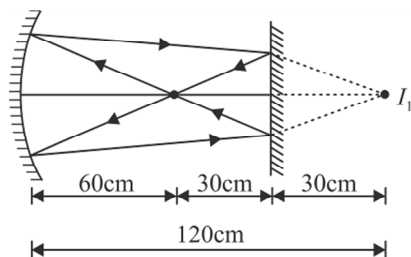
$$\therefore t = \frac{\text{Distance}}{\text{Speed}} = \frac{60}{10} = 6 \text{ s}$$

84.(90) Using mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for concave mirror first, we have

$$\frac{1}{v} - \frac{1}{60} = \frac{1}{-40} \quad \left(\because f = \frac{R}{2} \right)$$

$$\text{or } v = -120 \text{ cm}$$

First image I_1 at 120 cm concave mirror will act as virtual object for plane mirror. Plane mirror will form real image of I_1 at S .



Ray diagram is shown in figure.

Distance between two mirrors is 90 cm .

85.(14) For plane surface, $h_{\text{app}} = \mu h = \frac{3}{2} \times 10 = 15\text{ cm}$

This first image is at a distance, $(15+3)\text{ cm}$ from the plane mirror. So, mirror will make it second image at a distance 18 cm below the mirror or 21 cm below the plane surface.

Now further applying, $d_{\text{app}} = \frac{d}{\mu} = \frac{21}{1.5} = 14\text{ cm}$ below the plane surface.

86.(9) Given, $-\frac{dy}{dt} = 4\text{ cm/s}$

Distance of bird as observed by fish

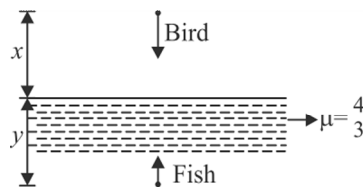
$$Z = y + \mu x = y + \frac{4}{3}x$$

$$\therefore \frac{-dZ}{dt} = \frac{-dy}{dt} + \frac{4}{3} \left(\frac{-dx}{dt} \right) \quad \dots (i)$$

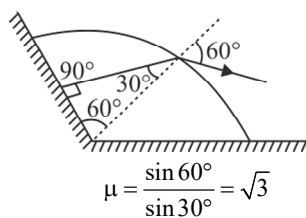
Given, $\frac{-dZ}{dt} = 16\text{ cm/s}$

Substituting in Eq. (i), we get

$$\frac{-dx}{dt} = 9\text{ cm/s}$$



87.(3)



88.(74) $i_1 = r_1 = 0^\circ$

$$r_2 = 90^\circ - \theta = 53^\circ$$

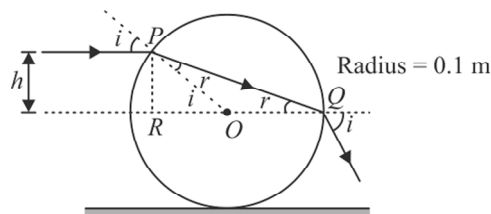
$$\theta_C = \sin^{-1} \left(\frac{1}{\mu} \right) = 37^\circ$$

Since, $r_2 > \theta_C$, T/R will take place on the face AB

$$\delta = \delta_{AC} + \delta_{AB} = 0 + 180^\circ - 2r_2 = 180^\circ - 2 \times 53^\circ = 74^\circ$$

89.(0.186) $PO = OQ$

$$\therefore \angle OPQ = \angle OQP = r \text{ (say)}$$



Also, $i = r + r = 2r$

In $\triangle POR$, $h = OP \sin i = 0.1 \sin 2r = 0.1 \sin 2r$

or $h = 0.2 \sin r \cos r \quad \dots (i)$

Also, $\sqrt{3} = \frac{\sin i}{\sin r} = \frac{2 \sin r \cos r}{\sin r} = 2 \cos r$

$$\therefore r = 30^\circ$$

Substituting in Eq. (i), we get

$$h = 0.2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = 0.086 \text{ m}$$

Hence, height from mirror is $0.1 + 0.086 = 0.186 \text{ m}$

90.(4) For the lens, $u = -2.0 \text{ m}$, $f = +1.5 \text{ m}$

$$\therefore \frac{1}{v} - \frac{1}{-2.0} = \frac{1}{1.5} \text{ or } v = 6.0 \text{ m}$$

$$m = \frac{6.0}{-2.0} = -3.0$$

Therefore, y-coordinate of image formed by lens is

$$m(0.1) = -0.3 \text{ m}$$

$$\frac{0.3}{NP} = \tan \theta = 0.3$$

$$\therefore NP = MP = 1.0 \text{ m}$$

or $d = 6.0 - 1.0 = 0.5 \text{ m}$

and x-coordinate of final image I_2 is

$$x = d - 1.0 = 4.0 \text{ m}$$

91.(16) $A_{\max} = 5 + 3 = 8 \text{ units}$

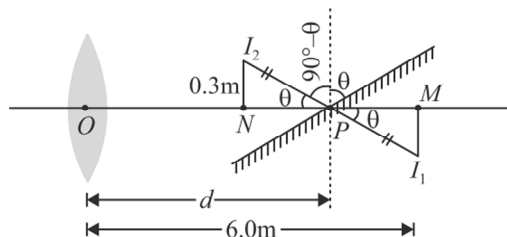
$$A_{\min} = 5 - 3 = 2 \text{ units}$$

$$\frac{A_{\max}}{A_{\min}} = 4 \quad \therefore \frac{I_{\max}}{I_{\min}} = (4)^2 = 16$$

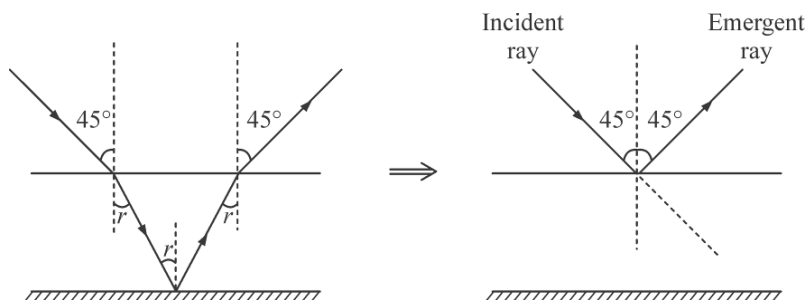
92.(1) Wavelength in water, $\lambda' = \frac{\lambda}{\mu}$

$$\text{Fringe width, } \omega = \frac{\lambda'D}{d} = \frac{\lambda D}{\mu d}$$

$$= \frac{(700 \times 10^{-6})(0.48)}{(4/3)(0.25 \times 10^{-3})} = 10^{-3} \text{ m} = 1 \text{ mm}$$



93.(90)



From the figure it is clear that the angle between the incident ray and the emergent ray is 90° .

94.(16) The apparent position of ball from the surface of water as seen by fish at any time t is.

$$y' = \mu y = \frac{4}{3} \left[20 - \frac{gt^2}{2} \right]$$

$$\frac{dy'}{dt} = \frac{4}{3} \left[-\frac{2gt}{2} \right]$$

To get t , use $20 - \frac{gt^2}{2} = 12.8$ or, $7.2 = \frac{10t^2}{2}$

or $\frac{14.4}{10} = t^2$ or $t = \sqrt{1.44} = 1.2 \text{ s}$

So, required speed $v = \frac{4}{3} [-10 \times 1.2] = 16 \text{ m/s}$

95.(20) When magnification is +2 ;

$$u = -x, \quad v = -2x, \quad f = +20$$

Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

We have $-\frac{1}{2x} + \frac{1}{x} = \frac{1}{20}$

Or $x = 10 \text{ cm}$

To have a magnification of -2

$$u = -y \text{ m}, \quad v = +2y, \quad \text{and} \quad f = +20$$

$$\therefore \frac{1}{2y} + \frac{1}{2} = \frac{1}{20} \quad \text{or} \quad y = 30 \text{ cm} \quad \therefore y - x = 20 \text{ cm}$$

Modern Physics

1.(B) When an electron moves in a circular path, then Radius, $r = \frac{mv}{eB} \Rightarrow \frac{r^2 e^2 B^2}{2} = \frac{m^2 v^2}{2}$

$$KE_{\max} = \frac{(mv)^2}{2m} \Rightarrow \frac{r^2 e^2 B^2}{2m} = (KE)_{\max}$$

Work function of the metal (W),

i.e., $W = h\nu - KE_{\max}$

$$1.89 - \phi = \frac{r^2 e^2 B^2}{2m} \frac{1}{2} eV = \frac{r^2 e^2 B^2}{2m} eV \quad [h\nu \rightarrow 1.89 \text{ eV}, \text{ for the transition on from third to second orbit of H-atom}]$$

$$= \frac{100 \times 10^{-6} \times 16 \times 10^{-19} \times 9 \times 10^{-8}}{2 \times 9.1 \times 10^{-31}}$$

$$\phi = 1.89 - \frac{1.6 \times 9}{2 \times 9.1} = 1.89 - 0.79 = 1.1 \text{ eV}$$

- 2.(C) Let ϕ_0 be the work function of the surface of the material. Then,
According to Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons in the first case is $K_{\max 1} = \frac{hc}{\lambda} - \phi_0$ and that in the second is $K_{\max 2} = \frac{hc}{\frac{\lambda}{2}} - \phi_0 = \frac{2hc}{\lambda} - \phi_0$

But $K_{\max 2} = K_{\max 1}$ (Given) $\therefore \frac{2hc}{\lambda} - \phi_0 = 3\left(\frac{hc}{\lambda} - \phi_0\right)$

$$\frac{2hc}{\lambda} - \phi_0 = \frac{3hc}{\lambda} - 3\phi_0 \quad ; \quad 3\phi_0 - \phi_0 = \frac{3hc}{\lambda} - \frac{2hc}{\lambda}$$

$$2\phi_0 = \frac{hc}{\lambda} \text{ or } \phi_0 = \frac{hc}{2\lambda}$$

- 3.(D) de-Broglie wavelength, $\lambda = \frac{h}{p}$ or $\lambda \propto \frac{1}{p}$, $\lambda p = \text{constant}$. This represents a rectangular hyperbola 4.(D)

5.(C) $K_{\max} = eV_0 = h(\nu - \nu_0)$

If $\nu' = 2\nu$

$\therefore K'_{\max} = eV'_0 = h(2\nu - \nu_0) = 2K_{\max} + h\nu_0 \quad \therefore K'_{\max} > 2K_{\max} \text{ and } \Rightarrow \nu'_0 > 2\nu_0$

- 6.(D) de-Broglie wavelength of neutrons in thermal equilibrium at temperature T is $\lambda = \frac{h}{\sqrt{2mk_B T}}$ where m is the mass of the neutron

k_B is the Boltzmann constant

h is the Planck's constant

Here, $m = 1.67 \times 10^{-27} \text{ kg}$

$k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$

$h = 6.63 \times 10^{-34} \text{ Js}$

$\therefore \lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 1.38 \times 10^{-23} \times T}} = \frac{3.08 \times 10^{-34} \times 10^{25}}{\sqrt{T}} m = \frac{30.8 \times 10^{-10}}{\sqrt{T}} = \frac{30.8}{\sqrt{T}} \text{ \AA}$

- 7.(B) The de-Broglie wavelength λ associated with the electrons is $\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$ where V is the accelerating potential in volts.

or $\lambda \propto \frac{1}{\sqrt{V}}$

$\therefore \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{V_2}{V_1}} = \sqrt{\frac{100 \times 10^3}{25 \times 10^3}} = 2 \text{ or } \lambda_2 = \frac{\lambda_1}{2}$

8.(B) $(KE)_{\max} = \frac{hc}{\lambda} - \phi \Rightarrow \phi = \frac{hc}{\lambda} - (KE)_{\max} \Rightarrow \phi = \frac{1240}{400} - 1.68 \Rightarrow \phi = 1.42 \text{ eV}$

- 9.(D) When a beam of cathode rays (or electrons) are subjected to crossed electric (E) and magnetic (B) fields, the beam is not deflected, if

Force on electron due to magnetic field = Force on electron due to electric field

$$Bev = eE \text{ or } v = \frac{E}{B} \quad \dots\dots (i)$$

If V is the potential difference between the anode and the cathode, then

$$\therefore \frac{1}{2}mv^2 = eV$$

$$\frac{e}{m} = \frac{v^2}{2V} \quad \dots\dots (ii)$$

Substituting the value of v from the equation (i) in equation (ii), we get $\frac{e}{m} = \frac{E^2}{2VB^2}$

Specific charge of the cathode rays $\frac{e}{m} = \frac{E^2}{2VB^2}$

- 10.(C) We know $\lambda = \frac{h}{mv}$

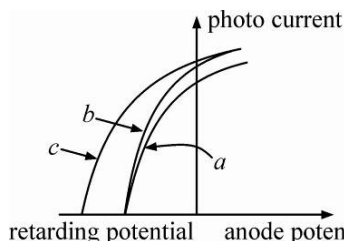
$$\text{and } K = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} \quad \Rightarrow \quad mv = \sqrt{2mK}$$

$$\text{Thus, } \lambda = \frac{h}{\sqrt{2mK}} \quad \Rightarrow \quad \lambda \propto \frac{1}{\sqrt{K}}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{\sqrt{K_1}}{\sqrt{K_2}} = \frac{\sqrt{K_1}}{\sqrt{2K_1}} \quad (\because K_2 = 2K_1)$$

$$\text{or } \frac{\lambda_2}{\lambda_1} = \frac{1}{\sqrt{2}} \quad \text{or } \lambda_2 = \frac{\lambda_1}{\sqrt{2}}$$

- 11.(A)



retarding potential

(a) and (b) represent radiations of the same frequency because their kinetic energies are the same. But saturation photocurrents are different. Therefore intensities are different.

- 12.(D) Power of monochromatic light beam is $P = Nh\nu$ where N is the number of photons emitted per second.

$$\text{Power } P = 2 \times 10^{-3} \text{ W}$$

$$\text{Energy of one photon } E = h\nu = 6.63 \times 10^{-34} \times 6 \times 10^{14} \text{ J}$$

Number of photons emitted per second,

$$N = P/E = \frac{2 \times 10^{-3}}{6.63 \times 10^{-34} \times 6 \times 10^{14}} = 0.05 \times 10^{17} = 5 \times 10^{15}$$

- 13.(C) K.E. = $h\nu - W$

$$\text{i.e. } \frac{1}{2}mv_{\text{max}}^2 = h\nu - W$$

$$\Rightarrow \frac{1}{2} m \times (4 \times 10^6)^2 = 2h\nu_0 - h\nu_0 \text{ or } \frac{1}{2} m \times (4 \times 10^6)^2 = 2h\nu_0$$

Another case, $2h\nu_0 \rightarrow 5h\nu_0$

$$\frac{1}{2} mv_{\max}^2 = 4h\nu_0$$

$$\Rightarrow \frac{1}{2} mv^2 = 4 \times \frac{1}{2} \times m \times (4 \times 10^6)^2 \Rightarrow v_{\max}^2 = 64 \times 10^{12} \Rightarrow v_{\max} = 8 \times 10^6 \text{ m/s}$$

14.(D) As λ is increased, there will be value of λ and which photoelectron will be cease to come out, so photocurrent will becomes zero.

15.(A)

16.(D) According to Einstein's photoelectric equation,

$$\frac{1}{2} mv^2 = \frac{hc}{\lambda} - W_0 \text{ or } \frac{hc}{\lambda} = \frac{1}{2} mv^2 + W_0 \quad \text{and} \quad \frac{1}{2} mv_1^2 = \frac{hc}{3\lambda/4} - W_0 = \frac{4}{3} \left(\frac{1}{2} mv^2 + W_0 \right) - W_0$$

So, v_1 is greater than $v(4/3)^{1/2}$.

17.(D) Einstein's photoelectric equation is

$$KE_{\max} = h\nu - \phi \quad \dots(i)$$

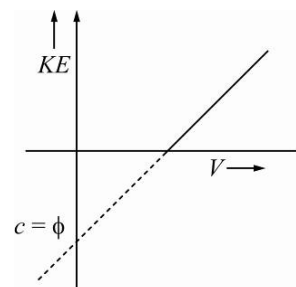
The equation of line is

$$y = mx + c$$

Comparing above two equations

$$m = h, \quad c = -\phi$$

Hence, slope of graph is equal to Planck's constant (non-variable) and does not depend on intensity of radiation.



18.(C) Kinetic energy (E) = 100 eV ;

Mass of electron (m) = $9.1 \times 10^{-31} \text{ kg}$;

$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ and Planck's constant (h) = $6.6 \times 10^{-34} \text{ J-s}$.

Energy of an electron (E) = $100 \times (1.6 \times 10^{-19}) \text{ J}$

$$\text{or } \lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 100 \times 1.6 \times 10^{-19}}} = 1.2 \times 10^{-10} \text{ m} = 1.2 \text{ \AA}$$

19.(A) The kinetic energy of an electron $\frac{1}{2} mv^2 = eV$ or final velocity of electron (v) = $\sqrt{\frac{2eV}{m}}$.

20.(D) Momentum of electrons (p_e) = $\sqrt{2meV}$ and momentum for proton (p_p) = $\sqrt{2MeV}$

$$\text{Therefore, } \frac{\lambda_p}{\lambda_e} = \frac{h/p_p}{h/p_e} = \sqrt{\frac{2meV}{2MeV}} = \sqrt{\left(\frac{m}{M}\right)}$$

$$\text{Therefore } \lambda_p = \lambda \sqrt{\left(\frac{m}{M}\right)}.$$

21.(B) The photoelectric effect is an instantaneous phenomenon (experimentally proved). It takes approximate time of the order of 10^{-10} s.

22.(B) Here, $\frac{hc}{\lambda} = 10^3 \text{ eV}$ and $h\nu = 10^6 \text{ eV}$. Hence, $\nu = \frac{10^3 c}{\lambda} = \frac{10^3 \times 3 \times 10^8}{1.24 \times 10^{-9}} = 2.4 \times 10^{20} \text{ Hz}$

23.(C) $\frac{hc}{\lambda_0} = \phi \Rightarrow \lambda_{max} = \frac{hc}{\phi} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4 \times 1.6 \times 10^{-19}}$

24.(A) Momentum of the photon $= \frac{h\nu}{c} \Rightarrow \frac{c}{\nu} = \frac{h}{p} = \lambda$

$$\nu = \frac{c}{\lambda} = \frac{cp}{h} = 3 \times 10^8 \times \frac{3.3 \times 10^{-29}}{6.6 \times 10^{-34}} = 1.5 \times 10^{13} \text{ Hz}$$

Where, ν = frequency of radiation

25.(C) For hydrogen atom, we get :

$$\begin{aligned} \frac{1}{\lambda} &= R Z^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \Rightarrow \frac{1}{\lambda_1} = R(1)^2 \left(\frac{3}{4} \right) \Rightarrow \frac{1}{\lambda_2} = R(1)^2 \left(\frac{3}{4} \right) \\ \Rightarrow \frac{1}{\lambda_3} &= R(2)^2 \left(\frac{3}{4} \right) \Rightarrow \frac{1}{\lambda_4} = R(3)^2 \left(\frac{3}{4} \right) \\ \therefore \frac{1}{\lambda_1} &= \frac{1}{4\lambda_3} = \frac{1}{9\lambda_4} = \frac{1}{\lambda_2} \end{aligned}$$

26.(A)

27.(B)

28.(B) $\frac{I_2}{I_1} = \frac{(r_1)^2}{(r_2)^2} \quad \left(\text{as } I \propto \frac{1}{r^2} \right)$

$$\Rightarrow \frac{I_2}{I_1} = \frac{(1)^2}{\left(\frac{1}{2} \right)^2} \Rightarrow I_2 = 4I_1$$

Now, since number of electrons emitted per second is directly proportional to intensity, so number of electrons emitted by photocathode would increase by a factor of 4.

29.(A) de-Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{mv}$

30.(B) The wavelength of a spectral line in the Lyman series is $\frac{1}{\lambda_L} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$, $n = 2, 3, 4, \dots$ and that in the

Balmer series is $\frac{1}{\lambda_B} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$, $n = 3, 4, 5, \dots$

For the longest wavelength in the Lyman series, $n = 2$

$$\therefore \frac{1}{\lambda_L} = R \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = R \left(\frac{1}{1} - \frac{1}{4} \right) = R \left(\frac{4-1}{4} \right) = \frac{3R}{4} \text{ or } \frac{1}{\lambda_L} = \frac{4}{3R}$$

For the longest wavelength in Balmer series, $n = 3$

$$\therefore \frac{1}{\lambda_B} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R \left(\frac{1}{4} - \frac{1}{9} \right) = R \left(\frac{9-4}{36} \right) = \frac{5R}{36} \text{ or } \lambda_B = \frac{36}{5R}$$

$$\text{Thus, } \frac{\lambda_L}{\lambda_B} = \frac{4}{3R} \times \frac{5R}{36} = \frac{5}{27}$$

$$31.(D) \text{ Radius of the nucleus } R = R_0 A^{1/3} \quad \therefore \frac{R_{Al}}{R_{Te}} = \left(\frac{A_{Al}}{A_{Te}} \right)^{1/3}$$

$$\text{Here, } A_{Al} = 27, A_{Te} = 125, R_{Te} = ?$$

$$\frac{R_{Al}}{R_{Te}} = \left(\frac{27}{125} \right)^{1/3} = \frac{3}{5} \Rightarrow R_{Te} = \frac{5}{3} R_{Al}$$

$$32.(D) \text{ Energy of electron in } He^+ 3^{\text{rd}} \text{ orbit } E_3 = -13.6 \times \frac{Z^2}{n^2} eV = -13.6 \times \frac{4}{9} eV = -13.6 \times \frac{4}{9} \times 1.6 \times 10^{-19} J = 9.7 \times 10^{-19} J$$

As per Bohr's model,

$$\text{Kinetic energy of electron in the } 3^{\text{rd}} \text{ orbit} = -E_3 \quad \therefore 9.7 \times 10^{-19} = \frac{1}{2} m_e v^2$$

$$v = \sqrt{\frac{2 \times 9.7 \times 10^{-19}}{9.1 \times 10^{-31}}} = 1.46 \times 10^6 \text{ ms}^{-1}$$

$$33.(C) \text{ Work function, } W = \frac{hc}{\lambda} \quad [\text{Here they are interested in asking threshold wavelength}]$$

where, h = Planck's constant,
 c = velocity of light

$$\text{Therefore, } \frac{W_{Na}}{W_{Cu}} = \frac{\lambda_{Cu}}{\lambda_{Na}} \quad \text{or} \quad \frac{\lambda_{Na}}{\lambda_{Cu}} = \frac{W_{Cu}}{W_{Na}} = \frac{4.5}{2.3} = 2 (\text{nearly})$$

$$34.(D) \text{ Binding energy of } {}^7_3\text{Li nucleus} = 7 \times 5.60 \text{ MeV} = 39.2 \text{ MeV}$$

$$\text{Binding energy of } {}^4_2\text{He nucleus} = 4 \times 7.06 \text{ MeV} = 28.24 \text{ MeV}$$

$$\text{The reaction is } {}^7_3\text{Li} + {}^1_1\text{H} \rightarrow 2({}^4_2\text{He}) + Q$$

$$\therefore Q = 2(\text{BE of } {}^4_2\text{He}) - (\text{BE of } {}^7_3\text{Li}) = 2 \times 28.24 \text{ MeV} - 39.2 \text{ MeV} \\ = 56.48 \text{ MeV} - 39.2 \text{ MeV} = 17.28 \text{ MeV}$$

$$35.(D) \Delta E = h\nu \quad \text{where, } \Delta E \text{ is energy of radiation, } h \text{ is Plank's constant and } \nu \text{ is frequency.}$$

$$\Rightarrow v = \frac{\Delta E}{h} = k \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right] = \frac{k2n}{n^2(n-1)^2} = \frac{2k}{n^3} \propto \frac{1}{n^3}$$

$$36.(C) \text{ Nuclear radius, } R = R_0 A^{1/3} \text{ where } R_0 \text{ is a constant and } A \text{ is the mass number}$$

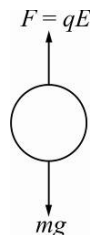
$$\therefore \frac{R_{Al}}{R_{Cu}} = \frac{(27)^{1/3}}{(64)^{1/3}} = \frac{3}{4} \text{ or } R_{Cu} = \frac{4}{3} \times R_{Al} = \frac{4}{3} \times 3.6 \text{ fermi} = 4.8 \text{ fermi}$$

37.(A) In steady state, electric force on drop = weight of drop

$$i.e., \quad qE = mg$$

$$\Rightarrow \quad q = \frac{mg}{E}$$

$$= \frac{9.9 \times 10^{-15} \times 10}{3 \times 10^4} = 3.3 \times 10^{-18} C$$



38.(B) According to radioactive decay law $N = N_0 e^{-\lambda t}$ where N_0 = Number of radioactive nuclei at time $t = 0$

N = Number of radioactive nuclei left undecayed at any time t

λ = decay constant

At time t_2 , $\frac{2}{3}$ of the sample had decayed

$$\therefore \quad N = \frac{1}{3} N_0 \quad \therefore \quad \frac{1}{3} N_0 = N_0 e^{-\lambda t_2} \quad \dots\dots (i)$$

At time t_1 , $\frac{1}{3}$ of the sample had decayed,

$$\therefore \quad N = \frac{2}{3} N_0 \quad \therefore \quad \frac{2}{3} N_0 = N_0 e^{-\lambda t_1} \quad \dots\dots (ii)$$

Divide (i) by (ii), we get

$$\frac{1}{2} = \frac{e^{-\lambda t_2}}{e^{-\lambda t_1}}$$

$$\frac{1}{2} = e^{-\lambda(t_2 - t_1)}$$

$$\lambda(t_2 - t_1) = \ln 2$$

$$t_2 - t_1 = \frac{\ln 2}{\lambda} = \frac{\ln 2}{\left(\frac{\ln 2}{T_{1/2}}\right)} \quad \left(\because \lambda = \frac{\ln 2}{T_{1/2}}\right)$$

$$= T_{1/2} = 50 \text{ days}$$

39.(C)

	P	Q
Number of nuclei, at $t = 0$	$4N_0$	N_0
Half-life	1 min	2 min
Number of nuclei after	N_P	N_Q

time t

Let after t min the number of nuclei of P and Q are equal.

$$\therefore \quad N_P = 4N_0 \left(\frac{1}{2}\right)^{t/1} \text{ and } N_Q = N_0 \left(\frac{1}{2}\right)^{t/2}$$

$$\text{As } N_P = N_Q$$

$$\therefore \quad 4N_0 \left(\frac{1}{2}\right)^{t/1} = N_0 \left(\frac{1}{2}\right)^{t/2}$$

$$\frac{4}{2^{t/1}} = \frac{1}{2^{t/2}} \text{ or } 4 = \frac{2^t}{2^{t/2}} \text{ or } 4 = 2^{t/2} \text{ or } 2^2 = 2^{t/2} \text{ or } \frac{t}{2} = 2 \text{ or } t = 4 \text{ min}$$

After 4 minutes, both P and Q have equal number of nuclei.

$$\therefore \text{Number of nuclei of } R = \left(4N_0 - \frac{N_0}{4}\right) + \left(N_0 - \frac{N_0}{4}\right) = \frac{15N_0}{4} + \frac{3N_0}{4} = \frac{9N_0}{2}$$

40.(A) Formation of covalent bonds due to the wave nature of particle is done in compounds.

41.(D) According to activity law

$$R = R_0 e^{-\lambda t} \quad \dots\dots (i)$$

Where,

R_0 = initial activity at $t = 0$

R = activity at time t

λ = decay constant

According to given problem,

$R_0 = N_0$ counts per minute

$R = \frac{N_0}{e}$ counts per minute

$t = 5$ minutes

Substituting these values in equation (i), we get

$$\frac{N_0}{e} = N_0 e^{-5\lambda}$$

$$e^{-1} = e^{-5\lambda}$$

$$5\lambda = 1 \text{ or } \lambda = \frac{1}{5} \text{ per minute}$$

At $t = T_{1/2}$, the activity R reduces to $\frac{R_0}{2}$.

Where $T_{1/2}$ = half life of a radioactive sample

From equation (i), we get :

$$\frac{R_0}{2} = R_0 e^{-\lambda T_{1/2}}$$

$$e^{\lambda T_{1/2}} = 2$$

Taking natural logarithms of both sides of above equation, we get

$$\lambda T_{1/2} = \log_e 2 \text{ or } T_{1/2} = \frac{\log_e 2}{\lambda} = \frac{\log_e 2}{\left(\frac{1}{5}\right)} = 5 \log_e 2 \text{ minutes}$$

42.(D) In emission spectrum, number of bright lines is given by $\frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$

43.(C) At the distance of closest approach d ,
Kinetic energy = potential energy

$$\frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{d}$$

Where,

Ze = charge of target nucleus

$2e$ = charge of alpha nucleus

$\frac{1}{2}mv^2$ = kinetic energy of alpha nucleus of mass m moving with velocity v

$$\text{or } d = \frac{2Ze^2}{4\pi\epsilon_0\left(\frac{1}{2}mv^2\right)} \quad \therefore d \propto \frac{1}{m}$$

44.(A)

$$45.(C) \quad ZM_P + (A-Z)M_n - M(A, Z) = \text{mass effect} = \frac{B.E}{c^2} \quad \Rightarrow \quad M(A, Z) = ZM_P + (A-Z)M_n - \frac{B.E}{c^2}$$

$$46.(A) \quad X_1 = N_0 e^{-\lambda_1 t}; \quad X_2 = N_0 e^{-\lambda_2 t}$$

$$\frac{X_1}{X_2} = e^{-1} = e^{(-\lambda_1 + \lambda_2)t}; \quad e^{-1} = e^{(-\lambda_1 - \lambda_2)t} \quad \therefore \quad t = \left| \frac{1}{\lambda_1 - \lambda_2} \right| = \frac{1}{(5\lambda - \lambda)} = \frac{1}{4\lambda}$$

$$47.(A) \quad hv_0 = 6.2 \text{ eV}, \quad eV_0 = 5 \text{ eV}$$

From Einstein's photoelectric equation $h\nu = h\nu_0 + eV_0 = 6.2 + 5 = 11.2 \text{ eV}$

$$\Rightarrow \quad \frac{hc}{\lambda} = 11.2 \text{ eV} \quad \text{or} \quad \lambda = \frac{hc}{11.2} = 1108.9 \text{ \AA}$$

Which belongs to ultraviolet region.

48.(A) According to Einstein's photoelectric effect maximum kinetic energy of a photoelectron.

$$KE = \frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi$$

According to question, for incident radiation of wavelength λ maximum speed of photoelectron is v .

$$\therefore \quad \frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi \quad \dots\dots(i)$$

Assume speed of fastest photoelectron is v' when incident photon has wavelength $\frac{3\lambda}{4}$

$$\therefore \quad \frac{1}{2}mv'^2 = \frac{4hc}{3\lambda} - \phi \quad \text{or,} \quad \frac{1}{2}mv'^2 = \frac{4}{3}\left(\frac{1}{2}mv^2 + \phi\right) - \phi$$

$$\text{or,} \quad \frac{1}{2}mv'^2 = \frac{2}{3}mv^2 + \frac{\phi}{3} \quad \text{or} \quad v' = \sqrt{\frac{4}{3}v^2 + \frac{2\phi}{3m}} \quad \therefore \quad v' > \sqrt{\frac{4}{3}}v$$

49.(D) Half life of A, $T_{1/2}(A) = 20 \text{ min}$; Half life of B, $T_{1/2}(B) = 40 \text{ min}$

Number of active nuclei after four half lives of A,

$$N_A = \frac{N}{2^4} = \frac{N}{16}$$

$$\therefore \quad \text{Number of decayed nuclei} = N - N_A = \frac{15}{16}N$$

Number of active nuclei after two half lives of B, $N_B = \frac{N}{2^2} = \frac{N}{4}$

$$\therefore \quad \text{Number of decayed nuclei} = N - N_B = \frac{3}{4}N \quad \therefore \quad \text{Required ratio} = \frac{\frac{15}{16}N}{\frac{3}{4}N} = \frac{5}{4}$$

50.(B) Energy of emitted photon

$$E = \left[\frac{1}{1^2} - \frac{1}{2^2} \right] \times 13.6 \text{ eV} = \frac{3}{4} \times 13.6 \text{ eV}$$

Energy required to completely remove the electron from n^{th} excited state of doubly ionized lithium,

$$E' = \frac{13.6Z^2}{n^2} \text{ eV} = \frac{13.6 \times 9}{n^2} \text{ eV}$$

As $E \geq E'$

$$\therefore \frac{3}{4} \times 13.6 \geq \frac{13.6 \times 9}{n^2} \Rightarrow n^2 \geq 3 \times 4 \text{ or } n \geq \sqrt{12} = 3.5$$

Least quantum number for the excited state = 4

51.(D) Let the threshold wavelength for sphere be λ_0 . According to Einstein's photoelectric equation

$$eV_s = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$\therefore eV = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_0} \quad \dots (i)$$

$$3eV = \frac{hc}{\lambda_2} - \frac{hc}{\lambda_0} \quad \dots (ii)$$

$$eV' = \frac{hc}{\lambda_3} - \frac{hc}{\lambda_0} \quad \dots (iii)$$

From eqns. (i) and (ii)

$$\frac{2hc}{\lambda_0} = \frac{3hc}{\lambda_1} - \frac{hc}{2\lambda_2} \text{ or } \frac{hc}{\lambda_0} = \frac{3hc}{2\lambda_1} - \frac{hc}{2\lambda_2}$$

Substituting in eqn. (iii)

$$eV' = \frac{hc}{\lambda_3} - \frac{3hc}{\lambda_1} + \frac{hc}{2\lambda_2} \text{ or } V' = \frac{hc}{e} \left[\frac{1}{\lambda_3} - \frac{3}{2\lambda_1} + \frac{1}{2\lambda_2} \right]$$

52.(A) Using conservation of linear momentum, Total momentum before collision = Total momentum after collision

$$mv = (m + m) v'$$

$$v' = \frac{v}{2}$$

$$\text{Loss in kinetic energy during the process, } \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}(2m) \left(\frac{v}{2} \right)^2 = \frac{1}{4}mv^2$$

$$\Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}(2m) \left(\frac{v}{2} \right)^2 = \frac{1}{4}mv^2$$

For minimum kinetic energy of neutron, lost kinetic energy should be used by the electron to jump from first orbit to second orbit.

$$\Rightarrow \frac{1}{4}mv^2 = (13.6 - 3.4) \text{ eV} = 10.2 \text{ eV}$$

$$\therefore \frac{1}{2}mv^2 = 20.4 \text{ eV} = \text{K.E. of neutron for inelastic collision.}$$

53.(A) According to Einstein's photoelectric equation, Maximum energy of photoelectrons

$$(KE)_{\text{max}} = h\nu - \phi_0$$

$$(KE)_{\max} = \frac{hc}{\lambda} - \phi_0$$

$$\text{First case. } K = \frac{hc}{\lambda} - \phi_0 \quad \dots (i)$$

$$\text{Second case, } 3K = \frac{2hc}{\lambda} - \phi_0 \quad \dots (ii)$$

From equations (i) and (ii)

$$3\left(\frac{hc}{\lambda} - \phi_0\right) = \frac{2hc}{\lambda} - \phi_0 \Rightarrow 2\phi_0 = \frac{3hc}{\lambda} - \frac{2hc}{\lambda} = \frac{hc}{\lambda} \quad \therefore \phi_0 = \frac{hc}{2\lambda}$$

54.(A) Franck – Hertz Experiment – Discrete energy levels of atom.

Photo – electric experiment – Particle nature of light.

Davision – Germer Experiment – Wave nature of electron.

55.(C) Momentum $p = \sqrt{2mE}$ and $E = eV$

So, de-Broglie wavelength of the electron is given by,

$$\begin{aligned} \lambda &= \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2meV}} \\ &= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 50}} = 1.7 \times 10^{-10} \text{ m} = 1.7 \text{ \AA} \end{aligned}$$

56.(B) de-Broglie wavelength of electron, $\lambda = \frac{h}{mv}$

$$\text{Also } mvr = \frac{nh}{2\pi}$$

$$\lambda = \frac{2\pi r}{n}$$

$$\therefore r \propto n^2 \quad \therefore l \propto n$$

For $n=4$, $\lambda_4 = 4\lambda$ i.e, the de-Broglie wavelength is four times that of ground state.

57.(C) For an electron in n^{th} excited state of hydrogen atom,

$$\text{kinetic energy} = \frac{e^2}{8\pi\epsilon_0 n^2 a_0}$$

$$\text{potential energy} = \frac{-e^2}{4\pi\epsilon_0 n^2 a_0}$$

$$\text{and total energy} = \frac{-e^2}{8\pi\epsilon_0 n^2 a_0}$$

where a_0 is Bohr radius.

As electron makes a transition from an excited state to the ground state, n decreases. Therefore kinetic energy increase but potential energy and total energy decrease.

$$\text{58.(B)} \quad mvR = \frac{nh}{2\pi} \quad \dots (i)$$

$$\text{and } qvB = \frac{mv^2}{R}; qB = \frac{mv}{R} \quad \dots (ii)$$

From eqns. (i) and (ii), we get

$$qB \left(\frac{nh}{2\pi mv} \right) = mv$$

$$\frac{1}{2}mv^2 = \frac{1}{4\pi m}nhqB$$

$$\frac{1}{2}mv^2 = \frac{1}{4\pi m}nhqB \quad \therefore \quad E = n \left(\frac{hqB}{4\pi m} \right)$$

$$59.(B) \quad \text{Half life} = 15 \text{ hrs.} = \frac{0.693}{\lambda} \quad \therefore \quad \lambda = 0.0462 \text{ hr}^{-1}$$

$$N_0 = \frac{1}{24} \text{ moles of Na, } t = 7.5 \text{ hrs.}$$

$$\text{Number of } \beta \text{ - particles disintegrated } N_\beta = N_0(1 - e^{-\lambda t})$$

$$= \left(\frac{1}{24} \text{ moles} \right) (1 - e^{-(0.0462) \times 7.5}) = \left(\frac{1}{24} \text{ moles} \right) (1 - e^{-0.35})$$

$$= 0.0122 \text{ moles} = 0.0122 \times 6.023 \times 10^{23} \quad \therefore \quad N_\beta = 7.4 \times 10^{21}$$

$$60.(7) \quad \frac{1}{2}m_\alpha v^2 = \frac{K(2e)2e}{r} = \frac{1}{4\pi\epsilon_0} \frac{(82e) \times 2e}{r}$$

$$r = 2.82 \times 10^{-14} \text{ m}$$

$$61.(2.86) \quad E = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] eV$$

$$62.(9) \quad A = \lambda N = \frac{\ln 2}{3.8 \times 24 \times 60} \times 8 \times 10^{12}$$

$$A = 1.01 \times 10^9 \text{ dps}$$

$$63.(13) \quad N = 8 \times 10^{14} \times e^{-\frac{\ln 2}{7} \times 24}$$

$$= 7.43 \times 10^{13}$$

$$64.(1.00) \quad N = N_0 e^{-\lambda t} \quad \Rightarrow \quad 0.25 = 2 \times e^{-\frac{\ln 2}{t/2} \times 3} \quad \Rightarrow \quad 0.25 = 2 \times 2^{-\frac{3}{t/2}} \quad \Rightarrow \quad 2^{-2} = 2^{1-\frac{3}{t/2}}$$

$$\Rightarrow \quad -2 = 1 - \frac{3}{t/2} \quad \Rightarrow \quad \frac{3}{t/2} = 3 \quad \Rightarrow \quad t/2 = 1 \text{ hr}$$

$$65.(1.6) \quad \text{The energy of photon of wavelength } 4.36 \times 10^{-7} \text{ m,}$$

$$E = h\nu = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34}) \times (3 \times 10^8)}{4.36 \times 10^{-7}}$$

$$= 4.54 \times 10^{-19} \text{ J}$$

$$\text{or} \quad E = \frac{4.54 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.84 \text{ eV}$$

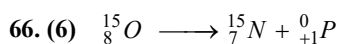
Now, according to Einstein equation for photoelectric effect

$$K_{\max} = h\nu - \phi = 2.84 - 1.24 = 1.6 \text{ eV}$$

And as in terms of stopping potential

$$K_{\max} = eV_S; \quad 1.6 \text{ eV} = eV_S$$

i.e., $V_S = 1.6 \text{ V}$



$$A = A_0 e^{-\lambda t} = \lambda N$$

$$A = 0.2 \times 10^{-6} \times 3.7 \times 10^{10} = \frac{\ln 2}{122} \times N \quad \Rightarrow \quad N = 1.30 \times 10^6$$

67. (2) $|TE| = (BE) \Rightarrow PE = 2[TE]$

$$TE = \frac{-13.6 \times 1^2}{2^2} (\text{eV}) = -3.4 \text{ eV}$$

$$PE = 2 \times (TE) = 2 \times (-3.4 \text{ eV}) = -6.8 \text{ eV}$$

$$KE = |TE| = 3.4 \text{ eV}$$

68. (6) $\frac{hc}{\lambda} = \phi_0 + (KE)_{\max} \quad \dots \text{ (i)}$

$$E = \frac{hc}{\lambda} = \frac{12400}{3000} = 4.13 \text{ eV}$$

From equation (i)

$$(KE) = 4.13 - 1 = 3.13 \text{ eV}$$

$$(KE) = \frac{1}{2} mv^2$$

69. (98.5) $A = \lambda N$

$$= \frac{\ln 2}{8.1 \times 24 \times 3600} \times \frac{0.80 \times 10^{-6}}{131} \times 6 \times 10^{23} = 98.5 \text{ mCi}$$

70. (2) $\frac{1}{2} mv_1^2 = h\nu_1 - \phi_0 = 1 - 0.5 = 0.5 \text{ eV}$

$$\frac{1}{2} mv_2^2 = h\nu_2 - \phi_0 = 2.5 - 0.5 = 2.0 \text{ eV}$$

Dividing,

$$\frac{v_1}{v_2} = \sqrt{\frac{0.5}{2.0}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

71. (2) $0.048 A_0 = A_0 e^{-\lambda t}$

$$0.048 = e^{-\frac{\ln 2}{5700} \times t}$$

$$\ln 0.048 = \frac{-\ln 2}{5700} \times t \quad \therefore \quad t = 2.5 \times 10^4 \text{ year}$$

72. (150) $\lambda = \frac{h}{\sqrt{2mqV}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \text{ V}}} = \sqrt{\frac{150}{V}} \text{ \AA}$

73. (4.2) $A = \frac{\ln 2}{t_{1/2}} \times N$

$$= \frac{\ln 2}{5.25 \times 365 \times 24 \times 3600} \times \frac{0.01}{60} \times 6.023 \times 10^{23}$$

$$= 4.2 \times 10^{11} Bq$$

74.(7.43) $N = N_0(1 - e^{-\lambda t})$

$$N_0 \times \lambda = 24 \times 10^4 \times 3.7$$

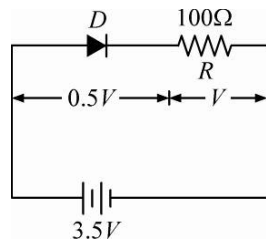
$$N_0 = \frac{24 \times 3.7 \times 10^4 \times 29}{\ln 2}$$

$$t = 365 \times 24 \times 60 \times 60 \text{ sec}$$

Hence, no nuclei decayed $= (N_0 - N)$

Semiconductor Electronic

1.(C)

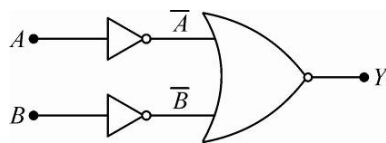


The potential difference across the resistance R is $V = 3.5V - 0.5V = 3V$

By Ohm's law,

The current in the circuit is $I = \frac{V}{R} = \frac{3V}{100\Omega} = 3 \times 10^{-2} A = 30 \times 10^{-3} A = 30 \text{ mA}$

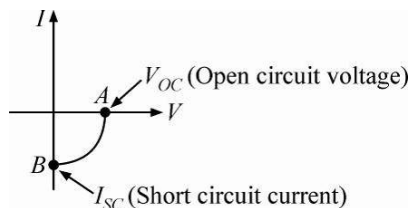
2.(A)



The Boolean expression of this arrangement is $Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B$

Thus, the combination represents AND gate.

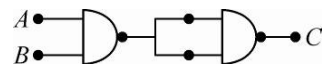
3.(A) The V - I characteristic for a solar cell is as shown in figure.



4.(D) The barrier potential depends on type of semiconductor (For Si , $V_b = 0.7V$ and for Ge , $V_b = 0.3V$), amount of doping and also on the temperature.

5.(A) The output of the given logic gate is $C = \overline{\overline{A} \cdot \overline{B}} = A \cdot B$

It is the Boolean expression AND gate. Hence, The resulting gate is AND gate.



6.(B) The emitter injects electrons into the base region of the $n-p-n$ transition and holes into the base region of $p-n-p$ transistor.

7.(D) In the given circuit the upper diode D_1 is forward biased and the lower diode D_2 is reverse biased. So, the current supplied by the battery is :

$$I = \frac{5V}{10\Omega} = \frac{1}{2}A = 0.5A$$

8.(C) The electronic configuration of carbon (6C) is $1s^2 2s^2 2p^2$. The electronic configuration of silicon (${}^{14}Si$) is $1s^2 2s^2 2p^6 3s^2 3p^2$. Hence, the four bonding electrons of C and Si respectively lie in second and third orbit.

9.(B) In the given graph

Region (I) – Cutoff region

Region (II) – Active region

Region (III) – Saturation region

Using transistor as a switch it is used in cutoff region or saturation region.

Using transistor as a amplifier it is used in active region.

10.(A) The truth table of given waveform is as shown in the table.

Time interval	Input A	Input B	Output C
0 to t_1	0	0	0
t_1 to t_2	1	0	1
t_2 to t_3	1	1	1
t_3 to t_4	0	1	1
t_4 to t_5	0	0	0
t_5 to t_6	1	0	1
$> t_6$	0	1	1

11.(A) Here,

Input resistance, $R_i = 100\Omega$

Change in base current, $\Delta I_B = 40\mu A$

Change in collector current, $\Delta I_C = 2mA$

Load resistance, $R_L = 4k\Omega = 4 \times 10^3\Omega$

$$\text{Current gain, } \beta = \frac{\Delta I_C}{\Delta I_B} = \frac{2mA}{40\mu A} = \frac{2 \times 10^{-3}A}{40 \times 10^{-6}A} = 50$$

$$\text{Voltage gain of the amplifier is } A_V = \beta \frac{R_L}{R_i} = 50 \times \frac{4 \times 10^3}{100} = 2000$$

12.(A) The voltage drop across $1k\Omega = V_Z = 15V$

The current through $1k\Omega$ is

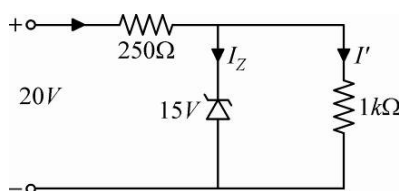
$$I' = \frac{15V}{1 \times 10^3 \Omega} = 15 \times 10^{-3} A = 15 mA$$

The voltage drop across $250 \Omega = 20V - 15V = 5V$

The current through 250Ω is

$$I = \frac{5V}{250 \Omega} = 0.02 A = 20 mA$$

The current through the zener diode is $I_Z = I - I' = (20 - 15) mA = 5 mA$



- 13.(A) *p*-type semiconductor is obtained when *Si* or *Ge* doped with a trivalent impurity like aluminium (*Al*), boron (*B*), indium (*In*) etc,

Here, $n_i = 1.5 \times 10^{16} m^{-3}$

$$n_h = 4.5 \times 10^{22} m^{-3}$$

As $n_e n_h = n_i^2$

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16} m^{-3})^2}{4.5 \times 10^{22} m^{-3}} = 5 \times 10^9 m^{-3}$$

- 14.(C) Here, Voltage gain = 50

Input resistance, $R_i = 100 \Omega$;

Output resistance, $R_o = 200 \Omega$

$$\text{Resistance gain} = \frac{R_o}{R_i} = \frac{200 \Omega}{100 \Omega} = 2 ; \quad \text{Power gain} = \frac{(\text{Voltage gain})^2}{\text{Resistance gain}} = \frac{50 \times 50}{2} = 1250$$

- 15.(D) It is clear from given logic circuit, that out put *Y* is low when both the inputs are high, otherwise it is high. Thus logic circuit is NAND gate.

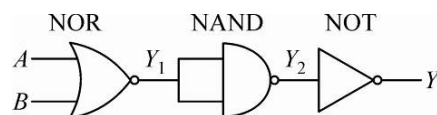
<i>A</i>	<i>B</i>	<i>Y</i>
1	1	0
0	0	1
0	1	1
1	0	1

- 16.(A)

NOR			NAND			NOT
<i>A</i>	<i>B</i>	<i>Y</i> ₁	<i>Y</i> ₁	<i>Y</i> ₁	<i>Y</i> ₂	<i>Y</i>
			<i>A</i>	<i>B</i>		
0	0	1	1	1	0	1
0	1	0	0	0	1	0
1	0	0	0	0	1	0
1	1	0	0	0	1	0

Same as NOR Gate

NOR Gate

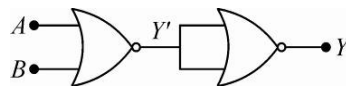


0	0	1
0	1	0
1	0	0
1	1	0

17.(C) $Y' = \overline{A+B} \cdot \overline{\overline{A+B}} = \overline{A+B} = A+B$.

Truth table of the given circuit is given by

A	B	Y'	Y
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1



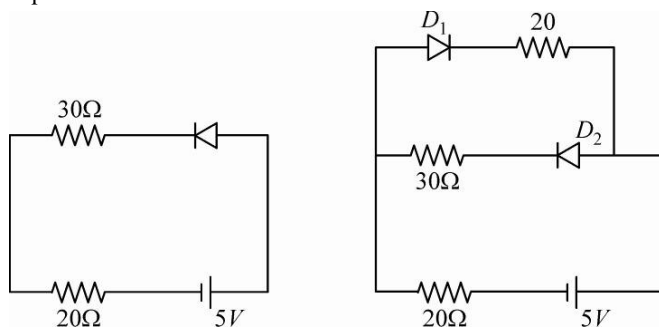
18.(B) The truth table corresponding to waveform is given by

A	B	C
1	1	1
0	1	0
1	0	0
0	0	0

∴ The given logic circuit gate is AND gate.

19.(B) $D_1 \rightarrow$ reverse biased & $D_2 \rightarrow$ forward biased.

Equivalent circuit is



$$I = \frac{5V}{(30+20)\Omega} = \frac{5}{50} A.$$

20.(A) $\frac{I_C}{I_E} = \alpha = 0.98, \frac{I_C}{I_B} = \beta = \frac{\alpha}{1-\alpha} = 49$ 21.(B) $\beta = \frac{I_c}{I_b} = \frac{I_c}{I_e - I_c} = \frac{I_c / I_e}{1 - (I_c / I_e)} = \frac{\alpha}{1-\alpha}$

22.(B) As soon as the $p-n$ junction is formed, there is an immediate diffusion of the charge carrier across the junction due to thermal agitation. After diffusion, these charge carriers combine with their counterparts and neutralise each other. Therefore correct direction of flow carriers is depicted in the figure (B).

23.(C) Output (x is high when atleast one of the inputs is high. Hence, x is the output of OR gate.

24.(A)

25.(BD) We know $\alpha = \frac{I_c}{I_e}, \beta = \frac{I_c}{I_b}$

$$\alpha = \frac{\beta I_b}{I_b + I_c} = \frac{\beta}{1 + I_c/I_b} = \frac{\beta}{1 + \beta} \text{ or, } \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

Hence options (b) and (d) are incorrect

26.(B) $p-n-p$ transistor :

Positive at terminal 2 and negative at terminal 1 implies $p-n$ junction is reverse biased and hence offers high resistance.

E(p)	B(n)	C(p)
1	2	3

27.(B) Potential drop across Zener diode

$$V_Z = V - I_R = V - 100I$$

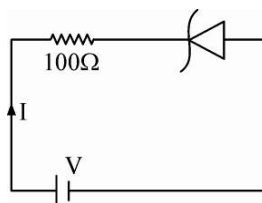
$$\therefore \text{Power, } P = V_Z I_Z = (V - 100I)I$$

$$\text{But } P = 1 \text{ W (given)}$$

$$\therefore (V - 100I)I = 1$$

$$\text{or } 100I^2 - VI + 1 = 0$$

$$\text{For } I \text{ to be real, } V^2 - 4 \times 100 \times 1 \geq 0$$



$$\text{or } V \geq 20V$$

28.(A) The given truth table represents OR gate

$$29.(B) R = \frac{\text{Output resistance}(r_o)}{\text{Input resistance}(r_i)} \equiv 1-10$$

30.(C) Output $Y = (a + b) \cdot c$

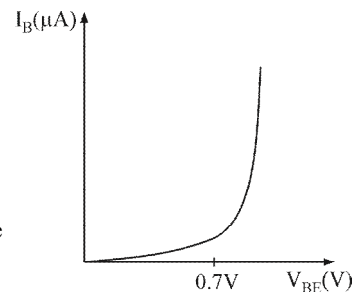
$$Y = 1 \text{ if } c = 1 \text{ and } a = 0, b = 1 \text{ or } a = 1, b = 0$$

31.(C) For common emitter configuration, the input characteristic :

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B}$$

$$\frac{1}{r_i} = \frac{dI_B}{dV_{BE}} = \text{Slope of } (I_B - V_{BE}) \text{ curve}$$

Slope of the input characteristic is almost constant upto knee voltage (0.7 V). Then it increases sharply. Hence option (C) is the correct choice.



32.(A) Carrier wave : $y_c = A_c \sin \omega_c t$

Message signal : $y_m = A_m \sin \omega_m t$

Amplitude modulated carrier wave :

$$y = (A_c + A_m \sin \omega_m t) \sin \omega_c t$$

33.(C)

34.(B) Here, $C_m(t) = 30 \sin(300\pi t) + 10(\cos(200\pi t) - \cos(400\pi t))$

Compare this equation with standard equation of amplitude modulated wave,

$$C_m(t) = A_c \sin \omega_c t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_m) t + \frac{\mu A_c}{2} \cos(\omega_c - \omega_m) t$$

$$A_c = 30V, \omega_c = 300\pi \Rightarrow 2\pi f_c = 300\pi \Rightarrow f_c = 150 \text{ Hz}$$

$$\omega_c - \omega_m = 200\pi \Rightarrow f_c - f_m = 100 \text{ Hz} \quad \therefore f_m = 150 - 100 = 50 \text{ Hz}$$

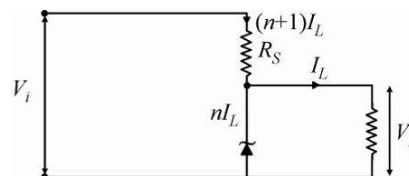
$$\frac{\mu A_c}{2} = 10, A_c = 30 \quad \therefore \mu = \frac{10}{15} = \frac{2}{3}$$

- 35.(C) Electron concentration in n -region is more as compared to that in p -region. So electrons diffuse from n -side to p -side.

- 36.(C) Voltage drop across Zener diode is V_L so voltage drop across R_s is

$$V_{R_s} = V_i - V_L = (n+1)I_L R_s$$

$$\therefore R_s = \frac{V_i - V_L}{(n+1)I_L}$$



- 37.(B) When positive terminal of battery is connected to A current passes through diode D_1 .

$$\therefore \text{Current supplied} = \frac{2V}{5\Omega} = 0.4 \text{ A}$$

$$\text{When positive terminal is connected to B current passes through } D_2. \quad \therefore \text{Current supplied} = \frac{2V}{10\Omega} = 0.2 \text{ A}$$

- 38.(D) Intensity of light, $I = u_{av}c$

$$\text{Also, } I = \frac{P}{4\pi r^2} \text{ and } u_{av} = \frac{1}{2}\epsilon_0 E_0^2 \quad \therefore \frac{P}{4\pi r^2} = \frac{1}{2}\epsilon_0 E_0^2 c \quad \text{or} \quad E_0 = \sqrt{\frac{2P}{4\pi\epsilon_0 r^2 c}}$$

$$\text{Here, } P = 0.1 \text{ W, } r = 1\text{ m, } c = 3 \times 10^8 \text{ ms}^{-1}$$

$$\frac{1}{4\pi r^2} = 9 \times 10^9 \text{ NC}^{-2} \text{ m}^2 \quad \therefore E_0 = \sqrt{\frac{2 \times 0.1 \times 9 \times 10^9}{1^2 \times 3 \times 10^8}} = \sqrt{6} = 2.45 \text{ V m}^{-1}$$

- 39.(A) Given, $v_m = 5\text{ kHz}$, $v_c = 2\text{ MHz} = 2000 \text{ kHz}$

The frequencies of the resultant signal are

$$v_c + v_m = (2000 + 5) \text{ kHz} = 2005 \text{ kHz} \quad ; \quad v_c = 2000 \text{ kHz} \text{ and } v_c - v_m = (2000 - 5) \text{ kHz} = 1995 \text{ kHz}$$

40.(C)

41.(D)

42.(B)

43.(C)

44.(B)

45.(B)

- 46.(348) The impurity provided impurity levels close to the conduction band and a number of electrons from the impurity level will populate the conduction band. Thus, the majority carriers are electrons and the material is n -type

According to the questions, $Kt = 30 \text{ meV}$

$$\begin{aligned} \text{Or } T &= \frac{30 \text{ meV}}{k} \\ &= \frac{0.03 \text{ eV}}{8.62 \times 10^{-5} \text{ eV/K}} = 348 \text{ K} \end{aligned}$$

- 47.(81) The energy of the photon is $E = \frac{hc}{\lambda}$

$$= \frac{1242 \text{ eV} \cdot \text{nm}}{589 \text{ nm}} = 2.1 \text{ eV}$$

Thus the band gap is 2.1 eV. This is also the minimum energy E required to push an electron from the valence band into the conduction band. Hence, the minimum energy required to create a hole – electron pair is 2.1 eV

At $T = 300\text{ K}$

$$kT = (8.62 \times 10^{-5} \text{ eV / K}) (300 \text{ K})$$

$$= 25.86 \times 10^{-3} \text{ eV}$$

$$\text{Thus } \frac{E}{kT} = \frac{2.1 \text{ eV}}{25.86 \times 10^{-3} \text{ eV}} = 81$$

So, it is difficult for the thermal energy to create the hole-electron pair but a photon of light can do it easily.

48.(127) The number of charge carriers in an intrinsic semiconductor is double the number of hole-electron pairs. If N_1 be the number of charge carriers at temperature T_1 and N_2 at T_2 we have.

$$N_1 = N_0 e^{-\Delta E / 2kT_1}$$

$$\text{and } N_2 = N_0 e^{-\Delta E / 2kT_2}$$

The percentage increase as the temperature is raised from T_1 to T_2 is

$$f = \frac{N_2 - N_1}{N_1} \times 100 = \left(\frac{N_2}{N_1} - 1 \right) \times 100$$

$$= 100 \left[e^{\frac{\Delta E}{k} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)} - 1 \right]$$

$$\text{Now } \frac{\Delta E}{2k} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$= \frac{0.68 \text{ eV}}{2 \times 8.62 \times 10^{-5} \text{ eV / K}} \left(\frac{1}{300 \text{ K}} - \frac{1}{320 \text{ K}} \right)$$

$$= 0.82$$

$$\text{Thus, } f = 100 \times [e^{0.82} - 1] = 127$$

Thus, the number of charge carriers increases by about 127%.

49.(18) The number of charge carriers before doping is equal to the number of holes plus the number of conduction electrons. Thus, the number of charge carriers per cubic meter before doping.

$$= 2 \times 7 \times 10^{15} = 14 \times 10^{15}$$

Since antimony is doped in a proportion of 1 in 10^7 , the number of antimony atoms per cubic meter is number of antimony atoms per cubic is $10^{-7} \times 5 \times 10^{28} = 5 \times 10^{21}$. As half of these atoms contribute electrons to the conduction band, the number of extra conduction electrons produced is 2.5×10^{21} per cubic metre. Thus, the number of charge carries per cubic meter after the doping is

$$2.5 \times 10^{21} + 14 \times 10^{15}$$

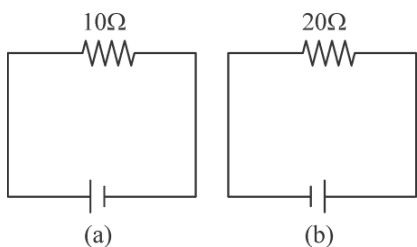
$$= 2.5 \times 10^{21}$$

The factor by which the number of charge carriers is increased

$$= \frac{2.5 \times 10^{21}}{14 \times 10^{16}} = 1.8 \times 10^8 = 18 \times 10^7$$

In fact, as the n -type impurity is doped, the number of holes will decrease. This is because the product of the concentrations of holes and conduction electrons remains almost the same. However, this does not affect, our result as the number of holes is anyway too small as compared to the number of conduction electrons.

- 50.(2)** When the positive terminal of the battery is connected to the point A, the diode D_1 is forward-biased and D_2 is reverse-biased. The resistance of the diode D_1 is zero, and it can be replaced by a resistanceless wire. Similarly, the resistance of the diode D_2 is infinity, and it can be replaced by a broken wire. The equivalent circuit is shown in figure. The current supplied by the battery is $i_1 = 2/10\Omega = 0.2 A$



- (b)** When the positive terminal of the battery is connected to the point B, the diode D_2 is forward-biased and D_1 is reverse biased. The equivalent circuit is shown in figure. The current through the battery is

$$i_2 = 2V / 20\Omega = 0.1 A \quad \frac{i_1}{i_2} = 2$$

51.(4.996) Using KVL ; $5 = iR + V_{diode}$; $V_{diode} = 5 - iR = 4.996V$

52.(20) 53.(50) 54.(5) 55.(1)

- 56.(1)** Forward bias $i = 1 A$

57.(125) $i = \frac{q}{t} = \frac{ne}{t} \Rightarrow \frac{t}{n} = \frac{i}{e} = \frac{20 \times 10^{-6}}{1.6 \times 10^{-19}} = \frac{5}{4} \times 10^{14} = 125 \times 10^{12}$

Electromagnetic waves

1.(C) As $\lambda = \frac{hc}{E}$

Where the symbols have their usual meanings

Here, $E = 15 \text{ keV} = 15 \times 10^3 \text{ V}$

And $hc = 1240 \text{ eV nm}$

$$\therefore \lambda = \frac{1240 \text{ eV nm}}{15 \times 10^3 \text{ eV}} = 0.083 \text{ nm}$$

As the wavelength range of X-rays is from 1 nm to 10^{-3} nm , so this wavelength belongs to X-rays.

2.(D) Energy of radiation, $E = h\nu = \frac{hC}{\lambda}$

Also, its momentum $p = \frac{h}{\lambda} = \frac{E}{C} = p_i$ $p_r = -p_i = -\frac{E}{C}$

So, momentum transferred to the surface $= p_i - p_r = \frac{E}{C} - \left(-\frac{E}{C}\right) = \frac{2E}{C}$

3.(B) Here,

Energy flux, $I = 25 \times 10^4 \text{ W m}^{-2}$

Area, $A = 15 \text{ cm}^2 = 15 \times 10^{-4} \text{ m}^2$

Speed of light, $c = 3 \times 10^8 \text{ m s}^{-1}$

For a perfectly reflecting surface, the average force exerted on the surface is

$$F = \frac{2IA}{c} = \frac{2 \times 25 \times 10^4 \text{ W m}^{-2} \times 15 \times 10^{-4} \text{ m}^2}{3 \times 10^8 \text{ m s}^{-1}} = 250 \times 10^{-8} \text{ N} = 2.50 \times 10^{-6} \text{ N}$$

4.(C) In microwave oven the frequency of the microwaves must match the resonant frequency of water molecules so that energy from the waves is transferred efficiently to the kinetic energy of the molecules.

5.(B) Frequency of electromagnetic wave does not change with change in medium but wavelength and velocity of wave changes with change in medium.

Velocity of electromagnetic wave in vacuum

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = v \lambda_{\text{vacuum}} \quad \dots (i)$$

$$\text{Velocity of electromagnetic wave in the medium } v_{\text{medium}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

Where μ_r and ϵ_r be relative permeability and relative permittivity of the medium.

For dielectric medium, $\mu_r = 1$ $\therefore v_{\text{medium}} = \frac{c}{\sqrt{\epsilon_r}}$

Here, $\epsilon_r = 4.0$ $\therefore v_{\text{medium}} = \frac{c}{\sqrt{4}} = \frac{c}{2} \quad \dots (ii)$

Wavelength of the wave in medium $\lambda_{\text{medium}} = \frac{v_{\text{medium}}}{\nu} = \frac{c}{2\nu} = \frac{\lambda_{\text{vacuum}}}{2}$ (Using (i) and (ii))

6.(A) Compare the given equation with

$$E = E_0 \cos(kz - \omega t)$$

We get, $\omega = 6 \times 10^8 \text{ s}^{-1}$

Wave vector, $k = \frac{\omega}{c} = \frac{6 \times 10^8 \text{ s}^{-1}}{3 \times 10^8 \text{ m s}^{-1}} = 2 \text{ m}^{-1}$

- 7.(B) The amplitude of magnetic field and electric field for an electromagnetic wave propagating in vacuum are related as $E_0 = B_0 c$

Where c is the speed of light in vacuum. $\therefore \frac{B_0}{E_0} = \frac{1}{c}$

- 8.(A) The electromagnetic wave is propagating along the $+z$ axis.
Since the electric and magnetic fields are perpendicular to each other and also perpendicular to the direction of propagation of wave.

Also, $\vec{E} \times \vec{B}$ given the direction of wave propagation.

$$\therefore \vec{E} = E_0 \hat{i}, B = B_0 \hat{j} \quad (\because \hat{i} \times \hat{j} = \hat{k})$$

- 9.(A) The decreasing order of wavelength of the given electromagnetic waves is as follows:

$$\lambda_{\text{Microwave}} > \lambda_{\text{Infrared}} > \lambda_{\text{Ultraviolet}} > \lambda_{\text{Gamma rays}}$$

- 10.(C) In an electromagnetic wave both electric and magnetic vectors are perpendicular to each other as well as perpendicular to the direction of propagation of wave.

- 11.(D) As given, $E = 10 \cos(10^7 t + kx)$... (i)

Comparing it with standard equation of *e.m.* wave,

$$E = E_0 \cos(\omega t + kx)$$

Amplitude $E_0 = 10 \text{ V/m}$ and $\omega = 10^7 \text{ rad/s}$ $\therefore c = v\lambda = \frac{\omega\lambda}{2\pi}$

Or $\lambda = \frac{2\pi c}{\omega} = \frac{2\pi \times 3 \times 10^8}{10^7} = 188.4 \text{ m}$; Also, $c = \frac{\omega}{k}$ or $k = \frac{\omega}{c} = \frac{10^7}{3 \times 10^8} = 0.033$

The wave is propagating along y direction.

12.(B) $E_y = 2.5 \left[\left(2\pi \times 10^6 \frac{\text{rad}}{\text{m}} \right) t - \left(\pi \times 10^{-2} \frac{\text{rad}}{\text{x}} \right) x \right]$

$$E_z = 0$$

The wave is moving in the positive direction of x .

This is the form $E_y = E_0 (\omega t - kx)$ $\omega = 2\pi \times 10^6$

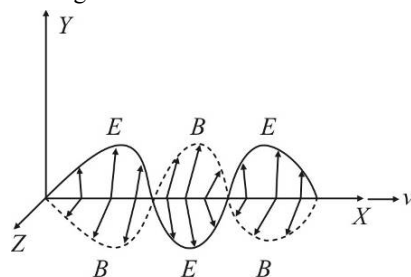
$$2\pi v = 2\pi \times 10^6 \Rightarrow v = 10^6 \text{ Hz}$$

$$\frac{2\pi}{\lambda} = k \Rightarrow \frac{2\pi}{\lambda} = \pi \times 10^{-2} \Rightarrow \lambda = \frac{2\pi}{\pi \times 10^{-2}} = 2 \times 10^2 = 200 \text{ m}$$

- 13.(A) The velocity of electromagnetic radiation in vacuum is $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$, where μ_0 and ϵ_0 are the permeability and permittivity of vacuum.

- 14.(C) In electromagnetic wave, electric and magnetic field are in phase and perpendicular to each other and also perpendicular to the direction of the propagation of the wave.

- 15.(B) $\lambda_m > \lambda_v > \lambda_x$. In spectrum X -rays has minimum wavelength and microwave has maximum wavelength
- 16.(A) Every body at all time. All at temperatures emit radiation (except at $T = 0$), which fall in the infrared region.
- 17.(C)
- 18.(B) According to Maxwell, the electromagnetic waves are those waves in which there are sinusoidal variation of electric and magnetic field vectors at to the direction of wave propagation.



Propagation of electro-magnetic waves

If the electric field ($\vec{E}$) and magnetic field ($\vec{B}$) and vibrating along Y and Z direction, propagation of electromagnetic wave will be along the X -axis. Therefore, the velocity of electromagnetic wave is parallel to $\vec{E} \times \vec{B}$.

- 19.(A) As the electro- magnetic radiations from sun pass through the atmosphere, some of them are absorbed by it while other reach the surface of earth, the range of wavelength which reaches earth lies in infrared region. This part of the radiation from the sun has shorter wavelength and can penetrate through the layer of gases line CO_2 and reach earth surface. But the radiation from the earth being of longer wavelength can't escapes through this layer. As a result the earth surface gets warm. This is known as green house effect.
- 20.(A) The ozone layer absorbs the harmful ultraviolet rays coming from sun. 21.(A)
- 22.(D) The range is from $380nm$ to even $200nm$ to $120nm$
- 23.(B) $\lambda = \frac{3 \times 10^8}{100Hz} = 3 \times 10^6 m$
- 24.(B) $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (free space) $v = \frac{1}{\sqrt{\mu \epsilon}}$ (medium) $\therefore \mu = \frac{c}{v} = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}}$
- 25.(B) 26.(D) 27.(B) X-rays are used for the investigation of structure of solids.
- 28.(B) The wave length of radiation used should be less than the size of the particle
- Size of particle = $\lambda = \frac{c}{\nu} \quad 3 \times 10^{-4} = \frac{3 \times 10^{10}}{\nu}$ or $\nu = 10^{14}$ hertz
- However, when frequency is higher than this, wavelength is still smaller, Resolution becomes better.
- 29.(D) Rays wave length [Range in m]
- | | |
|-----------------|--|
| X -rays | 1×10^{-11} to 3×10^{-8} |
| γ - rays | 6×10^{-14} to 1×10^{-11} |
| Microwaves | 10^{-3} to 0.3 |
| Radiowaves | 10 to 10^4 |
- 30.(A) $E_{radiowave} < E_{yellow} < E_{blue} < E_{x-ray}$ 31.(A)
- (D) (B) (A) (C)

- 32.(D) An electromagnetic wave propagating in + x direction means electric field and magnetic field should be function of x and t.

$$\text{Also, } \vec{E} \perp \vec{B} \text{ or } \hat{E} \perp \hat{B} \text{ i.e. } ; \quad (\hat{y} - \hat{z}) \cdot (\hat{y} + \hat{z}) = \hat{y} \cdot \hat{y} - \hat{z} \cdot \hat{z} = 0$$

- 33.(D) $\nu = 2 \times 10^{14} \text{ Hz}$

$$E_0 = 27 \text{ V m}^{-1}$$

$$\text{We know, } \frac{E_0}{B_0} = c ; \quad \text{So } B_0 = \frac{27}{3 \times 10^8} = 9 \times 10^{-8} \text{ T}$$

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{2 \times 10^{14}} = 1.5 \times 10^{-6} \text{ m}$$

$$B = B_0 \sin 2\pi \left(\frac{x}{\lambda} - \nu t \right) ; \quad B = (9 \times 10^{-8} \text{ T}) \sin 2\pi \left(\frac{x}{1.5 \times 10^{-6}} - 2 \times 10^{14} t \right)$$

Oscillation of B can be along either $\hat{j}$ or $\hat{k}$ direction.

- 34.(B) For electromagnetic wave, direction of propagation, $\vec{E}$ and $\vec{B}$ are transverse in nature.

According to questions,

$\vec{E} \times \vec{B}$ = direction of propagation = +z direction. Only option (b) satisfies both conditions

(i) $\vec{E} \cdot \vec{B} = 0$ (ii) $(\vec{E} \times \vec{B})$ directed along the z-axis.

- 35.(2) For partially absorbed and reflected

$$F = \frac{1}{c} \left(\frac{dE}{dt} \right) (2r + a)$$

$$\text{But } r + a = 1 \Rightarrow r = 1 - a$$

$$F = \frac{1}{c} \frac{dE}{dt} [2(1 - a) + a] = \frac{1}{c} \frac{dE}{dt} [2 - a]$$

$$\text{Radiation pressure} = \frac{E}{A} = \frac{SA}{c} \frac{(2 - f)}{A} = \frac{S}{c} (2 - f)$$

- 36.(13) $P = 100 \text{ W}$

$$r = 2 \text{ m}$$

$$D = \text{Diameter of Pupil} = 4 \times 10^{-3} \text{ m}$$

$$\lambda = 600 \times 10^{-9} \text{ m}$$

$$P = n \frac{hc}{\lambda}; \quad n = \text{no. of photons/sec}$$

$$\text{No. of photons per unit by source} = \frac{P}{A \frac{hc}{\lambda}} = \frac{\lambda P}{Ahc}$$

$$\therefore \text{No. of photons entering the eye} = \frac{\lambda P}{Ahc} \times \left(\frac{\pi D^2}{4} \right) \quad \text{Here, } A = \pi(2)^2$$

$$N = \text{no. of photons entering the eye} = \frac{600 \times 10^{-9} \times 100}{\pi(2)^2 \times 6.62 \times 10^{-34} \times 3 \times 10^8} \times \frac{\pi(4 \times 10^{-3})^2}{4} = 7.55 \times 10^{13}$$

- 37.(19) Energy absorbed = $mCdT$

$$n \frac{hc}{\lambda} = mCdT$$

Here, n = no. of photons

$$n \frac{hc}{\lambda} = Vol. \times \rho \times CdT$$

$$n = \frac{Vol. \times \rho \times CdT \times \lambda}{hc} = \frac{10^{-6} \times 1000 \times 1 \times 1 \times 575 \times 10^{-9}}{6.62 \times 10^{-34} \times 3 \times 10^8}$$

$$n = 1.2 \times 10^{19}$$

38.(2) Rad. Of capacitor = 0.12

Capacitance, $C = 5 \times 10^{-6} F$

Resistance, $R = 4 \times 10^5 \Omega$

$$I_D = \frac{\epsilon_0 d \phi E}{dt} = \frac{\epsilon_0 A}{d} \frac{dV}{dt} = \frac{CdV}{dt} = 5 \times 10^{-6} \times 2000 = 0.01 A$$

39.(5) $I = 220 W / m^2$

$R = 0.3 m$

$$F = \frac{dP}{dt} = \frac{1}{c} \frac{dE}{dt} (2r + a) = \frac{IA}{c} (2r + a)$$

$$dp = \frac{IA}{c} (2r + a) \times dt = \frac{220 \times \pi (0.3)^2}{3 \times 10^8} [2 \times 0.4 + 0.6] \times 5 \times 60 = 8.71 \times 10^{-5} kg m/s$$

40.(3) r = distance = 6m

E_0 = electric field amplitude = 10 V/m

$$\frac{E_0}{B_0} = c \Rightarrow B_0 = \frac{E_0}{c} = \frac{10}{3 \times 10^8} = 3.33 \times 10^{-8} T$$

41.(500)
$$\frac{P}{4\pi R^2} = \frac{1}{2} \epsilon_0 E^2 c$$

$$R = \sqrt{\frac{2P}{4\pi\epsilon_0 E_c^2}}$$

On substitution, we get $R \sim 5.00 m$

42.(714)
$$E_{rms} = \sqrt{\mu_0 CI}$$

43.(18)
$$F = \text{Projected Area} \times \text{radiation pressure} = \frac{\pi R^2 I}{C}$$

44.(13)
$$\frac{\text{Number of photons}}{\text{Area s}} = \frac{P\lambda}{4\pi hcr^2} = 6 \times 10^{13} cm^{-2} s^{-1}$$

45.(2)
$$E = \frac{12400}{\lambda_A} eV = \frac{12400}{6000} = 2 eV$$

46.(2)
$$E = 2.5 \cos(2\pi \times 10^6 t - \pi \times 10^{-2} x)$$

Comparing with

$$E = E_0 \cos(\omega t - kx)$$

$$k = \pi \times 10^{-2}$$

$$\frac{2\pi}{\lambda} = \pi \times 10^{-2}$$

$$\lambda = 200 \text{ m} = (100 \times) \text{ meter}$$

Where, $x = 2$

$$47.(2) I_{av} = \left(\frac{1}{2} \epsilon_0 E_0^2 \right) c = \frac{8.854 \times 10^{-12} \times (38.8)^2 \times 3 \times 10^8}{2}$$

$$\approx 2 \text{ watt / m}^2$$

$$48.(4) \lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.5 \times 10^6} = 40 \text{ m}$$

$$49.(4) B_0 = \frac{E_0}{c} = \frac{120}{3 \times 10^8} = 4 \times 10^{-7} \text{ T} \quad \therefore \quad n = 4$$